Exercises of A Course in Probability Theory

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Contents

1 Distribution function

1.1 Monotone functions

Exercise 1.1.1. Prove that for the f in Example 2 we have

$$f(-\infty) = 0, \quad f(+\infty) = 1.$$

Proof. Since the uniformly convergence of f, we have

$$\begin{aligned} f(-\infty) &= \lim_{x\to-\infty} f(x) = \sum_{n=1}^{\infty} b_n \lim_{x\to-\infty} \delta_{a_n}(x) = 0, \\ f(+\infty) &= \lim_{x\to+\infty} f(x) = \sum_{n=1}^{\infty} b_n \lim_{x\to+\infty} \delta_{a_n}(x) = \sum_{n=1}^{\infty} b_n = 1. \end{aligned}$$ □

Exercise 1.1.2. Construct an increasing function on $(-\infty, +\infty)$ with a jump of size one at, each integer, and constant between jumps. Such a function cannot be represented as $\sum_{n=1}^{\infty} b_n\delta_n(x)$ with $b_n = 1$ for each n, but a slight modification will do. Spell this out.

Proof. $f(x) = [x]$ is a function satisfying the requirement. In fact, by modifying the form of f to

$$f(x) = \sum_{n=1}^{\infty}[\delta_n(x) - \delta_n(-x)] + \delta_0(x),$$

then f satisfy the requirement. □

Exercise 1.1.3. Suppose that f is increasing and that there exist real numbers A and B such that $\forall x : A \le f(x) \le B$. Show that for each $\varepsilon > 0$, the number of jumps of size exceeding ε is at most $(B - A)/\varepsilon$, Hence prove (iv), first for bounded f and then in general.

Proof. If the number of jumps of size exceeding ε is more than $(B - A)/\varepsilon$, as f is increasing, the total increase of f is more than $(B - A)/\varepsilon \cdot \varepsilon = B - A$, which is a contradiction to the bounded condition of f. The point of discontinuity of f can be written as

$$\bigcup_{n=1}^{\infty}\{x|f(x+) - f(x-) \ge (B - A)/n\},$$

which is a countable set. For arbitrary unbounded f, define

$$f_n(x) = \begin{cases} f(-n) & x < -n, \\ f(x) & -n \le x < n, \\ f(n) & x \ge n. \end{cases}$$

Then f_n is bounded and increasing, let the discontinuity set of f_n is D_n, then D_n is countable. The discontinuity set of f can be represented as $\bigcup_{n=1}^{\infty} D_n$, which is countable. □

Exercise 1.1.4. Let f be an arbitrary function on $(-\infty, +\infty)$ and L be the set of x where f is right continuous but not left continuous. Prove that L is a countable set. [HINT: Consider $L \cap M_n$, where $M_n = \{x|O(f; x) > 1/n\}$ and $O(f; x)$ is the oscillation of f at x.]

Proof. Let $M_n = \{x|O(f;x) > 1/n\}, O(f;x)$ is the oscillation of f at x. Suppose $\exists x' \in L \cap M_n, \forall \delta > 0, \exists x'' \in (x', x''+\delta), x'' \in L \cap M_n$. According to the definition of L and M_n, for $\varepsilon_n = 1/(2n)$, there exists $\delta_n > 0$, such that for $x' < x''$, $x', x'' \in L \cap M_n, x'' - x' < \delta_n$, and $|f(x') - f(x)| < \varepsilon_n,\ x \in (x', x'+\delta_n)$. Then, $\exists \eta > 0$, s.t. $(x''-\eta, x''+\eta) \subset (x', x'+\delta_n)$, and $\forall x \in (x''-\eta, x''+\eta)$,

$$|f(x) - f(x'')| \le |f(x) - f(x')| + |f(x') - f(x'')| < \frac{1}{n},$$

which is a contradiction to the definition of M_n. Hence, $\forall x \in L \cap M_n, \exists \delta_x > 0, (x, x+\delta_x) \cap (L \cap M_n) = \varnothing$. Take q_x from each interval, then there is a one-to-one correspondence between $L \cap M_n$ and a subset of rational numbers, which implies that $L \cap M_n$ is countable. So

$$L \subset \bigcup_{n=1}^{\infty} (L \cap M_n)$$

is countable. (Notice $\bigcup_{n=1}^{\infty} M_n$ is the set of discontinuous point of f, and L is a subset of it.) □

Exercise 1.1.5. Let f and $\tilde{f}$ be as in (vii). Show that the continuity of f on D does not imply that of $\tilde{f}$ on $(-\infty, +\infty)$, but uniform continuity does imply uniform continuity.

Proof. Define $D = (-\infty, 0) \cup (0, +\infty)$,

$$f(x) = \begin{cases} -1, & x < 0, \\ 1, & x > 0, \end{cases}$$

then f is continuous on D. However, $\tilde{f}(0) = 1$, but $\tilde{f}(0^-) = -1$, which means $\tilde{f}$ is not continuous. The discontinuity is essentially caused by the way how D is defined. But for uniform continuity, $\forall \varepsilon > 0, \exists \delta > 0, \forall x', x'' \in D, |x' - x''| < \delta, |f(x') - f(x'')| < \varepsilon/2$. Let $x' < x''$ and $x'' - x' < \delta/2$ without loss of generality, because $\tilde{f}$ is increasing, $\tilde{f}$, we shall prove $\tilde{f}(x'') - \tilde{f}(x') < \varepsilon$. Now there exists $t_1 \in (x', +\infty) \cap D$, s.t. $\tilde{f}(t_1) < f(x') + \varepsilon/2$. If $t_1 > x''$, then

$$\tilde{f}(x'') - \tilde{f}(x') \le \tilde{f}(t_1) - \tilde{f}(x') < \varepsilon/2 < \varepsilon.$$

If $t_1 \le x''$, take $t_2 \in (x'', x''+\delta/2)$, and $|t_2 - t_1| \le |t_1 - x''| + |t_2 - x''| < \delta$. Then

$$\begin{aligned} f(x'') \le f(t_2) \le f(t_1) + \frac{\varepsilon}{2} \quad & (\text{uniform continuity of} f) \\ < f(x') + \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = f(x') + \varepsilon. & \end{aligned}$$

Hence, the uniform continuity of f implies the uniform continuity of $\tilde{f}$. □

Exercise 1.1.6. Given any extended-valued f on $(-\infty, +\infty)$, there exists a countable set D with the following property. For each t, there exist $t_n \in D, t_n \to t$ such that $f(t) = \lim_{n\to\infty} f(t_n)$. This assertion remains true if "$t_n \to t$" is replaced by "$t_n \downarrow t$" or "$t_n \uparrow t$". [This is the crux of "separability" for stochastic process. Consider the graph of $(t, f(t))$ and introduce a metric.]

Proof. Let

$$G = \{(t, f(t)) | t \in (-\infty, +\infty)\},$$

which is the graph of f. Let

$$\mathscr{B} = \{(p, q) \times (r, s) | p, q, r, s \in \mathbb{Q}, p < q, r < s\},$$

and $\mathscr{B}$ is countable. Consider $B_k \in \mathscr{B}$ such that $B_k \cap G \neq \varnothing$. Then, for each k, choose a point $(t_k, f(t_k)) \in B_k \cap G$. The set of all such points is countable, $\forall (t, f(t)) \in G$, let

$$U_n = \left(t - \frac{1}{n}, t + \frac{1}{n}\right) \times \left(f(t) - \frac{1}{n}, f(t) + \frac{1}{n}\right),$$

as rational numbers are dense in real numbers, there exists p_n, q_n, r_n, s_n, s.t.

$$t - \frac{1}{n} < p_n < t < q_n < t + \frac{1}{n},$$
$$f(t) - \frac{1}{n} < r_n < f(t) < s_n < f(t) + \frac{1}{n}.$$

Let $B_{k_n} = (p_n, q_n) \times (r_n, s_n) \in \{B_k\}$, then $B_{k_n} \cap G \neq \varnothing$. Choose $(t_{k_n}, f(t_{k_n})) \in B_{k_n} \cap G$, then $|t_{k_n} - t| < 1/n$, $|f(t_{k_n}) - f(t)| < 1/n$, which implies

$$t_{k_n} \to t,\ f(t_{k_n}) \to f(t)(n \to \infty).$$

Let $D = \{t_{k_n}\}$, which is a desirable result. □

1.2 Distribution functions

Exercise 1.2.1. Let F be a d.f. Then for each x,

$$\lim_{\varepsilon \downarrow 0} [F(x + \varepsilon) - F(x - \varepsilon)] = 0$$

unless x is a point of jump of F, in which case the limit is equal to the size of the jump.

Proof. Note that F has a unique decomposition

$$F = F_c + F_d,$$

where F_c is continuous and F_d is discrete, then

$$\begin{aligned}\lim_{\varepsilon \downarrow 0} [F(x + \varepsilon) - F(x - \varepsilon)] &= \lim_{\varepsilon \downarrow 0} [F_c(x + \varepsilon) - F_c(x - \varepsilon) + F_d(x + \varepsilon) - F_d(x - \varepsilon)] \\ &= F_d(x) - F_d(x-). \quad \text{(since } F_d \text{ is right continuous)}\end{aligned}$$

only at the point of jump of F (also the point of jump of F_d), $F_d(x) - F_d(x-) \neq 0$ and equal to the size of jump. □

Exercise 1.2.2. Let F be a d.f. with points of jump $\{a_j\}$. Prove that the sum

$$\sum_{x - \varepsilon < a_j < x} [F(a_j) - F(a_j-)]$$

converges to zero as $\varepsilon \downarrow 0$, for every x. What if the summation above is extended to $x - \varepsilon < a_j \leq x$ instead? Give another proof of the continuity of F_c in Theorem 1.2.1 by using this problem.

Proof. For any finite number of points in $\{a_j\}$, say $a_1 < a_2 < \cdots < a_k$, we have

$$\begin{aligned}\sum_{i=1}^{k} [F(a_i) - F(a_i-)] &\leq \sum_{i=1}^{k} [F(a_i) - F(a_{i-1})] \\ &= F(a_k) - F(a_1) \leq F(x-) - F(x - \varepsilon).\end{aligned}$$

Notice that the inequality holds for all $k \in \mathbb{N}_+$, as $\{a_j\}$ is countable, we have

$$0 \le \sum_{x-\varepsilon<a_j<x} [F(a_j) - F(a_j-)] \le F(x-) - F(x-\varepsilon).$$

Let $\varepsilon \downarrow 0$, as $F(x-) - F(x-\varepsilon) \to 0$, by squeeze theorem, we have this summation converge to 0. But if the summation range is extended, it is possible that $a_i = x$ for one i and then the summation converges to the jump size at a_i .(i.e. $F(x) - F(x-)$) But no matter whether x is a jump or not, we always have the summation converging to $F(x) - F(x-)$.

Now we use this to prove F_c is continuous. Notice that

$$\begin{aligned} F_d(x) - F_d(x-\varepsilon) &= \sum_j b_j \delta_{a_j}(x) - \sum_j b_j \delta_{a_j}(x-\varepsilon) \\ &= \sum_{x-\varepsilon<a_j\le x} b_j = \sum_{x-\varepsilon<a_j\le x} [F(a_j) - F(a_j-)]. \end{aligned}$$

then

$$F_c(x) - F_c(x-\varepsilon) = [F(x) - F(x-\varepsilon)] - \sum_{x-\varepsilon<a_j\le x} [F(a_j) - F(a_j-)].$$

Let $\varepsilon \downarrow 0$, using the lemma in this question, we conclude that F_c is left continuous. The right continuity of F_c is obvious. □

Exercise 1.2.3. A plausible verbal definition of a discrete d.f. may be given thus: "It is a d.f. that has jumps and is constant between jumps." [Such a function is sometimes called a "step function", though the meaning of this term does not seem to be well established.] What is wrong with this? But suppose that the set of points of jumps is "discrete" in the Euclidean topology, then the definition is valid. (apart from our convention of right continuity).

Proof. If the jumps is condense in $\mathbb{R}$, then we cannot find such step function. For instance, let $\{r_1, r_2, \cdots\}$ is a enumeration of $\mathbb{Q}$, then

$$F(x) = \sum_{k:r_k\le x} \frac{1}{2^k}$$

is a valid d.f. However, for any number x and any given $\delta > 0$, $\exists r_k \in U(x,\delta)$, which means $F(x)$ is strictly increasing. But in Euclidean topology, if the jumps are discrete, i.e.

$$\forall a_i \in D, \exists \delta_i > 0, |a_i - a_j| \ge \delta_i, \forall j \ne i,$$

then the definition is valid. □

Exercise 1.2.4. For a general increasing function F there is a similar decomposition $F = F_c + F_d$, where both F_c and F_d is increasing, F_c is continuous, and F_d is "purely jumping". [HINT: Let a be a point of continuity, put $F_d(a) = F(a)$, add jumps in (a,∞) and subtract jumps in $(-\infty, a)$ to define F_d. Cf. Exercise 2 in Sec. 1.1.]

Proof. We can assume F is a right continuous function without loss of generality. To avert the possible infinity in the summation, let $D = \{a_1, a_2, \cdots\}$ be the discontinuous point set of F, choose a continuous point of F (name it a) and define

$$F_d(x) = \begin{cases} F(a) + \sum_{a<a_j\le x} [F(a_j) - F(a_j-)] & x > a, \\ F(a) & x = a, \\ F(a) - \sum_{x<a_j<a} [F(a_j) - F(a_j-)] & x < a, \end{cases}$$

and put $F_c = F - F_d$. Actually, F_d can be rewritten as

$$F_d(x) = F(a) + \sum_{j=1}^{\infty}[F(a_j) - F(a_j-)][\delta_{a_j}(x) - \delta_{a_j}(a)].$$

The proof of basic properties of F_c (positive, increasing, continuous) quite resembles the proof of Theorem 1.2.1. □

Exercise 1.2.5. Theorem 1.2.2 can be generalized to any bounded increasing function. More generally, let f be the difference of two bounded increasing functions on $(-\infty, +\infty)$; such a function is said to be *of bounded variation* there. Define its purely discontinuous and continuous parts and prove the corresponding decomposition theorem.

Proof. Suppose g and h are two bounded increasing function and $f = g - h$ (without loss of generality we assume g and h are right continuous). Let D_g and D_h represent the set of discontinuous points of g and h, and then $D_f = D_g \cup D_h = \{x_1, x_2, \cdots\}$ is countable and is the discontinuous point set of f. Now we put

$$f_d(x) = \sum_{n=1}^{\infty}[f(x_n) - f(x_n-)][\delta_{x_n}(x) - \delta_{x_n}(x_0)] \text{ and } f_c = f - f_d,$$

where x_0 is any given continuous point of both g and h. As is discussed in Exercise 1.2.4, g and h has a decomposition

$$\begin{aligned} g &= g_c + g_d, \\ h &= h_c + h_d. \end{aligned}$$

Then f can be written as

$$f = (g_c - h_c) + (g_d - h_d) = \tilde{f}_c + \tilde{f}_d.$$

Now we shall prove $\tilde{f}_d = f_d$. First

$$\begin{aligned} \sum_{n=1}^{\infty}|f(x_n) - f(x_n-)| &= \sum_{n=1}^{\infty}|[g(x_n) - g(x_n-)] - [h(x_n) - h(x_n-)]| \\ &\leq \sum_{n=1}^{\infty}[|g(x_n) - g(x_n-)| + |h(x_n) - h(x_n-)|] < \infty, \end{aligned}$$

which means we can arbitrarily change the order of summation. For $x \in D_g \backslash D_h, h(x_n) - h(x_n-) = 0$ do not influence the summation, vice versa. As g_d and h_d has the form of

$$\begin{aligned} g_d(x) &= \sum_{x_n \in D_g}[g(x_n) - g(x_n-)][\delta_{x_n}(x) - \delta_{x_n}(x_0)], \\ h_d(x) &= \sum_{x_n \in D_h}[h(x_n) - h(x_n-)][\delta_{x_n}(x) - \delta_{x_n}(x_0)], \end{aligned}$$

using the fact above, the difference is

$$\tilde{f}_d(x) = \sum_{x_n \in D_g \cup D_h}[g(x_n) - g(x_n-) - h(x_n) + h(x_n-)][\delta_{x_n}(x) - \delta_{x_n}(x_0)] = f_d(x).$$

The uniqueness can be proved like what is done in Theorem 1.2.2. □

Exercise 1.2.6. A point x is said to belong to the support of the d.f. F iff for every $\varepsilon > 0$ we have $F(x+\varepsilon) - F(x-\varepsilon) > 0$. The set of all such x is called *the support of F*. Show that each point of jump belongs to the support, and that each isolated point of the support is a point of jump. Give an example of a discrete d.f. whose support is the whole line.

Proof. For each point of jump of d.f. F, say a_j, we know

$$F(a_j) - F(a_j-) = b_j > 0.$$

As F is a increasing function, $\forall \varepsilon > 0$, we have

$$F(a_j+\varepsilon) - F(a_j-\varepsilon) \geq F(a_j) - F(a_j-) > 0.$$

So a_j is the support. For a point of support x which is isolated, there exists an interval $U = (x-\delta, x+\delta)$ and $\forall x' \in U, \exists \varepsilon_{x'} > 0, F(x'+\varepsilon_{x'}) - F(x'-\varepsilon_{x'}) = 0$, that means F is continuous at x'. So x can only be a jump according to the condition. The example of a d.f. which support is the whole line is given in Exercise 1.2.3. □

Exercise 1.2.7. Prove that the support of any d.f. is a closed set, and the support of any continuous d.f. is a perfect set.

Proof. Let S represents the set of the support of a d.f. Let $s \in D'$, then $\exists s_n \in S, s_n \to s (n \to \infty)$. Now $\forall \varepsilon > 0, \exists n \in \mathbb{N}_+, |s - s_n| < \varepsilon/2$. As s_n is the support of F, $F(s_n+\varepsilon/2) - F(s_n-\varepsilon/2) > 0$. As

$$\left|s - \left(s_n + \frac{\varepsilon}{2}\right)\right| \leq |s - s_n| + \frac{\varepsilon}{2} < \varepsilon,$$
$$\left|s - \left(s_n - \frac{\varepsilon}{2}\right)\right| \leq |s - s_n| + \frac{\varepsilon}{2} < \varepsilon,$$

so $(s_n - \varepsilon/2, s_n + \varepsilon/2) \subset (s-\varepsilon, s+\varepsilon)$, which leads to

$$F(s+\varepsilon) - F(s-\varepsilon) \geq F\left(s_n + \frac{\varepsilon}{2}\right) - F\left(s_n - \frac{\varepsilon}{2}\right) > 0.$$

So $s \in S$, which means S is a closed set.

For a continuous d.f. F, according to Exercise 1.2.6, if F has an isolated point of support, then it is a point of jump, a contradiction to the continuity. Furthermore, the support of F is a perfect set. □

1.3 Absolutely continuous and singular distributions

Exercise 1.3.1. A d.f. f is singular if and only if $F = F_s$; it is absolutely continuous if and only if $F \equiv F_{ac}$.

Proof. If F is singular, then $F' = 0$ a.e. and F is not identically zero. So

$$\forall x \in \mathbb{R}, F_{ac}(x) = \int_{-\infty}^{x} F'(t)\mathrm{d}t = 0,$$

which implies $F = F_s$. If $F = F_s$, then $F_{ac} = 0$, so $F' = 0$ a.e. Further to that, F is a d.f., so $F(+\infty) = 1$, which means F is not identically zero, so F is singular.

If F is absolutely continuous, then there exists f,

$$f \geq 0 \text{ a.e. and } \int_{-\infty}^{+\infty} f(t)\mathrm{d}t = 1,$$

and use the definition we have

$$F(x) = \int_{-\infty}^{x} f(t)\mathrm{d}t = F_{ac}(x)$$

if we put $f(x) = F'(x)$. Reversely, if $F = F_{ac}$, Then

$$\forall x' > x, F(x') - F(x) = \int_{x}^{x'} F'(t)\mathrm{d}t,$$

and obviously F is absolutely continuous. □

Exercise 1.3.2. Prove Theorem 1.3.2.

Proof. As we have $F = F_{ac} + F_s$, and the discrete part of F is contained in F_s, call it F_d, then F_s can be written as

$$F_s = F_d + F_{sc},$$

where F_{sc} is the singular continuous part of F. Note that $F - F_d = F_c$ and F_{ac} is continuous, so F_{sc} is indeed continuous. And because $F'_d = 0$ a.e., we have $F'_{sc} = 0$ a.e. Thus F_{sc} is singular. Now put

$$\alpha_1 = \lim_{x\to\infty} F_s(x), \quad \alpha_2 = \lim_{x\to\infty} F_{sc}(x), \quad \alpha_3 = \lim_{x\to\infty} F_{ac}(x),$$

then $\alpha_1 + \alpha_2 + \alpha_3 = 1$, and let

$$F_1(x) = \frac{1}{\alpha_1}F_s(x), \quad F_2(x) = \frac{1}{\alpha_2}F_{sc}(x), \quad F_3(x) = \frac{1}{\alpha_3}F_{ac}(x),$$

Then F can be written as

$$F = \alpha_1 F_1 + \alpha_2 F_2 + \alpha_3 F_3,$$

which is a form of convex combination. Suppose there exists another set of functions G_d, G_{sc}, G_{ac} such that $F = G_d + G_{ac} + G_{sc}$, by the uniqueness of decomposition $F = F_d + F_c$, we have

$$F_d = G_d, \quad F_c = F_{ac} + F_{sc} = G_c = G_{ac} + G_{sc}.$$

as $F'_{sc} = 0$ a.e. and $F'_{ac} = 0$ a.e., we have $F'_{ac} = G'_{ac}$ a.e. As F_{ac} and G_{ac} is absolutely continuous, intergraling from $-\infty$ to x yields

$$F_{ac}(x) = \int_{-\infty}^{x} F'_{ac}(t)\mathrm{d}t = \int_{-\infty}^{x} G'_{ac}(t)\mathrm{d}t = G_{ac}(x).$$

So $F_{ac} = G_{ac}$, which leads to $F_{sc} = G_{sc}$. Uniqueness confirmed. □

Exercise 1.3.3. If the support of a d.f. (see Exercise 6 of Sec. 1.2) is of measure zero, then F is singular. The converse is false.

Proof. Let S represent the support of d.f. As is proved before, S is a closed set, so S^c is an open set. $\forall x \in S^c, \exists \delta > 0, F(x+\delta) - F(x-\delta) = 0$, which means $F'(x) = 0, \forall x \in S^c$. As $m(S) = 0$, we have $F'(x) = 0$ a.e. So F is singular. However, if $\{r_n\}$ is a enumeration of rational numbers in $[0, 1]$, and set

$$F(x) = \sum_{n=1}^{\infty} \frac{\delta_{r_n}(x)}{2^n},$$

as is proved before, the support of F is $[0, 1]$ and $F'(x) = 0$ at irrational points in $[0, 1]$ and $(-\infty, 0) \cup (1, +\infty)$, which means $F' = 0$ a.e. However, $m(S) = 1$. □

Exercise 1.3.4. Suppose that F is a d.f. and (3) holds with a continuous f. Then $F' = f \geq 0$ everywhere.

Proof. Now that f is continuous, we can use basic calculus theorem to solve the problem. As

$$F(x) = \int_{-\infty}^{x} f(t)\mathrm{d}t = \int_{-\infty}^{a} f(t)\mathrm{d}t + \int_{a}^{x} f(t)\mathrm{d}t = \int_{a}^{x} f(t)\mathrm{d}t + C,$$

$F' = f$ everywhere is obvious by derivating on both side of (3). Suppose $\exists x_0$ such that $f(x_0) < 0$, as f is continuous, $\exists \delta > 0, \forall x \in (x_0 - \delta, x_0 + \delta), f(x) < 0$, which leads to

$$F(x_0 + \delta) = F(x_0 - \delta) + \int_{x_0-\delta}^{x_0+\delta} f(t)\mathrm{d}t < F(x_0 - \delta),$$

which is a contradiction to the increasing feature of F. So $f \geq 0$. □

Exercise 1.3.5. Under the conditions in the preceding exercise, the support of F is the clousure of the set $\{t|f(t) > 0\}$; the complement of the support is the *interior* of the set $\{t|f(t) = 0\}$.

Proof. Still denote S as the support of F. Notice that

$$F(x+\varepsilon) - F(x-\varepsilon) = \int_{x-\varepsilon}^{x+\varepsilon} f(t)\mathrm{d}t,$$

Suppose $x \in S$, then $\int_{x-\varepsilon}^{x+\varepsilon} f(t)\mathrm{d}t > 0$, which means $\forall \varepsilon > 0, \exists t_\varepsilon \in U(x;\varepsilon), f(t_\varepsilon) > 0$. Now denote $G = \{t|f(t) > 0\}$. If $f(x) > 0$, then $x \in G$; If $f(x) = 0$, take $\varepsilon_n = 1/n$ and $\exists t_n \in U(x;\varepsilon_n), t_n \to x$, which means $x \in G'$. In both cases, $x \in \overline{G}$. And for any $x \in S^c, \exists \delta > 0, \int_{x-\delta}^{x+\delta} f(t)\mathrm{d}t = 0$, which means $f(t) = 0, t \in U(x;\delta)$, so x is an interior point of G^c. □

Exercise 1.3.6. Prove that a discrete distribution is singular. [Cf. Exercise 13 of Sec. 2.2.]

Proof. The discrete distribution F has $F'(x) = 0$ a.e. but is not identically zero, so is singular. □

Exercise 1.3.7. Prove that a singular function as defined here is (Lebesgue) measurable but need not be of bounded variation even locally. [HINT: Such a function is continuous except on a set of Lebesgue measure zero; use the completeness of the Lebesgue measure.]

Proof. Denote E as the continuous part of a singular function F, and $m(E^c) = 0$. Now

$$\{x|F(x) < a\} = \{x \in E|F(x) < a\} \cup \{x \in E^c|F(x) < a\},$$

as $\{E^c\}$ is of measure zero, the second set is measurable, and the first set is an open set, which is also measurable, so F is measurable. Let K represents the cantor set on $[0, 1]$, then $m(K) = 0$. On $[0, 1]\backslash K$, define $I_1, I_2, I_3, \cdots$, which is the interval removed in order when constructing K, we define F as follow:

(1) Set $F(x) = 1$ on $I_1 = (1/3, 2/3)$;

(2) Set $F(x) = -1$ on $I_2 = (1/9, 2/9), I_2 = (7/9, 8/9)$;

(3) Set $F(x) = (-1)^{n+1}$ on $I_{2^{n-1}}, I_{2^{n-1}+1}, \cdots, I_{2^n-1}$;

(4) $\cdots$

We can easily verify that F is measurable but not of bounded variation, since we can choose a path through those intervals. □

The F in these exercises is the $\tilde{F}$ defined above (see $\tilde{F}$ before these exercises).

Exercise 1.3.8. Prove that the support of F is exactly C.

Proof. As is stated in the book, we can guarantee that $S \subset C$, where S is the support of F. For $x \in C$, suppose x is not the support of F, then $\exists \delta > 0, F(t) \equiv F(x), t \in U(x; \delta)$. However, as C is a perfect set, $\forall \varepsilon > 0$, we can always find two different intervals in D on two sides of x. The values taken on these two intervals is different according to the construction. Furthermore, F is increasing, so we can assert that $F(x + \varepsilon) - F(x - \varepsilon) > 0$. So $S = C$. □

Exercise 1.3.9. It is well known that any point x in C has a ternary expansion without the digit 1:

$$x = \sum_{n=1}^{\infty} \frac{a_n}{3^n}, \quad a_n = 0 \text{ or } 2.$$

Prove that for this x we have

$$F(x) = \sum_{n=1}^{\infty} \frac{a_n}{2^{n+1}}.$$

Proof. For $x \in J_{n,k}$, $F(x) = c_{n,k} = k/2^n$. Now we can construct a series x_n, such that

$$x_n = \sum_{k=1}^{n} \frac{a_k}{3^k}, \quad n = 1, 2, \cdots$$

then obviously $x_n \uparrow x, F(x_n) \uparrow F(x) (n \to \infty)$. Now for x_1, the equation can be easily verified. Now suppose for x_n the equation holds, which means

$$F(x_n) = \sum_{k=1}^{n} \frac{a_n}{2^{n+1}}, \text{ and } x_{n+1} = x_n + \frac{a_{n+1}}{3^{n+1}}.$$

We should notice the construction step of F. In n th step, actually we are choosing the next end point of one new interval. If $a_{n+1} = 0$, we don't make any change; If $a_{n+1} = 2$, then we move to the second interval that is not removed, and we should add $1/2^n = a_{n+1}/2^{n+1}$. In each case, we always have

$$F(x_{n+1}) = \sum_{k=1}^{n} \frac{a_k}{2^{k+1}} + \frac{a_{n+1}}{2^{n+1}} = \sum_{k=1}^{n+1} \frac{a_k}{2^{k+1}}.$$

Finally, using the continuity of F, we have

$$F(x) = F\left(\lim_{n\to\infty} x_n\right) = \lim_{n\to\infty} F(x_n) = \sum_{n=1}^{\infty} \frac{a_n}{2^{n+1}}.$$

□

Exercise 1.3.10. For each $x \in [0,1]$, we have

$$2F\left(\frac{x}{3}\right) = F(x), \;\; 2F\left(\frac{2}{3}+\frac{x}{3}\right) - 1 = F(x).$$

Proof. Using the conclusion in Exercise 1.3.9, as

$$\frac{x}{3} = \sum_{n=1}^{\infty} \frac{a_n}{3^{n+1}} = 0 + \sum_{n=2}^{\infty} \frac{a_{n-1}}{3^n},$$

we have

$$\begin{aligned} 2F\left(\frac{x}{3}\right) &= 2\sum_{n=2}^{\infty} \frac{a_{n-1}}{2^{n+1}} = \sum_{n=1}^{\infty} \frac{a_n}{2^{n+1}} = F(x), \\ 2F\left(\frac{2}{3}+\frac{x}{3}\right) - 1 &= 1 + \sum_{n=2}^{\infty} \frac{a_{n-1}}{2^{n+1}} - 1 = F(x). \end{aligned}$$

Now we have proved the equation when $x \in C$. For $x \in [0,1]\backslash C, x \in U, x/3 \in U, (2+x)/3 \in U$. And in each interval, the value of F is identical. Suppose $x \in (a,b), a, b \in C$, and $x/3 \in (a/3, b/3)$. So

$$F(x) = F(a) = 2F\left(\frac{a}{3}\right) = 2F\left(\frac{x}{3}\right),$$

the second equality can be similarly proved. □

Exercise 1.3.11. Calculate

$$\int_0^1 x\mathrm{d}F(x), \;\; \int_0^1 x^2\mathrm{d}F(x), \;\; \int_0^1 \mathrm{e}^{\mathrm{i}tx}\mathrm{d}F(x).$$

[HINT: This can be done directly or by using Exercise 10; for a third method see Exercise 9 of Sec. 5.3.]

Proof. We shall use Exercise 1.3.10. On interval $(1/3, 2/3)$, $\mathrm{d}F(x) = 0$. Now calculate the first integral I_1. We have

$$\begin{aligned} I_1 &= \int_0^1 x\mathrm{d}F(x) = \int_0^{\frac{1}{3}} x\mathrm{d}F(x) + \int \frac{2}{3}1x\mathrm{d}F(x) \\ &= \int_0^1 \frac{t}{3}\mathrm{d}F\left(\frac{t}{3}\right) + \int_0^1 \left(\frac{2+t}{3}\right)\mathrm{d}F\left(\frac{2+t}{3}\right) \\ &= \frac{1}{6}I_1 + \frac{1}{2}\left(\frac{2}{3}+\frac{1}{3}I_1\right). \end{aligned}$$

So $I_1 = 1/2$. For the second,

$$\begin{aligned} I_2 &= \int_0^1 \frac{t^2}{9}\mathrm{d}F\left(\frac{t}{3}\right) + \int_0^1 \frac{t^2+4t+4}{9}\mathrm{d}F\left(\frac{2+t}{3}\right) \\ &= \frac{1}{18}I_2 + \frac{1}{2}\left(\frac{1}{9}I_2 + \frac{4}{9}I_1 + \frac{4}{9}\right). \end{aligned}$$

So $I_2 = 3/8$. For the third, let $\varphi(t) = \int_0^1 \mathrm{e}^{\mathrm{i}tx}\mathrm{d}F(x)$, then

$$\begin{aligned}\varphi(t) &= \frac{1}{2}\int_0^1 \mathrm{e}^{\mathrm{i}tx/3}\mathrm{d}F(x) + \frac{1}{2}\mathrm{e}^{2\mathrm{i}t/3}\int_0^1 \mathrm{e}^{\mathrm{i}tx/3}\mathrm{d}F(x)\\ &= \frac{1}{2}\left(1+\mathrm{e}^{2\mathrm{i}t/3}\right)\varphi\left(\frac{t}{3}\right)\\ &= \mathrm{e}^{\mathrm{i}t/3}\cdot\cos\left(\frac{t}{3}\right)\varphi\left(\frac{t}{3}\right)\\ &= \mathrm{e}^{\mathrm{i}t/3}\cos\left(\frac{t}{3}\right)\mathrm{e}^{\mathrm{i}t/9}\cos\left(\frac{t}{9}\right)\varphi\left(\frac{t}{9}\right) = \cdots\\ &= \mathrm{e}^{\mathrm{i}t/2}\prod_{n=1}^{\infty}\cos\left(\frac{t}{3^n}\right). \text{ (using the continuity of } \varphi)\end{aligned}$$

□

Exercise 1.3.12. Extend the function F on $[0,1]$ trivially to $(-\infty,+\infty)$. Let $\{r_n\}$ be an enumeration of the rationals and

$$G(x) = \sum_{n=1}^{\infty}\frac{1}{2^n}F(r_n+x).$$

Show that G is a d.f. that is strictly increasing for all x and singular. Thus we have a singular d.f. with support $(-\infty,+\infty)$.

Proof. Clearly G is uniformly converged. When $x\to\infty$, $F(r_n+x)\to 1, \forall n\in\mathbb{N}_+$, so $G(+\infty)=1$, similarly $G(-\infty)=0$. Due to the increasing and continuous property of F, G is also increasing and continuous (recall the uniformly convergence), thus G is a d.f. $\forall x<x'$, and we can always find n_0 such that $x'+r_{n_0}$ and $x+r_{n_0}$ falls into two different intervals of D. In fact, choose $x_0\in C$, and we hope $x+r_{n_0}<x_0<x'+r_{n_0}$, which is equivalent to $x_0-x'<r_{n_0}<x_0-x$. As $\mathbb{Q}$ is dense in $\mathbb{R}$, r_{n_0} do exist. Thus

$$\begin{aligned}G(x')-G(x) &= \sum_{n=1}^{\infty}\frac{1}{2^n}\left[F(r_n+x')-F(r_n+x)\right]\\ &\geq \frac{1}{2^{n_0}}\left[F(r_{n_0}+x')-F(r_{n_0}+x)\right] > 0.\end{aligned}$$

So the increasing is strict. For each n, the set of those x such that $F'(r_n+x)\neq 0$, call it E_n, is of measure zero, then $\bigcup_{n=1}^{\infty}E_n$ is also of measure zero, so G is singular. □

Exercise 1.3.13. Consider F on $[0,1]$. Modify its inverse F^{-1} suitably to make it single-valued in $[0,1]$. Show that F^{-1} so modified is a discrete d.f. and find its points of jump and their sizes.

Proof. Now F^{-1} should have a jump between two points of C. We can define

$$F^{-1}(x) = \begin{cases}\inf\{y\in[0,1]:F(y)>x\}, & x<1,\\ 1, & x\geq 1,\end{cases}$$

then F^{-1} is right continuous and is single valued in $[0,1]$, and it is a d.f, and is discrete because it ony has jump on $k/2^n, n=0,1,\cdots,k=1,2,\cdots,2^n-1.$, and the number of points of jump is countable. The size of jump is essentially the length of intervals removed when constructing C. More specifically, when n is given, the new removed intervals have a common length of $1/3^n$. Now we conclude as follow: for the point

$$\frac{2k-1}{2^n}, k=1,2,\cdots,2^{n-1}, n\in\mathbb{N}_+,$$

it has a jump size of $1/3^n$. □

Exercise 1.3.14. Given any closed set C in $(-\infty, +\infty)$, there exists a d.f. whose support is exactly C. [HINT: Such a problem becomes easier when the corresponding measure is considered; see Sec. 2.2 below.]

Proof. Now we use the terms of measure to prove. Take a countable dense set in C, call it $S = \{x_1, x_2, \cdots\}$, and set $\mu\{x_n\} = 1/2^n, n \geq 1$, then $\forall x \in C, \exists\{x_k\} \subset S, x_k \to x(k \to \infty)$, hence it is easy to verify each $x \in C$ is the support. As C^c is an open set with $\mu(C^c) = 0$, the support can only be in C, hence the corresponding d.f. F to this p.m. has the support exactly C. □

Exercise 1.3.15. The Cantor d.f. F is a good building block of "pathological" examples. For example, let H be the inverse of the homeomorphic map of $[0, 1]$ onto itself: $x \to \frac{1}{2}[F(x)+x]$; and E a subset of $[0, 1]$ which is not Lebesgue measurable. Show that

$$1_{H(E)} \circ H = 1_E$$

where $H(E)$ is the image of E, 1_B is the indicator function of B, and $\circ$ denotes the composition of functions. Hence deduce: (1) a Lebesgue measurable function of a strictly increasing and continuous function need not be Lebesgue measurable; (2) there exists a Lebesgue measurable function that is not Borel measurable.

Proof. H is a bijection as defined. $\forall y \in [0, 1]$, we have

$$(1_{H(E)} \circ H)(y) = 1_{H(E)}(H(y))$$

When $H(y) \in H(E)$, the value above equals to 1. As H is a bijection, $H(y) \in H(E) \Leftrightarrow y \in E$. So $1_{H(E)} \circ H = 1_E$. For the function in this problem ($G(x) = \frac{1}{2}[F(x) + x]$), it is strictly increasing and continuous, so $H = G^{-1}$ is also strictly increasing and continuous. Now we shall observe that G maps the removed intervals U when constructing C to a set of half measure. take $I_1 = (1/3, 2/3)$ as an example,

$$G(I_1) = \frac{1}{2}\left[\frac{1}{2} + \left(\frac{1}{3}, \frac{2}{3}\right)\right] = \left(\frac{5}{12}, \frac{7}{12}\right),$$

and $m(G(I_1)) = 1/6$. As $m(U) = 1$, we can deduce that $m(G(U)) = 1/2$, which means $m(G(C)) = 1 - m(G(U)) = 1/2$. Let $M = G(C)$, and there exists $E \subset M$ which is not measurable, but $H(E) \subset C$ is measurable, but the composition $1_{H(E)} \circ H = 1_E$ is not measurable, which is (1).

Assume $1_{H(E)}$ is Borel measurable, then we can conclude that 1_E is Borel measurable because H is continuous, so is Borel measurable, and the composition of Borel measurable functions is Borel measurable. However, 1_E is not Lebesgue measurable, thus not Borel measurable, so $1_{H(E)}$ cannot be Borel measurable, which proves (2). □

2 Measure Theory

2.1 Classes of sets

Exercise 2.1.1. $(\bigcup_j A_j)\backslash(\bigcup_j B_j) \subset \bigcup_j(A_j\backslash B_j).(\bigcap_j A_j)\backslash(\bigcap_j B_j) \subset \bigcup_j(A_j\backslash B_j)$. When is there equality?

Proof. If $x \in (\bigcup_j A_j)\backslash(\bigcup_j B_j)$, then x does not belong to any B_j, and x belongs to at least one A_{j_0}, the combination of two assertions leads to $x \in \bigcup_j (A_j\backslash B_j)$. If $x \in (\bigcap_j A_j)\backslash(\bigcap_j B_j)$, then $x \in A_j, x \notin B_j, j = 1, 2, \cdots$, and now the second relation is obvious. For the first equality, we should observe that when adding $x \notin B_j$ on the right side, then equality holds, which is

$$\forall j, k, \ \ (A_j\backslash B_j) \cap B_k = \varnothing.$$

Similarly, for the second, the condition for equality is

$$\forall j, k, \ \ (A_j\backslash B_j) \subset A_k. \qquad \square$$

Exercise 2.1.2. The best way to define the symmetric difference is through indicators of sets as follows:

$$1_{A\triangle B} = 1_A + 1_B \pmod 2$$

where we have arithmetical addition modulo 2 on the right side. All properties of $\triangle$ follow easily from this definition, some of which are rather tedious to verify otherwise. As examples:

$$\begin{aligned}
&(A\triangle B)\triangle C = A\triangle(B\triangle C),\\
&(A\triangle B)\triangle(B\triangle C) = A\triangle C,\\
&(A\triangle B)\triangle(C\triangle D) = (A\triangle C)\triangle(B\triangle D),\\
&A\triangle B = C \Leftrightarrow A = B\triangle C,\\
&A\triangle B = C\triangle D \Leftrightarrow A\triangle C = B\triangle D.
\end{aligned}$$

Proof. Equation 1,2,3 is easy to verify. For the fourth, again notice that $2 \cdot 1_A = 0 \pmod 2$, thus

$$1_A + 1_B = 1_C \pmod 2 \Leftrightarrow 1_A + 2 \cdot 1_B = 1_A = 1_B + 1_C \pmod 2$$

Again use this property we get the fifth equation. $\square$

Exercise 2.1.3. If Ω has exactly n points, then $\mathscr{J}$ has 2^n members. The B.F. generated by n given sets "without relations among them" has 2^{2^n} members. (**Note: The book made a mistake on the superscript for** 2^{2^n}.)

Proof. The first is trivial. The second means any $k(k = 2, \cdots, n-1)$ sets cannot have a relation "contain" or "mutual exclusion". Thus, we can divide Ω into 2^n "atoms", which has the form of

$$\bigcap_{i=1}^{n} B_i, \quad B_i = A_i \text{ or } A_i^c.$$

Then using these atoms to generate $\mathscr{J}$, we get 2^{2^n} members. $\square$

Exercise 2.1.4. If Ω is countable, then $\mathscr{J}$ is generated by the singletons, and conversely. [HINT: All countable subsets of Ω and their complements form a B.F.]

Proof. If Ω is countable, let $A \subset \Omega$ is any subset of Ω, then A is countable, and A can be written in a form of $\bigcup_{x\in A}\{x\}$, so $A \in \mathscr{J}$, thus $\mathscr{J}$ is generated by singletons. Conversely, assume Ω is not countable, consider

$$\mathscr{C} = \{A \subset \Omega : \ A \text{ or } A^c \text{ is countable}\},$$

then $\forall A_i \in \mathscr{C}, i = 1, 2, \cdots, \bigcup_{i=1}^{\infty} A_i \in \mathscr{C}$, which implies $\mathscr{C}$ is a B.F. As $\mathscr{J}$ is generated by the singletons, and $\mathscr{C}$ contains all singletons, so $\mathscr{J} \subset \mathscr{C}$. However, $\exists E \in \Omega, E$ and E^c is both uncountable, which leads to $\mathscr{J} \neq \mathscr{C}$, a contradiction, so Ω must be countable. $\square$

Exercise 2.1.5. The intersection of any collection of B.F.'s $\{\mathscr{F}_\alpha, \alpha \in A\}$ is the maximal B.F. contained in all of them; it is indifferently denoted by $\bigcap_{\alpha\in A} \mathscr{F}_\alpha$ or $\wedge_{\alpha\in A}\mathscr{F}_\alpha$. (**Note: The book made a mistake on the notation "$\wedge$".**)

Proof. The key is to prove $\bigcap_{\alpha\in A} \mathscr{F}_\alpha$ is a B.F. and there cannot be a bigger B.F. having the property. $\forall E \in \bigcap_{\alpha\in A} \mathscr{F}_\alpha, E \in \mathscr{F}_\alpha, \alpha \in A$, so $E^c \in \mathscr{F}_\alpha, \alpha \in A$, which means $E^c \in \bigcap_{\alpha\in A} \mathscr{F}_\alpha$. For $E_j \in \bigcap_{\alpha\in A} \mathscr{F}_\alpha, j = 1, 2, \cdots$, similarly we have

$$\bigcup_{j=1}^{\infty} E_j \in \bigcap_{\alpha\in A} \mathscr{F}_\alpha.$$

So $\bigcap_{\alpha\in A} \mathscr{F}_\alpha$ is a B.F. Assume that there exists $\mathscr{G}$ such that $\mathscr{G} \subset \mathscr{F}_\alpha, \alpha \in A$, then $\mathscr{G} \subset \bigcap_{\alpha\in A} \mathscr{F}_\alpha$. So $\bigcap_{\alpha\in A}$ is the maximal B.F. contained in all given B.F.s. □

Exercise 2.1.6. The union of a countable collection of B.F.'s $\{\mathscr{F}_j\}$ such that $\mathscr{F}_j \subset \mathscr{F}_{j+1}$ need not be a B.F., but there is a minimal B.F. containing all of them, denoted by $\vee_j\mathscr{F}_j$. In general $\vee_{\alpha\in A}\mathscr{F}_\alpha$ denotes the minimal B.F. containing all $\mathscr{F}_\alpha, \alpha \in A$. [HINT: Ω = the set of positive integers; $\mathscr{F}_j$ = the B.F. generated by those up to j.]

Proof. Let $\Omega = \mathbb{N}_+$, and $\mathscr{F}_j$ is the generated B.F. on $\{\{1\}, \{2\}, \cdots, \{j\}\}$, then it is easy to verify $\mathscr{F}_j \subset \mathscr{F}_{j+1}$, but the even number set $\{2, 4, 6, \cdots\}$ is not in the union of these $\mathscr{F}_j$, because $\{j+1, j+2, \cdots\} \cap \{2, 4, 6, \cdots\} \neq \emptyset, \forall j$. Now denote $\mathscr{P}$ is the B.F. generated by Ω, and it contains all $\mathscr{F}_\alpha$ defined on subset of Ω. Then denote

$$\mathscr{T} = \{\mathscr{H} : \mathscr{H} \text{ is a B.F. and } \bigcup_{\alpha\in A} \mathscr{F}_\alpha \subset \mathscr{H}\},$$

we know $\mathscr{T} \neq \emptyset$ as the existence of $\mathscr{P}$. Define

$$\bigvee_{\alpha\in A} \mathscr{F}_\alpha = \bigcap_{\mathscr{H}\in\mathscr{T}} \mathscr{H},$$

and similar to Exercise 2.1.5 we can verify this is the minimal B.F. containing all $\mathscr{F}_\alpha$. □

Exercise 2.1.7. A B.F. is said to be countably generated iff it is generated by a countable collection of sets. Prove that if each $\mathscr{F}_j$ is countably generated, then so is $\vee_{j=1}^{\infty}\mathscr{F}_j$.

Proof. In the following exercises we use $\sigma(\cdot)$ to represent the B.F. generated by a collection of sets. Let $\mathscr{F}_j = \sigma(\mathscr{C}_j), j = 1, 2, \cdots$, and $\mathscr{C} = \bigcup_{j=1}^{\infty} \mathscr{C}_j$, then $\mathscr{C}$ is countable. Now we prove $\sigma(\mathscr{C}) = \vee_{j=1}^{\infty}\mathscr{F}_j = \mathscr{G}$. $\forall A \in \mathscr{C}, \exists j_0, A \in \mathscr{C}_{j_0}$, so $A \in \mathscr{F}_{j_0}$, thus $A \in \mathscr{G}$. As A can be picked arbitrarily and $\mathscr{G}$ is a B.F., $\sigma(\mathscr{C}) \subset \mathscr{G}$. Conversely, as $\mathscr{C}_j \subset \mathscr{C}, \sigma(\mathscr{C}_j) = \mathscr{F}_j \subset \sigma(\mathscr{C})$. But $\mathscr{G}$ is the smallest B.F. containing all $\mathscr{F}_j$, so $\mathscr{G} \subset \sigma(\mathscr{C})$. So $\mathscr{G}$ is countably generated by $\mathscr{C}$. □

Exercise 2.1.8. Let $\mathscr{F}$ be a B.F. generated by an arbitrary collection of sets $\{E_\alpha, \alpha \in A\}$. Prove that for each $E \in \mathscr{F}$, there exists a countable subcollection $\{E_{\alpha_j}, j \geq 1\}$ (depending on E) such that E belongs already to the B.F. generated by this subcollection. [HINT: Consider the class of all sets with the asserted property and show that it is a B.F. containing each E_α.]

Proof. Let $\mathscr{C}$ denote a countable subset of $\{E_\alpha\}$, and

$$\mathscr{G} = \{E \in \mathscr{F} : \exists\mathscr{C}, E \in \sigma(\mathscr{C})\}.$$

Clearly $\mathscr{G} \subset \mathscr{F}$, $\{E_\alpha, \alpha \in A\} \subset \mathscr{G}$. $\forall E \in \mathscr{G}$, assume $E = \bigcup_{i=1}^\infty E_{\alpha_i}, E_{\alpha_i} \in \sigma(\mathscr{C}_E)$, then

$$E_{\alpha_i}^c \in \sigma(\mathscr{C}_E), E^c = \bigcap_{i=1}^\infty E_{\alpha_i}^c \in \sigma(\mathscr{C}_E) \Rightarrow E^c \in \mathscr{G}.$$

Then for $E_j \in \mathscr{G}, j = 1, 2, \cdots, \exists \mathscr{C}_j, E_j \subset \sigma(\mathscr{C}_j)$. Let $\mathscr{C} = \bigcup_{j=1}^\infty \mathscr{C}_j$ is the countable union, then

$$E_j \in \sigma(\mathscr{C}_j) \subset \sigma(\mathscr{C}), \quad \forall j \in \mathbb{N}_+.$$

So $\bigcup_{j=1}^\infty E_j \in \mathscr{G}$, and $\mathscr{G}$ is a B.F. Thus $\mathscr{F} \subset \mathscr{G}$, then $\mathscr{F} = \mathscr{G}$. □

Exercise 2.1.9. If $\mathscr{F}$ is a B.F. generated by a countable collection of disjoint sets $\{\Lambda_n\}$, such that $\bigcup_n \Lambda_n = \Omega$, then each member of $\mathscr{F}$ is just the union of a countable subcollection of these Λ_n's.

Proof. This is the direct corollary of Exercise 2.1.8. □

Exercise 2.1.10. Let $\mathscr{D}$ be a class of subsets of Ω having the closure property (iii); let $\mathscr{A}$ be a class of sets containing Ω as well as $\mathscr{D}$, and having the closure properties (vi) and (x). Then $\mathscr{A}$ contains the B.F. generated by $\mathscr{D}$. (This is Dynkin's form of a monotone class theorem which is expedient for certain applications. The proof proceeds as in Theorem 2.1.2 by replacing $\mathscr{F}_0$ and $\mathscr{G}$ with $\mathscr{D}$ and $\mathscr{A}$ respectively.)

Proof. As $\Omega \in \mathscr{A}$ and property (x) holds for $\mathscr{A}$, we can deduce that property (i) holds for $\mathscr{A}$, and thus $\mathscr{A}$ is an M.C. Now put $\mathscr{L}$ as the minimal M.C. containing $\mathscr{D}$ and has the same property with $\mathscr{A}$, and let

$$\begin{aligned}
\mathscr{C}_1 &= \{E \in \mathscr{L} : E \cap D \in \mathscr{L} \text{ for all } D \in \mathscr{D}\}, \\
\mathscr{C}_2 &= \{E \in \mathscr{L} : E \cap L \in \mathscr{L} \text{ for all } L \in \mathscr{L}\},
\end{aligned}$$

the indentities

$$\left(\bigcup_{n=1}^\infty E_n\right) \cap D = \bigcup_{n=1}^\infty (E_n \cap D)$$
$$\left(\bigcap_{n=1}^\infty E_n\right) \cap D = \bigcap_{n=1}^\infty (E_n \cap D)$$

shows that $\mathscr{C}_1, \mathscr{C}_2 \subset \mathscr{L}$ is an M.C. when setting $E_n \subset E_{n+1}$ or $E_n \supset E_{n+1}, E_n \in \mathscr{C}_1, n = 1, 2, \cdots$ and $\mathscr{D} \subset \mathscr{C}_1$. However, $\mathscr{L} \subset \mathscr{C}_1$, so $\mathscr{L} = \mathscr{C}_1$, and in turn means $\mathscr{D} \subset \mathscr{C}_2$, then $\mathscr{L} = \mathscr{C}_2$ and thus $\mathscr{C}_2$ is closed under intersection. Now we have proved that $\mathscr{L}$ is a B.F. containing $\mathscr{D}$, so $\sigma(\mathscr{D}) \subset \mathscr{L} \subset \mathscr{A}$. □

Exercise 2.1.11. Take $\Omega = \mathscr{R}^n$ or a separable metric space in Exercise 10 and let $\mathscr{D}$ be the class of all open sets. Let $\mathscr{H}$ be a class of real-valued functions on Ω satisfying the following conditions.

(a) $1 \in \mathscr{H}$ and $1_D \in \mathscr{H}$ for each $D \in \mathscr{D}$;

(b) $\mathscr{H}$ is a vector space, namely: if $f_1 \in \mathscr{H}, f_2 \in \mathscr{H}$ and c_1, c_2 are any two real numbers, then $c_1 f_1 + c_2 f_2 \in \mathscr{H}$;

(c) $\mathscr{H}$ is closed with respect to increasing limits of positive functions, namely: if $f_n \in \mathscr{H}, 0 \leq f_n \leq f_{n+1}$ for all n, and $f = \lim_n \uparrow f_n < \infty$, then $f \in \mathscr{H}$.

Then $\mathscr{H}$ contains all Borel measurable functions on Ω, namely all finite-valued functions measurable with respect to the topological Borel field (= the minimal B.F. containing all open sets of Ω). [HINT: let $\mathscr{C} = \{E \subset \Omega : 1_E \in \mathscr{H}\}$; apply Exercise 10 to show that $\mathscr{C}$ contains the B.F. just defined. Each positive Borel measurable function is the limit of an increasing sequence of simple (finitely-valued) functions.]

Proof. Let $\mathscr{B}$ represent the topological Borel field, and $\mathscr{C} = \{E \subset \Omega : 1_E \in \mathscr{H}\}$. From condition (a), we know $\Omega \in \mathscr{C}$ and $\mathscr{D} \subset \mathscr{C}$. Now we show that $\mathscr{C}$ has the closure property (vi) and (x). For (x), consider $E_1 \subset E_2, E_1, E_2 \subset \Omega$, and $1_{E_1}, 1_{E_2} \in \mathscr{H}$, then

$$1_{E_2} - 1_{E_1} = 1_{E_2 \backslash E_1} \in \mathscr{H}$$

according to condition (b), so $E_2 \backslash E_1 \in \mathscr{C}$. For $E_n \in \mathscr{C}, E_n \subset E_{n+1}$, we have $1_{E_n} \leq 1_{E_{n+1}}$, and according to (c), we know that

$$f = \lim_{n \to \infty} 1_{E_n} = 1_{\bigcup_{n=1}^{\infty} E_n} \in \mathscr{H}.$$

So $\bigcup_{n=1}^{\infty} E_n \in \mathscr{C}$. So we can deduce that $\mathscr{C} \supset \mathscr{B}$. Each Borel measurable simple function s can be written as the linear combination of indicator functions in $\mathscr{H}$, which is

$$s(x) = \sum_{i=1}^{n} c_i 1_{E_i}(x),$$

where $E_i \in \mathscr{B}$. Now suppose f is a nonnegative Borel measurable function which is defined on Ω, then according to (c) and the conclusion in the hint, we can find a nonnegative increasing series of $\{s_n\}$ such that

$$f = \lim_n \uparrow s_n < \infty \in \mathscr{H}.$$

Then suppose f is any Borel measurable function, then $f = f^+ - f^-$, where $f^+ = \max\{f, 0\}, f^- = \max\{-f, 0\} \in \mathscr{H}$, and according to condition (b), we know that $f \in \mathscr{H}$. The proof is complete. □

Exercise 2.1.12. Let $\mathscr{C}$ be an M.C. of subsets of $\mathscr{R}^n$ (or a separable metric space) containing all the open sets and closed sets. Prove that $\mathscr{C} \supset \mathscr{B}^n$ (the topological Borel field defined in Exercise 11). [HINT: Show that the minimal such class is a field.]

Proof. Assume $\mathscr{C}_0$ is just the minimal M.C. having the properties mentioned above. Notice that $\mathscr{R}^n \in \mathscr{C}_0$, and let $\mathscr{D}$ denote all the open sets and closed set in $\mathscr{R}^n$, then $\mathscr{B}^n = \sigma(\mathscr{D})$. Denote

$$\mathscr{T} = \{E \in \mathscr{C} : E^c \in \mathscr{C}\},$$

Then $\mathscr{T}$ is an M.C. and $\mathscr{D} \subset \mathscr{T}$. The next proof procedure is alike Exercise 2.1.10, and shall use the conclusion: Any closed set can be written in a form of the intersection of countable decreasing open sets. After this, we can prove $\mathscr{C}_0$ is closed under intersection, then $\mathscr{C}_0$ is a field, thus a B.F. containing $\mathscr{D}$. Finally we have $\mathscr{C} \supset \mathscr{C}_0 \supset \mathscr{B}^n$. □

2.2 Probability measures and their distribution functions

Exercise 2.2.1. For any countably infinite set Ω, the collection of its finite subsets and their complements forms a field $\mathscr{F}$. If we define $\mathscr{P}(E)$ on $\mathscr{F}$ to be 0 or 1 according as E is finite or not, then $\mathscr{P}$ is finitely additive but not countably so.

Proof. The set in $\mathscr{F}$ is either finite or complement-finite. $\forall E, F \in \mathscr{F}$, if one of two is finite, then $E \cap F$ is finite. If both are countable, then E^c and F^c is finite, thus $(E \cap F)^c = E^c \cup F^c$ is finite, so $\mathscr{F}$ is closed under intersection, so $\mathscr{F}$ is a field. If $E_1, \cdots, E_n$ are pairwise disjoint, then we can deduce that at most one E_k is countable. Otherwise, assume $E_1, E_2 \in \mathscr{F}$ is countably infinite and disjoint, then $E_1^c \cup E_2^c = (E_1 \cap E_2)^c = \Omega$ is finite, which is a contradiction. So

$$\mathscr{P}\left(\bigcup_{k=1}^{n} E_k\right) = \sum_{k=1}^{n} \mathscr{P}(E_k).$$

However, when each $E_n \in \mathscr{F} (n = 1, 2, \cdots)$ is finite and pairwise disjoint, $\bigcup_{n=1}^{\infty}$ can be countable, so $\mathscr{P}$ is not countably additive. $\square$

Exercise 2.2.2. Let Ω be the space of natural numbers. For each $E \subset \Omega$ let $N_n(E)$ be the cardinality of the set $E \cap [0, n]$ and let $\mathscr{C}$ be the collection of E's for which the following limit exists:

$$\mathscr{P}(E) = \lim_{n\to\infty} \frac{N_n(E)}{n}.$$

$\mathscr{P}$ is finitely additive on $\mathscr{C}$ and is called the "asymptotic density" of E. Let $E = \{\text{all odd integers}\}$, $F = \{\text{all odd integers in } [2^{2n}, 2^{2n+1}] \text{ and all even integers in } [2^{2n+1}, 2^{2n+2}] \text{ for } n \geq 0\}$. Show that $E \in \mathscr{C}, F \in \mathscr{C}$, but $E \cap F \notin \mathscr{C}$. Hence $\mathscr{C}$ is not a field.

Proof. For pairwise disjoint $E_j \in \mathscr{C}, j = 1, 2, \cdots, k$, we have

$$N_n\left(\bigcup_{j=1}^{k} E_j\right) = \sum_{j=1}^{k} N_n(E_j).$$

As we can define $\mathscr{P}$ for each E_k, we have

$$\lim_{n\to\infty} \frac{N_n\left(\bigcup_{j=1}^{k} E_j\right)}{n} = \lim_{n\to\infty} \sum_{j=1}^{k} \frac{N_n(E_j)}{n} = \sum_{j=1}^{k} \lim_{n\to\infty} \frac{N_n(E_j)}{n},$$

which means $\mathscr{P}\left(\bigcup_{j=1}^{k} E_j\right)$ exists, and $\mathscr{P}$ is finitely additive. Now for E, obviously $\mathscr{P}(E) = 1/2$. For F, there are 2^{2k-1} odd integers in $[2^{2k}, 2^{2k+1}] (k \geq 1)$, and $2^{2k+1} + 1$ even integers in $[2^{2k+1}, 2^{2k+2}]$, and $\mathscr{P}(F) = 1/2$ as it behaves "like" E. But for $E \cap F$, the result is all odd integers in $[2^{2k}, 2^{2k+1}], k \geq 0$. For the subsequence $\{1, 4, \cdots, 2^{2n}, \cdots\}$,

$$\lim_{n\to\infty} \frac{N_{2^{2n}}(E \cap F)}{2^{2n}} = \lim_{n\to\infty} \frac{1 + \sum_{k=1}^{n-1} 2^{2k-1}}{2^{2n}} = \frac{1}{6},$$

and for the subsequence $\{2, 8, \cdots, 2^{2k+1}, \cdots\}$,

$$\lim_{n\to\infty} \frac{N_{2^{2n+1}}(E \cap F)}{2^{2n+1}} = \lim_{n\to\infty} \frac{1 + \sum_{k=1}^{n} 2^{2k-1}}{2^{2n+1}} = \frac{1}{3},$$

so the limit does not exist, hence $E \cap F \notin \mathscr{C}$. $\square$

Exercise 2.2.3. In the preceeding example show that for each real number α in [0,1] there is an E such that $\mathscr{P}(E) = \alpha$. Is the set of all primes in $\mathscr{C}$? Give an example of E that is not in $\mathscr{C}$.

Proof. If $\alpha = 0$, let $E_0 = \varnothing$; Else let sequence $a_n = [n/\alpha], n = 1, 2, \cdots$ and $E_\alpha = \bigcup_{n=1}^{\infty}\{a_n\}$, then

$$N_n(E_\alpha) = \#k : \left[\frac{k}{\alpha}\right] \leq n = \max\{[\alpha(n-1)], [\alpha n]\}.$$

By squeeze theorem,

$$\mathscr{P}(E_\alpha) = \lim_{n\to\infty}\frac{N_n(E_\alpha)}{n} = \alpha.$$

So $\forall\alpha \in [0, 1], \exists E_\alpha, \mathscr{P}(E_\alpha) = \alpha$. For the prime set, we shall use a wide-known conclusion, called Prime Number Theorem, denote $\mathbb{P}$ the set of all primes, we have

$$N_n(\mathbb{P}) \sim \frac{n}{\log n},$$

thus $\mathscr{P}(\mathbb{P}) = 0, \mathbb{P} \in \mathscr{C}$. The example of $E \notin C$ is given in Exercise 2.2.2. □

Exercise 2.2.4. Prove the nonexistence of a p.m. on $(\Omega, \mathscr{I})$, where $(\Omega, \mathscr{I})$ is as in Example 1, such that the probability of each singleton has the same value. Hence criticize a sentence such as: "Choose an integer at random".

Proof. Assume that such p.m. exists, then if $p_i = c$, it leads to

$$\mathscr{P}(\Omega) = \mathscr{P}\left(\bigcup_{n=1}^{\infty}\omega_n\right) = \sum_{n=1}^{\infty} c = 1.$$

When $c = 0$, it is a contradiction; When $c > 0$, the summation is infinite, also a contradiction. So such p.m. does not exist. So we cannot choose an integer at random as the same-likely probability on each integer is impossible. □

Exercise 2.2.5. Prove that the trace of a B.F. $\mathscr{F}$ on any subset Δ of Ω is a B.F. Prove that the trace of $(\Omega, \mathscr{F}, \mathscr{P})$ on any Δ in $\mathscr{F}$ is a probability space, if $\mathscr{P}(\Delta) > 0$.

Proof. Let $\mathscr{T}$ denote the trace of $\mathscr{F}$ on subset Δ. $\forall E \in \mathscr{T}, \exists F \in \mathscr{F}, E = \Delta \cap F$. Then $E^c = F^c = \Delta \cap F^c \in \mathscr{T}$ (notice that the complement operation should be done on Δ). Then for $E_i \in \mathscr{T}, i = 1, 2, \cdots$, as

$$\bigcup_{i=1}^{\infty} E_i = \bigcup_{i=1}^{\infty}(\Delta \cap F_i) = \Delta \cap \bigcup_{i=1}^{\infty} F_i,$$

and $\mathscr{F}$ is a B.F., we know $\mathscr{T}$ is a B.F. Then we verify $(\Delta, \mathscr{T}, \mathscr{P}_\Delta)$ is a probability space. When $\mathscr{P}(\Delta) > 0$ we can define $\mathscr{P}_\Delta(E) = \mathscr{P}(E)/\mathscr{P}(\Delta)$. It is trivial for the axiom (i) and (iii). For (ii), as $\mathscr{P}$ has the countable additivity, we have

$$\mathscr{P}_\Delta\left(\bigcup_{n=1}^{\infty} E_n\right) = \frac{\mathscr{P}\left(\bigcup_{n=1}^{\infty} E_n\right)}{\mathscr{P}(\Delta)} = \sum_{n=1}^{\infty}\frac{\mathscr{P}(E_n)}{\mathscr{P}(\Delta)} = \sum_{n=1}^{\infty}\mathscr{P}_\Delta(E_n).$$

So it is a probability space. □

Exercise 2.2.6. Now let $\Delta \notin \mathscr{F}$ be such that

$$\Delta \subset F \in \mathscr{F} \Rightarrow \mathscr{P}(F) = 1.$$

Such a set is called *thick* in $(\Omega, \mathscr{F}, \mathscr{P})$. If $E = \Delta \cap F, F \in \mathscr{F}$, define $\mathscr{P}^*(E) = \mathscr{P}(F)$. Then $\mathscr{P}^*$ is a well-defined (what does it mean?) p.m. on $(\Delta, \Delta \cap \mathscr{F})$. This procedure is called the *adjunction* of Δ to $(\Omega, \mathscr{F}, \mathscr{P})$.

Proof. First we prove $\mathscr{P}^*$ is a p.m. The axiom (1) is trivial. $\mathscr{P}^*(\Delta) = 1$ since Δ is thick, so (iii) is verified. Assume $E_n \subset \Delta \cap \mathscr{F}$, then $\exists F_n \in \mathscr{F}, E_n = \Delta \cap F_n$. We have

$$\mathscr{P}^*\left(\bigcup_{n=1}^{\infty} E_n\right) = \mathscr{P}^*\left(\Delta \cap \bigcup_{n=1}^{\infty} F_n\right) = \mathscr{P}\left(\bigcup_{n=1}^{\infty} F_n\right) = \sum_{n=1}^{\infty} \mathscr{P}(F_n) = \sum_{n=1}^{\infty} \mathscr{P}^*(E_n).$$

So axiom (ii) satisfies, hence $\mathscr{P}^*$ is a p.m. Consider two different sets $F_1, F_2 \in \mathscr{F}$ such that $\Delta \cap F_1 = \Delta \cap F_2$, we shall prove $\mathscr{P}(F_1) = \mathscr{P}(F_2)$. We can show that $\Delta \cap (F_1 \triangle F_2) = \varnothing$, so $\Delta \subset (F_1 \triangle F_2)^c$, which implies $\mathscr{P}(F_1 \triangle F_2) = \mathscr{P}(F_1 \backslash F_2) + \mathscr{P}(F_2 \backslash F_1) = 0$. However $\mathscr{P}(\cdot) \geq 0$, so two parts equal to 0. Hence

$$\mathscr{P}(F_1) = \mathscr{P}(F_1 \backslash F_2) + \mathscr{P}(F_1 \cap F_2) = \mathscr{P}(F_2 \backslash F_1) + \mathscr{P}(F_1 \cap F_2) = \mathscr{P}(F_2),$$

which means $\mathscr{P}^*$ is well defined, it does not depend on the selection of F. □

Exercise 2.2.7. The B.F. $\mathscr{B}^1$ on $\mathscr{R}^1$ is also generated by the class of all open intervals or all closed intervals, or all half-lines of the form $(-\infty, a]$ or $(a, +\infty)$, or these intervals with rational endpoints. But it is not generated by all the singletons of $\mathscr{R}^1$ nor by any finite collection of subsets of $\mathscr{R}^1$.

Proof. It is sufficient to prove $(a, b], -\infty < a < b < +\infty$ can be generated by open intervals. Let $\{b_n\}$ is a sequence such that $b_n \downarrow b$, then we have

$$\bigcap_{n=1}^{\infty} (a, b_n) = (a, b].$$

Since the complement of all open intervals are all closed intervals, and

$$(-\infty, b] \cap (a, +\infty) = (a, b], \quad -\infty < a < b < \infty,$$

$\mathscr{B}^1$ can also be generated using these kinds of collection of sets. However, the cardinality of $[0, 1]$ is c, which is uncountable, so it cannot be generated by singletons, and if $\mathscr{B}^1$ can be generated by finite collection of subsets of $\mathscr{R}^1$, then $\mathscr{B}^1$ has finite elements, but the interval having the form of $(-\infty, a]$ is uncountable, which is a contradiction. □

Exercise 2.2.8. $\mathscr{B}^1$ contains every singleton, countable set, open set, closed set. G_δ set, F_σ set. (For the last two kinds of sets see, e.g., Natanson [3].)

Proof. Suppose $\{(a_n, b_n)\}$ is a collection of open sets where $a_n \uparrow c, b_n \downarrow c$, then $\bigcap_{n=1}^{\infty} (a_n, b_n) = \{c\}$, so $\mathscr{B}^1$ contains every singleton, and using singletons we get all countable set. Consider open intervals in $\mathscr{R}^1$ having the form (r, s), where r, s are rationals, then for any open set G, $\forall x \in G, \exists \delta > 0, r_x, s_x, (r_x, s_x) \subset U(x, \delta) \subset G$. So

$$G = \bigcup_{x \in G} (r_x, s_x).$$

Notice all rational intervals is countable, so the union above is mostly countable, hence all open sets are in $\mathscr{B}^1$, then all closed sets are in $\mathscr{B}^1$. A G_δ set is a countable intersection of open sets, and a F_σ set is a countable union of closed sets, so both of them are in $\mathscr{B}^1$. □

Exercise 2.2.9. Let $\mathscr{C}$ be a countable collection of pairwise disjoint subsets $\{E_j, j \geq 1\}$ of $\mathscr{R}^1$, and let $\mathscr{F}$ be the B.F. generated by $\mathscr{C}$. Determine the most general p.m. on $\mathscr{F}$ and show that the resulting probability space is "isomorphic" to that discussed in Example 1.

Proof. We can assign any sequence of numbers $\{p_j, j \geq 1\}$ to each E_j such that

$$\forall j \geq 1 : p_j \geq 0; \quad \sum_{j=1}^{\infty} p_j = 1;$$

and define a set function $\mathscr{P}$ on $\mathscr{F}$ as follows:

$$\forall E \in \mathscr{F} : \mathscr{P}(E) = \sum_{E_j \subset E} p_j.$$

Such $\mathscr{P}$ is clearly a p.m. We can choose $\omega_j \in E_j$ as the representation for each E_j, and alternate probability space to what is considered in Example 1, we can show the two is isomorphic, with a bijection $E_j \Leftrightarrow j$. □

Exercise 2.2.10. Instead of requiring that the E_j's be pairwise disjoint, we may make the broader assumption that each of then intersects only a finite numebr in the collection. Carry through the rest of the problem.

Proof. Consider sets

$$A_S = \left(\bigcap_{j \in S} E_j \right) \cap \left(\bigcap_{k \notin S} E_k^c \right),$$

where S is a finite index set, then $\{A_S\}$ is a collection such that $A_{S_1} \cap A_{S_2} = \varnothing, S_1 \neq S_2$, then we can use Exercise 2.2.9 to construct a p.m. □

In the following, μ is a p.m. on $\mathscr{B}^1$ and F is its d.f.

Exercise 2.2.11. An *atom* of any measure μ on $\mathscr{B}^1$ is a singleton $\{x\}$ such that $\mu(\{x\}) > 0$. The number of atoms of any σ-finite measure is countable. For each x we have $\mu(\{x\}) = F(x) - F(x-)$.

Proof. As we have constrained the measure to a p.m., then $\mu(\{x\}) \leq 1$. The proof can be finished as what is done in Exercise 1.1.3. For general σ-finite measure, we can find $E_n \to \Omega$ and $\mu(E_n) < \infty$. And on each E_n, the number of atoms is countable, thus the whole is countable. Using the identity

$$\{x\} = \bigcap_{n=1}^{\infty} \left(x - \frac{1}{n}, x \right]$$

and the continuity of p.m., we have

$$\mu(\{x\}) = \mu\left(\lim_{n\to\infty} \bigcap_{k=1}^{n} \left(x - \frac{1}{n}, x \right] \right) = \lim_{n\to\infty} \left[F(x) - F\left(x - \frac{1}{n} \right) \right] = F(x) - F(x-).$$

□

Exercise 2.2.12. μ is called *atomic* iff its value is zero on any set not containing any atom. This is the case if and only if F is discrete. μ is without any atom or *atomless* if and only if F is continuous.

Proof. From the previous exercise, we know μ has atom at the point of jump of F, vice versa. Denote $\{a_j\}$ all atoms of μ, and $\mu(\{a_j\}) = b_j$, then by countable additivity of p.m.,

$$F(x) = \mu((-\infty, x]) = \sum_{j \le x} b_j \mu(\{a_j\}) = \sum_j b_j \delta_{a_j}(x),$$

which is the form of a discrete d.f. Conversely, paying attention to jumps of F, $\sum_j b_j = 1$ holds, which implies $\mu(\mathscr{R} \backslash \{a_j, j \ge 1\}) = 0$. So μ can be nothing but zero on any set not containing any atom. If μ is atomless, then at any point x we have $F(x) - F(x-) = 0$, which is equivalent to F is continuous. □

Exercise 2.2.13. μ is called *singular* iff there exists a set Z with $m(Z) = 0$ such that $\mu(Z^c) = 0$. This is the case if and only if F is singular. [HINT: One half is proved by using Theorems 1.3.1 and 2.1.2 to get $\int_B F'(x)\mathrm{d}x \le \mu(B)$ for $B \in \mathscr{B}^1$; the other requires Vitali's covering theorem.]

Proof. For an interval $(x, x']$, according to (b) of Theorem 1.3.1, we have

$$\int_{x'}^{x} F'(t)\mathrm{d}t \le F(x') - F(x) = \mu((x, x']).$$

For $B \in \mathscr{B}_0$, the inequality above also holds. From Theorem 2.1.2 we know $\mathscr{B}$ is also a M.C. So $\forall B \in \mathscr{B}, \exists E_1 \subset E_2 \subset \cdots \subset E_n \subset \cdots$, such that $B = \bigcup_{n=1}^{\infty} E_n$. Define $f_n(x) = F'(t)1_{E_n}$, then $f_n(x) \le f_{n+1}(x)$. Using Levi theorem and the continuity of μ, we have

$$\int_B F'(x)\mathrm{d}x = \lim_{n\to\infty} \int f_n(x)\mathrm{d}x \le \lim_{n\to\infty} \mu(E_n) = \mu(B).$$

If μ is singular, then $\exists Z, m(Z) = 0$ and $\mu(Z^c) = 0$, and $B \in \mathscr{B}$, $Z \subset B, m(B) = 0$. We have

$$\int_{B^c} F'(x)\mathrm{d}x \le \mu(B^c) \le \mu(Z^c) = 0,$$

which implies $F'(x) = 0$ a.e. on B^c, and due to $m(B) = 0$, we have $F' = 0$ a.e. on $\mathscr{R}$, so F is singular.

Conversely, if F is singular, then we have $F' = 0$ a.e. Assume S is the set where $F' = 0$, and $\{I_x\}$ is a collection of set such that $\mu(I_x)/m(I_x) < \varepsilon, x \in S$, such interval exists by $F'(x) = 0, x \in S$. By Vitali's covering theorem, $\exists I_n \in \mathscr{B}, n = 1, 2, \cdots$ and I_n pairwise disjoint, such that $m\left(S \backslash \left(\bigcup_{n=1}^{\infty} I_n\right)\right) = 0$. Let $Z = S^c \backslash \left(\bigcup_{n=1}^{\infty} I_n\right)$, then $m(Z) = 0$, and

$$\mu(Z^c) = \mu\left(\bigcup_{n=1}^{\infty} I_n\right) < \varepsilon \cdot m\left(\bigcup_{n=1}^{\infty} I_n\right).$$

There we suppose $m(S) < \infty$. But ε is arbitrary, it leads to $\mu\left(\bigcup_{n=1}^{\infty} I_n\right) = 0$. Actually $m(S) = \infty$, for $E_n = Z^c \cap [-n, n]$, we can use the same method and find $\{I_x\}$ to cover E_n and use Vitali's covering theorem. By continuity of μ,

$$\mu(Z^c) = \mu\left(\bigcup_{n=1}^{\infty} (Z^c \cap [-n, n])\right) = \lim_{n\to\infty} \mu(E_n) = 0.$$

Hence we proved μ is singular. □

Exercise 2.2.14. Translate Theorem 1.3.2 in terms of measures.

Proof. Every p.m. μ can be written as the convex combination of an atomic, a singular atomless, and an absolutely atomless p.m. Such a decomposition is unique. □

Exercise 2.2.15. Translate the construction of a singular continuous d.f. in Sec. 1.3 in terms of measures. [It becomes clearer and easier to describe!] Generalize the construction by replacing the Cantor set with any perfect set of Borel measure zero. What if the latter has positive measure? Describe a probability scheme to realize this.

Proof. We still use the same notation in Sec. 1.3. After n steps we leave 2^n closed intervals in $[0, 1]$, name them $I_{n,k}, k = 1, 2, \cdots, 2^n$. Now we shall define μ_n for n th step. On those removed open intervals U_n, we set $\mu(U_n) = 0$, also on $(-\infty, 0) \cup (1, \infty)$. And on each $I_{n,k}$, set $\mu_n(I_{n,k}) = 1/2^n$. When $n \to \infty, U_n \cup (-\infty, 0) \cup (1, \infty) \to D, \mu_n \to \mu$, which is an open set, and $\mu(D) = 0$. And $D^c = C$ is Cantor set with $m(C) = 0$. Hence μ is singular. Now for any $x \in C$, $\mu(\{x\}) \leq 1/2^n$ after n steps. Let $n \uparrow \infty$, we get $\mu(\{x\}) = 0$, so μ is also atomless.

For general pefect set S with Borel measure zero, the complement is an open set, which can be constructed by disjoint open intervals, and these intervals have no common end, otherwise S has at least one isolated point. We just consider the condition when these open intervals is countable, and we just need to guarantee when constructing S, we remove 2^{n-1} open intervals in n th step (do not contain ∞ in these intervals) and assign $1/2^n$ for each closed interval left.

When the latter has positive measure, $\forall B \in \mathscr{B}$, define

$$\mathscr{P}(B) = \frac{m(B \cap S)}{m(S)},$$

where m is the Lebesgue measure, it is easy to verify $\mathscr{P}$ is a p.m. □

Exercise 2.2.16. Show by a trivial example that Theorem 2.2.3 becomes false if the field $\mathscr{F}_0$ is replace by an arbitrary collection that generates $\mathscr{F}$.

Proof. Consider the collection $\mathscr{F}_0 = \{\{1, 2\}, \{2, 3\}, \Omega\}, \Omega = \{1, 2, 3, 4\}$ and

$$\mu(\{1\}) = 0, \mu(\{2\}) = 0.5, \mu(\{3\}) = 0, \mu(\{4\}) = 0.5,$$
$$\nu(\{1\}) = 0.5, \nu(\{2\}) = 0, \nu(\{3\}) = 0.5, \nu(\{4\}) = 0,$$

Then $\mu = \nu$ on $\mathscr{F}_0$, but $\mu \neq \nu$ on $\mathscr{F}$. □

Exercise 2.2.17. Show that Theorem 2.2.3 may be false for measures that are σ-finite on $\mathscr{F}$. [HINT: Take Ω to be $\{1, 2, \cdots, \infty\}$ and $\mathscr{F}_0$ to be the finite sets excluding ∞ and their complements, $\mu(E)$ = number of points in E, $\mu(\infty) \neq \nu(\infty)$.]

Proof. Let $\Omega = \{1, 2, \cdots, \infty\}$, and $\mathscr{F}_0$ be the finite sets excluding ∞ and their complements. Define $\mu(E)$ = number of points in E, also ν, then $\mu = \nu$ on $\mathscr{F}_0$ (those infinite must have ∞ but $\mu = \nu = \infty$) and are σ-finite on $\mathscr{F}$, but we can define $\mu(\infty) = 1 \neq 2 = \nu(\infty)$. □

Exercise 2.2.18. Show that the $\overline{\mathscr{F}}$ in (9) | is also the collection of sets of the form $F \cup N$[or $F \backslash N$] where $F \in \mathscr{F}$ and $N \in \mathscr{N}$.

Proof. we verify bilaterally. For the set with the form $F \cup N, F \in \mathscr{F}, N \in \mathscr{N}$, $(F \cup N) \triangle F = N \backslash F \subset N$, so $F \cup N \in \overline{\mathscr{F}}$. Conversely, suppose $E \in \overline{\mathscr{F}}, F_0 \in \mathscr{F}, F_0 \backslash E \subset Z \in \mathscr{F}, \mathscr{P}(Z) = 0$, let $F = F_0 \backslash Z$, then $F \in \mathscr{F}$. As $E = (E \backslash F) \cup (E \cap F)$, we show $F \subset E$. $\forall x \in F, x \in F_0, x \in Z^c \subset (F_0^c \cup E)$, so $x \in E$, which means $F \subset E, F \in \mathscr{F}$. The case "$F \backslash N$" is similar. □

Exercise 2.2.19. Let $\mathscr{N}$ be as in the proof of Theorem 2.2.5 and $\mathscr{N}_0$ be the set of all null sets in $(\Omega, \mathscr{F}, \mathscr{P})$. Then both these collections are monotone classes, and closed with respect to the operation "$\backslash$".

Proof. Clearly $\mathscr{N}_0 \subset \mathscr{N}$ and $\mathscr{N}$ is an M.C. closed with respect to "$\backslash$". The key is to prove properties on $\mathscr{N}_0$. Assume $N_1, N_2, \cdots \in \mathscr{N}_0$ and $N_k \subset N_{k+1}, k \geq 1$, then by countably additive of $\mathscr{P}$,

$$\mathscr{P}\left(\bigcup_{k=1}^{\infty} N_k\right) = \lim_{k\to\infty} \mathscr{P}(N_k) = 0,$$

so $\bigcup_{k=1}^{\infty} N_k \in \mathscr{N}_0$. And $\forall N_1, N_2 \in \mathscr{N}_0$,

$$0 \leq \mathscr{P}(N_1 \backslash N_2) \leq \mathscr{P}(N_1) = 0,$$

so $N_1 \backslash N_2 \in \mathscr{N}_0$, so $\mathscr{N}_0$ is closed with respect to "$\backslash$" and hence is an M.C. □

Exercise 2.2.20. Let $(\Omega, \mathscr{F}, \mathscr{P})$ be a probability space and $\mathscr{F}_1$ a Borel subfield of $\mathscr{F}$. Prove that there exists a minimal B.F. $\mathscr{F}_2$ satisfying $\mathscr{F}_1 \subset \mathscr{F}_2 \subset \mathscr{F}$ and $\mathscr{N}_0 \subset \mathscr{F}_2$, where $\mathscr{N}_0$ is as in Exercise 19. A set E belongs to $\mathscr{F}_2$ if and only if there exists a set F in $\mathscr{F}_1$ such that $E \triangle F \in \mathscr{N}_0$. This $\mathscr{F}_2$ is called the *augmentation* of $\mathscr{F}_1$ with respect to $(\Omega, \mathscr{F}, \mathscr{P})$.

Proof. Let

$$\mathscr{C} = \{\mathscr{G} : \mathscr{F}_1 \subset \mathscr{G} \subset \mathscr{F}, \mathscr{N}_0 \subset \mathscr{F}\},$$

and

$$\mathscr{F}_2 = \bigcap_{\mathscr{G}\in\mathscr{C}} \mathscr{G}.$$

By Exercise 2.1.5, $\mathscr{F}_2$ is the minimal B.F. satisfying the condition. The intersection operation is valid as $\mathscr{F} \in \mathscr{G}$. Now define the subset of $\mathscr{F}_2$

$$\mathscr{H} = \{E \in \mathscr{F}_2 : \exists F \in \mathscr{F}_1, E \triangle F \in \mathscr{N}_0\}.$$

First $\mathscr{F}_1 \subset \mathscr{H}$, then $\mathscr{N}_0 \subset \mathscr{H}$ as we can take $F = \varnothing$. Now $\forall E \in \mathscr{H}$, let $F \in \mathscr{F}_1$ such that $E \triangle F \in \mathscr{N}_0$, then $E^c \in \mathscr{F}_2, F^c \in \mathscr{F}_1, E^c \triangle F^c = E \triangle F \in \mathscr{N}_0$, so $E^c \in \mathscr{H}$. Then by Exercise 2.2.19, we know property (viii) in Sec. 2.1 also holds for $\mathscr{N}_0$, then use Exercise 2.1.1, we know $\mathscr{H}$ is a B.F., but $\mathscr{F}_2$ is the minimal B.F. satisfying conditions, so $\mathscr{F}_2 \subset \mathscr{H}$, hence $\mathscr{F}_2 = \mathscr{H}$. □

Exercise 2.2.21. Suppose that $\tilde{F}$ has all the defining properties of a d.f. except that it is not assumed to be right continuous. Show that Theorem 2.2.2 and Lemma remain valid with F replaced by $\tilde{F}$, provided that we replace $F(x), F(b), F(a)$ in (4) and (5) by $\tilde{F}(x+), \tilde{F}(b+), \tilde{F}(a+)$, respectively. What modification is necessary in Theorem 2.2.4?

Proof. The only part we adjust is when $x_n \downarrow x, F(x_n) \downarrow \mu(I_x) = F(x+)$, and the same modification can be applied in the proof, so Theorem 2.2.2 and Lemma remain valid. In Theorem 2.2.4, we should modify "unique d.f. F" to "set of F at each point x its left and right limit are determined". □

Exercise 2.2.22. For an arbitrary measure $\mathscr{P}$ on a B.F. $\mathscr{F}$, a set E in $\mathscr{F}$ is called an atom iff $\mathscr{P}(E) > 0$ and $F \subset E, F \in \mathscr{F}$ imply $\mathscr{P}(F) = \mathscr{P}(E)$ or $\mathscr{P}(F) = 0$. $\mathscr{P}$ is called *atomic* iff its value is zero over any set in $\mathscr{F}$ that is disjoint from all the atoms. Prove that for a measure μ on $\mathscr{B}^1$ this new definition is equivalent to that given in Exercise 12 above provided we identify two sets which differ by a $\mathscr{P}$-null set.

Proof. We shall prove the definition of "atom" is equivalent. For $E \in \mathscr{B}^1$, consider $f(x) = \mathscr{P}(E \cap (-\infty, x])$, then f is increasing and can only take value on $\mathscr{P}(E)$ and 0. So there must exist x_0 such that $\mathscr{P}(\{x_0\}) > 0$, which is $\sup\{t : f(t) = 0\}$. And if two sets which differ by a $\mathscr{P}$-null set are viewed as the same, then $\{x_0\}$ can be the representation of atom. Conversely, set $E = \{x\}$ where $\mu(\{x\}) > 0$, then the new definition is obvious. The property atomic is automatically inherited. □

Exercise 2.2.23. Prove that if the p.m. $\mathscr{P}$ is atomless, then given any α in $[0, 1]$ there exists a set $E \in \mathscr{F}$ with $\mathscr{P}(E) = \alpha$. [HINT: Prove first that there exists E with "arbitrarily small" probability. A quick proof then follows from Zorn's lemma by considering a maximal collection of disjoint sets, the sum of whose probabilities does not exceed α. But an elementary proof without using any maximality principle is also possible.]

Proof. As $\mathscr{P}$ is atomless, there $\exists A, B \in \mathscr{F}, B \subset A, 0 < \mathscr{P}(B) < \mathscr{P}(A)$. Otherwise, $\mathscr{P}(B)$ can only be $\mathscr{P}(A)$ or 0, but this is the case where A is an atom, which is a contradiction. Assume $\exists \varepsilon_0, \forall E \in \mathscr{F}, \mathscr{P}(E) \geq \varepsilon_0$. Now $\mathscr{P}(E_1) > 0, \mathscr{P}(F_1) = \mathscr{P}(E_1^c) = 1 - \mathscr{P}(E_1) > 0$. Then $\exists E_2 \subset F_1, \mathscr{P}(E_2) \geq \varepsilon_0$. Let $F_2 = F_1 \backslash E_1$ and take E_3 from F_2, we have $\mathscr{P}(E_3) \geq \varepsilon_0$. Notice $E_1, E_2, \cdots$ are disjoint, after n steps, we have

$$\mathscr{P}\left(\bigcup_{k=1}^{n} E_k\right) = \sum_{k=1}^{n} \mathscr{P}(E_k) \geq n \cdot \varepsilon_0.$$

But $\mathscr{P} \leq 1$, after $[1/\varepsilon_0] + 1$ steps, the left side of the term above will exceed 1, which is a contradiction. So $\forall \varepsilon > 0, \exists E \in \mathscr{F}, 0 < \mathscr{P}(E) < \varepsilon$. Consider a collection $\mathscr{C}$ in which the sets are disjoint and their sum of probabilities do not exceed α, if the sum is always less than α, and by Zorn's lemma, we can find a maximal collection $\mathscr{C}$ with property mentioned, and the sum of probabilities of sets equal to α. □

Exercise 2.2.24. A point x is said to be in the support of a measure μ on $\mathscr{B}^n$ iff every open neighborhood of x has strictly positive measure. The set of all such points is called the *support* of μ. Prove that the support is a closed set whose complement is the maximal open set on which μ vanishes. Show that the support of a p.m. on $\mathscr{B}^1$ is the same as that of its d.f., defined in Exercise 6 of Sec. 1.2.

Proof. The proof for closed set is similar to what is done in Exercise 1.2.7. On its complement F^c, $\forall x \in F^c, \exists \delta_x, \mu(U(x; \delta_x)) = 0$, and clearly μ vanishes on F^c. As any p.m. on $\mathscr{B}^1$ has an one-to-one mapping to d.f., and $F(x + \varepsilon) - F(x - \varepsilon) = \mu((x - \varepsilon, x + \varepsilon])$, we shall prove $\forall \varepsilon > 0, \mu((x - \varepsilon, x + \varepsilon]) > 0$ implies $\mu(U(x; \varepsilon)) > 0$. For the former we can take $\varepsilon' = \varepsilon/2$, and we have

$$\mu\left(U\left(x; \varepsilon\right)\right) \geq \mu\left(\left(x - \frac{\varepsilon}{2}, x + \frac{\varepsilon}{2}\right]\right) = \mu((x - \varepsilon', x + \varepsilon']) > 0.$$

The converse is trivial. □

Exercise 2.2.25. Let f be measurable with respect to $\mathscr{F}$, and Z be contained in a null set. Define

$$\tilde{f} = \begin{cases} f & \text{on } Z^c, \\ K & \text{on } Z, \end{cases}$$

where K is a constant. Then $\tilde{f}$ is measurable with respect to $\mathscr{F}$ provided that $(\Omega, \mathscr{F}, \mathscr{P})$ is complete. Show that the conclusion may be false otherwise.

Proof. We shall verify $\{E : \tilde{f}(E) > t\} \in \mathscr{F}$. As

$$\{E : \tilde{f}(E) > t\} = \{\omega \subset Z^c : f(\omega) > t\} \cup \{\omega \subset Z : K > t\},$$

Notice the latter can only be $Z \in \mathscr{N}$ or $\varnothing$, and as f is measurable with respect to $\mathscr{F}$, so the former set is in $\mathscr{F}$, furthermore, $(\Omega, \mathscr{F}, \mathscr{P})$ is complete, so $Z \in \mathscr{F}$. Hence $\tilde{f}$ is measurable with respect to $\mathscr{F}$. But if we set $(\Omega, \mathscr{F}, \mathscr{P}) = (\mathscr{R}, \mathscr{B}^1, \mathscr{P})$, and a set Z with Lebesgue measure zero but is not Borel measurable, then $\tilde{f} = 1_Z$ is not measurable with $\mathscr{B}^1$. □

3 Random variable. Expectation. Independence

3.1 General definitions

Exercise 3.1.1. Prove Theorem 3.1.1. For the "direct mapping" X, which of these properties of X^{-1} holds?

Proof. $\forall \omega \in X^{-1}(A^c), X(\omega) \notin A$, which is equivalent to $\omega \in (X^{-1}(A))^c$. $\forall \omega \in X^{-1}\left(\bigcup_\alpha A_\alpha\right)$, $\exists \alpha_0, \omega \in X^{-1}(A_{\alpha_0})$, which is equivalent to $\omega \in \bigcup_\alpha X^{-1}(A_\alpha)$. The third equality is similar. For "direct mapping" X, none of these properties hold, unless it is a injection. □

Exercise 3.1.2. If two r.v.'s are equal a.e., then they have the same p.m.

Proof. Assume $X_1 = X_2$ a.e. Let $N = \{\omega : X_1(\omega) \neq X_2(\omega)\}$, Then

$$\mathscr{P}_1\{X_1 \leq x\} = \mathscr{P}_1(\{X_1 \leq x\} \cap N^c) + \mathscr{P}_1(\{X_1 \leq x\} \cap N),$$
$$\mathscr{P}_2\{X_2 \leq x\} = \mathscr{P}_2(\{X_2 \leq x\} \cap N^c) + \mathscr{P}_2(\{X_2 \leq x\} \cap N).$$

As the term behind "+" are subsets of N, their probabilities are all equals to 0. The term in front of "+" is obviously equivalent, hence $\mathscr{P}_1\{X_1 \leq x\} = \mathscr{P}_2\{X_2 \leq x\}$, thus $\mathscr{P}_1 = \mathscr{P}_2$. □

Exercise 3.1.3. Given any p.m. μ on $(\mathscr{R}^1, \mathscr{B}^1)$, define an r.v. whose p.m. is μ. Can this be done in an arbitrary probability space?

Proof. We only need to define X on $(\mathscr{R}^1, \mathscr{B}^1)$, with $X(B) = B, B \in \mathscr{B}^1$, then X has the p.m. μ. But it cannot be done in arbitrary probability space. For example, we want to define an r.v. on $(\{1\}, \{\varnothing, \{1\}\})$ which is uniformly distributed on $\mathscr{U}$. but this cannot be done as $\mathscr{P}(\{1\})$ has to be 1 under the given space, and $\mu(\{1\}) = 0$ if setting μ the Lebesgue measure on $\mathscr{U}$. □

Exercise 3.1.4. Let θ be uniformly distributed on $[0, 1]$. For each d.f. F, define $G(y) = \sup\{x : F(x) \leq y\}$. Then $G(\theta)$ has the d.f. F.

Proof. First we have

$$\mathscr{P}\{G(\theta) \leq x\} = \mathscr{P}\{\omega : \sup\{y : F(y) \leq \omega\} \leq x\}.$$

Now we show $\{\omega : G(\omega) \leq x\} = \{\omega : \omega \leq F(x)\}$ a.e. Denote the first set A and the second B, $\forall \omega \in A$, if $\omega > F(x)$, then according to the right continuity of F, $\exists \delta > 0, F(x + \delta) < \omega$, hence $G(\omega) \geq x + \delta > x$, which is a contradiction. Now $\forall \omega \in B \backslash \{\omega : \omega = F(x)\}$, as F is increasing, clearly $G(\omega) \leq x$. So

$$B \backslash \{\omega : \omega = F(x)\} \subset A \subset B.$$

Then we have $\mathscr{P}(B) - 0 \leq \mathscr{P}(A) \leq \mathscr{P}(B)$, which implies $\mathscr{P}(A) = F(x)$, as $\mathscr{P}$ is the Lebesgue measure on $[0, 1]$. □

Exercise 3.1.5. Suppose X has the continuous d.f. F, then $F(X)$ has the uniform distribution on [0,1]. What if F is not continuous?

Proof. We have

$$\begin{aligned}\mathscr{P}\{\omega : F(X(\omega)) \le u\} &= \mathscr{P}\{\omega : X(\omega) \le \sup\{t : F(t) = u\}\} \\ &= F(\sup\{t : F(t) = u\}) = u, \ \ u \in [0,1].\end{aligned}$$

The last equality is guaranteed by the continuity of F. When $u < 0$, it turns out $\{F(X) \le u\} = \varnothing$; When $u > 1$, it becomes Ω, thus $F(X)$ has the uniform distribution on [0,1]. But if F is not continuous, then $\mathscr{P}\{F(X) \le u\} \le u$, as F has jump points. □

Exercise 3.1.6. Is the range of an r.v. necessarily Borel or Lebesgue measurable?

Proof. Not really. Consider $\Omega = V$ is a Vitali's set in $\mathscr{R}^1$, and $\mathscr{F} = \mathscr{I}$. An identical mapping $X : X(\omega) = \omega$ on $(\Omega, \mathscr{I})$ is obviously an r.v., but the range is not Lebesgue measurable. □

Exercise 3.1.7. The sum, difference, product, or quotient (denominator nonvanishing) of the two discrete r.v.'s is discrete.

Proof. These forms can be uniformly written as $Z = f(X,Y)$, where f is a continuous function and X, Y are discrete r.v.'s. Now $\exists$ countable set $B_1, B_2 \subset \mathscr{R}^1$, such that $\mathscr{P}(X \in B_1) = 1, \mathscr{P}(Y \in B_2) = 1$. Consider $B = f(B_1, B_2)$, obviously B is countable. For the new r.v. Z, we have

$$\begin{aligned}1 \ge \mathscr{P}\{Z \in B\} \ge \mathscr{P}\{X \in B_1, Y \in B_2\} &= 1 - \mathscr{P}\{X \in B_1^c \text{ or } Y \in B_2^c\} \\ &\ge 1 - \mathscr{P}\{X \in B_1^c\} - \mathscr{P}\{Y \in B_2^c\} = 1.\end{aligned}$$

Hence $f(X,Y)$ is a discrete r.v. □

Exercise 3.1.8. If Ω is discrete (countable), then every r.v. is discrete. Conversely, every r.v. in a probability space is discrete if and only if the p.m. is atomic. [HINT: Use Exercise 23 of Sec. 2.2.]

Proof. We shall let

$$B = \bigcup_{n=1}^{\infty} \{X(\omega_n)\},$$

where $\omega_n (n \ge 1)$ is all singletons of Ω. Then $\mathscr{P}\{X \in B\} = 1$ is obvious. Conversely, suppose an arbitrary r.v. X is discrete, then there exists countable set Z, $\mathscr{P}\{X \in Z\} = 1$, and $\mathscr{P}\{X \in Z^c\} = 0$, and all atoms of the p.m. denoted by A is in $X^{-1}(Z)$. If the p.m. is not atomic, then $\exists S = A^c, \mathscr{P}(S) = \beta > 0$. However, according to Exercise 2.2.23, we can similarly prove that $\forall \alpha \in (0, \beta], \exists E_\alpha \subset S, E_\alpha \in \mathscr{F}, \mathscr{P}(E_\alpha) = \alpha$. We define an r.v. X_0 such that $X_0^{-1}(\alpha) = E_\alpha$, then this r.v. is obviously not discrete, which is a contradiction.

If the p.m. of r.v. is atomic, then $\mathscr{P}\left(\bigcup_{n=1}^{\infty} \{\omega_n\}\right) = 1$, where $\omega_n, n \ge 1$ is all the atoms of the p.m. Similar to the case when Ω is discrete we can finish the proof. □

Exercise 3.1.9. If f is Borel measurable, and X and Y are identically distributed, then so are $f(X)$ and $f(Y)$.

Proof. As X and Y are identically distributed, we have

$$\mathscr{P}\{X \in B\} = \mathscr{P}\{Y \in B\}, \quad \forall B \in \mathscr{B}^1.$$

Because f is Borel measurable, we have $f^{-1}(B) \in \mathscr{B}^1$, so

$$\mathscr{P}\{f(X) \in B\} = \mathscr{P}\{X \in f^{-1}(B)\} = \mathscr{P}\{f(Y) \in f^{-1}(B)\} = \mathscr{P}\{Y \in B\}, \quad \forall B \in \mathscr{B}^1,$$

hence $f(X)$ and $f(Y)$ is identically distributed. □

Exercise 3.1.10. Express the indicators of $\Lambda_1 \cup \Lambda_2, \Lambda_1 \cap \Lambda_2, \Lambda_1 \backslash \Lambda_2, \Lambda_1 \triangle \Lambda_2, \limsup \Lambda_n, \liminf \Lambda_n$ in terms of those of Λ_1, Λ_2 or Λ_n. [For the definitions of the limits see Sec. 4.2.]

Proof. We have $1_{\Lambda_1 \cup \Lambda_2} = 1_{\Lambda_1} + 1_{\Lambda_2} - 1_{\Lambda_1} \cdot 1_{\Lambda_2}$, $1_{\Lambda_1 \cap \Lambda_2} = 1_{\Lambda_1} \cdot 1_{\Lambda_2}$, $1_{\Lambda_1 \backslash \Lambda_2} = 1_{\Lambda_1}(1 - 1_{\Lambda_2})$, $1_{\Lambda_1 \triangle \Lambda_2} = 1_{\Lambda_1} + 1_{\Lambda_2} - 2 \cdot 1_{\Lambda_1} 1_{\Lambda_2}$,

$$1_{\limsup \Lambda_n} = 1_{\bigcap_{n=1}^\infty \bigcup_{m=n}^\infty \Lambda_m} = 1 - \lim_{n\to\infty} \prod_{m=n}^{\infty} (1 - 1_{\Lambda_m}) = \limsup 1_{\Lambda_n},$$

$$1_{\liminf \Lambda_n} = 1_{\bigcup_{n=1}^\infty \bigcap_{m=n}^\infty \Lambda_m} = \lim_{n\to\infty} \prod_{m=n}^{\infty} 1_{\Lambda_m} = \liminf 1_{\Lambda_n}.$$

□

Exercise 3.1.11. Let $\mathscr{F}\{X\}$ be the minimal B.F. with respect to which X is measurable. Show that $\Lambda \in \mathscr{F}\{X\}$ if and only if $\Lambda = X^{-1}(B)$ for some $B \in \mathscr{B}^1$. Is this B unique? Can there be a set $A \notin \mathscr{B}^1$ such that $\Lambda = X^{-1}(A)$?

Proof. if $\Lambda = X^{-1}(B)$ for some $B \in \mathscr{B}^1$, then Λ is a set such that X is measurable, so $\Lambda \in \mathscr{F}\{X\}$. The proof for converse is similar. B is usually not unique. For example, let X be uniformly distributed on $[0, 1]$, and $X^{-1}([0, 1]) = X^{-1}((-\infty, 1]) = \Omega$. Now assume $X \equiv 0$ on $\mathscr{F}\{X\}$, and $A = L$ is a Lebesgue measurable but not Borel measurable set and $0 \notin A$, then $\Lambda = \varnothing = X^{-1}(A) \in \mathscr{F}\{X\}$. □

Exercise 3.1.12. Generalize the assertion in Exercise 11 to a finite set of r.v.'s. [It is possible to generalize even to an arbitrary set of r.v.'s.]

Proof. Let $\mathscr{F}\{X_1, X_2, \cdots, X_n\}$ be the minimal B.F. with respect to which $(X_1, X_2, \cdots, X_n)$ is measurable. Then $(\Lambda_1, \Lambda_2, \cdots, \Lambda_n) \in \mathscr{F}\{X_1, X_2, \cdots, X_n\}$ if and only if $(\Lambda_1, \Lambda_2, \cdots, \Lambda_n) = (X_1, X_2, \cdots, X_n)^{-1}(B)$ for some $B \in \mathscr{B}^n$. The B is not unique in general and $\exists A \notin \mathscr{B}^n$, such that $(\Lambda_1, \Lambda_2, \cdots, \Lambda_n) = X^{-1}(A)$. □

3.2 Properties of mathematical expectation

Exercise 3.2.1. If $X \geq 0$ a.e. on Λ and $\int_\Lambda X \mathrm{d}\mathscr{P} = 0$, then $X = 0$ a.e. on Λ.

Proof. Define sequence

$$E = \{\omega : X(\omega) > 0\}, E_n = \left\{\omega : X(\omega) \geq \frac{1}{n}\right\},$$

then $E = \bigcup_{n=1}^\infty E_n$. We have

$$0 = \int_\Lambda X \mathrm{d}\mathscr{P} \geq \int_{E_n} X \mathrm{d}\mathscr{P} \geq \frac{1}{n} \mathscr{P}(E_n) \geq 0,$$

so $\mathscr{P}(E_n) = 0, \forall n \geq 1$. Hence

$$0 \leq \mathscr{P}(E) = \mathscr{P}\left(\bigcup_{n=1}^{\infty} E_n\right) \leq \sum_{n=1}^{\infty} \mathscr{P}(E_n) = 0,$$

which implies $X = 0$ a.e. on Λ. $\square$

Exercise 3.2.2. If $\mathscr{E}(|X|) < \infty$ and $\lim_{n\to\infty} \mathscr{P}(\Lambda_n) = 0$, then $\lim_{n\to\infty} \int_{\Lambda_n} X \mathrm{d}\mathscr{P} = 0$. In particular

$$\lim_{n\to\infty} \int_{\{|X|>n\}} X \mathrm{d}\mathscr{P} = 0.$$

Proof. First we have

$$0 \leq \left| \lim_{n\to\infty} \int_{\Lambda_n} X \mathrm{d}\mathscr{P} \right| \leq \lim_{n\to\infty} \int_{\Lambda_n} |X| \mathrm{d}\mathscr{P}.$$

Then we discuss the limit concerning $|X|$.

$$\begin{aligned} \int_{\Lambda_n} |X| \mathrm{d}\mathscr{P} &= \int_{\Lambda_n \cap \{|X| \leq M\}} |X| \mathrm{d}\mathscr{P} + \int_{\Lambda_n \cap \{|X| > M\}} |X| \mathrm{d}\mathscr{P} \\ &\leq M\mathscr{P}(\Lambda_n) + \int_{\{|X|>M\}} |X| \mathrm{d}\mathscr{P}, \quad \forall M > 0, n \geq 1. \end{aligned}$$

First we let $n \to \infty$ and $M\mathscr{P}(\Lambda_n) \to 0$. Then use the dominated convergence theorem, we know the second term converges to 0 when $M \to \infty$. Hence $\lim_{n\to\infty} \int_{\Lambda_n} X \mathrm{d}\mathscr{P} = 0$. Let $\Lambda_n = \{|X| > n\}$ and we get the particular case. $\square$

Exercise 3.2.3. Let $X \geq 0$ and $\int_\Omega X \mathrm{d}\mathscr{P} = A, 0 < A < \infty$. Then the set function ν defined on $\mathscr{F}$ as follows:

$$\nu(\Lambda) = \frac{1}{A} \int_\Lambda X \mathrm{d}\mathscr{P},$$

is a probability measure on $\mathscr{F}$.

Proof. As $X \geq 0$, $\nu(\cdot) \geq 0$. The countably additivity is easy to verify. Finally, $\nu(\Omega) = (\int_\Omega X \mathrm{d}\mathscr{P})/A = 1$. Hence ν is a p.m. on $\mathscr{F}$. $\square$

Exercise 3.2.4. Let c be a fixed constant, $c > 0$. Then $\mathscr{E}(|X|) < \infty$ if and only if

$$\sum_{n=1}^{\infty} \mathscr{P}(|X| \geq cn) < \infty.$$

In particular, if the last series converges for one value of c, it converges for all values of c.

Proof. The proof is same as the proof of Theorem 3.2.1, and substitute n with cn is enough. The particular case is because we know $\mathscr{E}(|X|) < \infty$ from the convergence, and then we can set arbitrary $c > 0$ and the series is converged. $\square$

Exercise 3.2.5. For any $r > 0, \mathscr{E}(|X|^r) < \infty$ if and only if

$$\sum_{n=1}^{\infty} n^{r-1} \mathscr{P}(|X| \geq n) < \infty.$$

Proof. Imitate the proof of Theorem 3.2.1, we get

$$\sum_{n=0}^{\infty} n^r \mathscr{P}(\Lambda_n) \leq \mathscr{E}(|X|) \leq \sum_{n=0}^{\infty} (n+1)^r \mathscr{P}(\Lambda_n),$$

where $\Lambda_n = \{n \leq |X| < n+1\}$. Then use Abel's method, we get

$$\sum_{n=0}^{N} n^r \mathscr{P}(\Lambda_n) = \sum_{n=1}^{N} \{n^r - (n-1)^r\} \mathscr{P}(|X| \geq n) - N^r \mathscr{P}(|X| \geq N+1).$$

As $\lim_{n\to\infty} \{n^r - (n-1)^r\}/n^{r-1} = r$, we can find constant c_1, c_2 such that

$$c_1 \cdot n^{r-1} \leq n^r - (n-1)^r \leq c_2 \cdot n^{r-1}, \quad \forall n \geq 1.$$

Hence

$$\begin{aligned} c_1 \cdot \sum_{n=1}^{N} n^{r-1} \mathscr{P}(|X| \geq n) - N^r \mathscr{P}(|X| \geq N+1) &\leq \mathscr{E}(|X|) \\ &\leq c_2 \cdot \sum_{n=1}^{N+1} n^{r-1} \mathscr{P}(|X| \geq n) - N^r \mathscr{P}(|X| \geq N+1) - \mathscr{P}(|X| \geq 1). \end{aligned}$$

Then the sufficient and necessary condition in the question is obvious. □

Exercise 3.2.6. Suppose that $\sup_n |X_n| \leq Y$ on Λ with $\int_\Lambda Y \mathrm{d}\mathscr{P} < \infty$. Deduce from Fatou's lemma:

$$\int_\Lambda \left(\overline{\lim_{n\to\infty}} X_n \right) \mathrm{d}\mathscr{P} \geq \overline{\lim_{n\to\infty}} \int_\Lambda X_n \mathrm{d}\mathscr{P}.$$

Show this is false if the condition involving Y is omitted.

Proof. We have $Y - X_n \geq 0$ on Λ for all n. By Fatou's lemma, we have

$$\int_\Lambda \underline{\lim}_{n\to\infty} (Y - X_n) \mathrm{d}\mathscr{P} \leq \underline{\lim}_{n\to\infty} \int_\Lambda (Y - X_n) \mathrm{d}\mathscr{P}.$$

As $\int_\Lambda Y \mathrm{d}\mathscr{P} < \infty$, we can minus it on the both sides and get the desired inequality. If the condition involving Y is ommited, consider $X_n = n \times 1_{(0,\frac{1}{n}]}, n \geq 1$ defined on $(\mathscr{U}, \mathscr{B}, m)$. Then LHS equals to 0 and RHS equals to 1, do not satisfy the equality. □

Exercise 3.2.7. Given the r.v. X with finite $\mathscr{E}(X)$, and $\varepsilon > 0$, there exists a simple r.v. X_ε (see the end of Sec. 3.1) such that

$$\mathscr{E}(|X - X_\varepsilon|) < \varepsilon.$$

Hence there exists a sequence of simple r.v.'s X_m such that

$$\lim_{m\to\infty} \mathscr{E}(|X - X_m|) = 0.$$

We can choose $\{X_m\}$ so that $|X_m| \leq |X|$ for all m.

Proof. This is similar to the construction of arbitrary r.v. Let

$$\Lambda_{m,n} = \left\{\omega : \frac{n}{2^m} \le X(\omega) < \frac{n+1}{2^m}\right\}, \quad n = 0, 1, \cdots, m \cdot 2^m - 1,$$

and $\Lambda_{m,m\cdot 2^m} = \{\omega : X(\omega) \ge n\}$, and let X_m denote the r.v. belonging to the weighted partition $\bigcup_{n=1}^{m\cdot 2^m} \{\Lambda_{m,n}; n/2^m\}$, then $\forall\omega : X_m(\omega) \le X_{m+1}(\omega)$ and $X_m \uparrow X$. Hence by monotone convergence theorem, we have

$$\lim_{m\to\infty} \mathscr{E}(X_n) = \mathscr{E}(X).$$

This is the case when $X \ge 0$, and as $\mathscr{E}(X) < \infty$, we know $\mathscr{E}(X_n) < \infty, \forall n \ge 1$, so $\forall\varepsilon > 0, \exists N \in \mathbb{N}_+, \mathscr{E}(|X - X_n|) = \mathscr{E}(X - X_n) < \varepsilon$. Let $X_\varepsilon = X_{N+1}$ is the simple r.v. satisfying the condition and it finishes the proof for $X \ge 0$. For general X, it can be written as $X^+ - X^-$, then we know $\mathscr{E}(X^+) < \infty, \mathscr{E}(X^-) < \infty$. Hence for X^+ and X^-, there exists $X_\varepsilon^+, X_\varepsilon^-$ such that

$$\mathscr{E}(X^+ - X_\varepsilon^+) < \frac{\varepsilon}{2}, \mathscr{E}(X^- - X_\varepsilon^-) < \frac{\varepsilon}{2}.$$

Let $X_\varepsilon = X_\varepsilon^+ - X_\varepsilon^-$, then we have

$$\begin{aligned}\mathscr{E}(|X - X_\varepsilon|) &= \mathscr{E}(|X^+ - X_\varepsilon^+ - (X^- - X_\varepsilon^-)|)\\ &\le \mathscr{E}(|X^+ - X_\varepsilon^+|) + \mathscr{E}(|X^- - X_\varepsilon^-|) < \varepsilon.\end{aligned}$$

This finishes the whole proof. choose $\varepsilon = 1/m$ and X_m corresponding to it we get the particular case. □

Exercise 3.2.8. For any two sets Λ_1 and Λ_2 in $\mathscr{F}$, define

$$\rho(\Lambda_1, \Lambda_2) = \mathscr{P}(\Lambda_1 \triangle \Lambda_2);$$

then ρ is a pseudo-metric in the space of sets in $\mathscr{F}$; call the resulting metric space $\mathbf{M}(\mathscr{F}, \mathscr{P})$. Prove that for each integrable r.v. X the mapping of $\mathbf{M}(\mathscr{F}, \mathscr{P})$ to $\mathscr{R}^1$ given by $\Lambda \to \int_\Lambda X \mathrm{d}\mathscr{P}$ is continuous. Similarly, the mappings on $\mathbf{M}(\mathscr{F}, \mathscr{P}) \times \mathbf{M}(\mathscr{F}, \mathscr{P})$ to $\mathbf{M}(\mathscr{F}, \mathscr{P})$ given by

$$(\Lambda_1, \Lambda_2) \to \Lambda_1 \cup \Lambda_2, \Lambda_1 \cap \Lambda_2, \Lambda_1 \backslash \Lambda_2, \Lambda_1 \triangle \Lambda_2$$

are all continuous. If (see Sec. 4.2 below)

$$\limsup_n \Lambda_n = \liminf_n \Lambda_n$$

modulo a null set, we denote the common equivalence class of these two sets by $\lim_n \Lambda_n$. Prove that in this case $\{\Lambda_n\}$ converges to $\lim_n \Lambda_n$ in the metric ρ. Deduce Exercise 2 above as a special case.

Proof. First $\rho \ge 0$ and $\rho = 0$ when $\Lambda_1 \triangle \Lambda_2$ is a null-set. As $\Lambda_1 \triangle \Lambda_2 = (\Lambda_1 \triangle \Lambda_3) \triangle (\Lambda_2 \triangle \Lambda_3)$, we have

$$\rho(\Lambda_1, \Lambda_2) \le \rho(\Lambda_1, \Lambda_3) + \rho(\Lambda_2, \Lambda_3).$$

Hence ρ is a pseudo-metric in the space of sets in $\mathscr{F}$. Now $\forall\varepsilon > 0$,

$$\begin{aligned}\left|\int_{\Lambda_1} X \mathrm{d}\mathscr{P} - \int_{\Lambda_2} X \mathrm{d}\mathscr{P}\right| &= \left|\int_{\Lambda_1\backslash\Lambda_2} X \mathrm{d}\mathscr{P} - \int_{\Lambda_2\backslash\Lambda_1} X \mathrm{d}\mathscr{P}\right|\\ &\le \int_{\Lambda_1\backslash\Lambda_2} |X| \mathrm{d}\mathscr{P} + \int_{\Lambda_2\backslash\Lambda_1} |X| \mathrm{d}\mathscr{P} = \int_{\Lambda_1\triangle\Lambda_2} |X| \mathrm{d}\mathscr{P}.\end{aligned}$$

Use Exercise 3.2.2, $\exists\delta > 0, \rho(\Lambda_1, \Lambda_2) = \mathscr{P}(\Lambda_1 \triangle \Lambda_2) < \delta$, and $\int_{\Lambda_1\triangle\Lambda_2} |X| \mathrm{d}\mathscr{P} < \varepsilon$. Hence we proved the continuity. As for the mapping $(\Lambda_1, \Lambda_2) \to \Lambda_1 \cup \Lambda_2$, $\forall (A_1, A_2), (B_1, B_2) \in \mathbf{M}(\mathscr{F}, \mathscr{P}) \times \mathbf{M}(\mathscr{F}, \mathscr{P})$, we have

$$\begin{aligned}\rho(A_1 \cup A_2, B_1 \cup B_2) &= \mathscr{P}((A_1 \cup A_2) \triangle (B_1 \cup B_2)) \\ &\leq \mathscr{P}(A_1 \triangle B_1) + \mathscr{P}(A_2 \triangle B_2) \\ &= \rho(A_1, B_1) + \rho(A_2, B_2).\end{aligned}$$

We use the pseudo-metric $\rho + \rho$ in the two-dimensional space. From equality above we can easily see the continuity. The proof for continuity of other mappings is like what is done for union. Denote $\Lambda^* = \limsup_n \Lambda_n, \Lambda_* = \liminf_n \Lambda_n, \Lambda = \lim_n \Lambda_n$, and we know $\rho(\Lambda^*, \Lambda_*) = 0$. Define

$$A_n = \bigcap_{m=n}^{\infty} \Lambda_m, B_n = \bigcup_{m=n}^{\infty} \Lambda_m,$$

then obviously $A_n \subset \Lambda_n \subset B_n, A_n \uparrow \Lambda_*, B_n \downarrow \Lambda^*$. As

$$\Lambda_n \triangle \Lambda = (\Lambda_n \backslash \Lambda) \cup (\Lambda \backslash \Lambda_n) \subset (\Lambda_n \backslash A_n) \cup (B_n \backslash \Lambda_n),$$

we have

$$\rho(\Lambda_n, \Lambda) \leq \mathscr{P}(\Lambda_n \backslash A_n) + \mathscr{P}(B_n \backslash \Lambda_n) = \mathscr{P}(B_n) - \mathscr{P}(A_n), \quad \forall n \geq 1.$$

Use the continuity of $\mathscr{P}$, we have

$$\lim_{n\to\infty} \rho(\Lambda_n, \Lambda) \leq \mathscr{P}(\Lambda^*) - \mathscr{P}(\Lambda_*) = 0.$$

Hence $\Lambda_n \to \Lambda$ under ρ. Then we take $\Lambda = \varnothing$, then the condition of Exercise 3.2.2 if just $\rho(\Lambda_n, \varnothing) \to 0 (n \to \infty)$. As $f(\Lambda) = \int_\Lambda X \mathrm{d}\mathscr{P}$ is a continuous function concerning Λ, we have $\lim_{n\to\infty} f(\Lambda_n) = f(\varnothing) = 0$. □

Exercise 3.2.9. Another proof of (14): verify it first for simple r.v.'s and then use Exercise 7 of Sec. 3.2.

Proof. Let $B \in \mathscr{B}^2$ and $f = 1_B$, then let side is $\mathscr{P}((X, Y) \in B)$ and the right side is $\mu(B)$, and they are equal. Then by the linearity of both integrals in (14), it will also hold if

$$f = \sum_j b_j 1_{B_j}.$$

As $f(X, Y)$ is also an r.v., by Exercise 3.2.7, there exists simple $\{f_m(X, Y)\}$ such that $\lim_{m\to\infty} \mathscr{E}(|f - f_m|) = 0$. Hence

$$\begin{aligned}\int_\Omega f(X(\omega), Y(\omega)) \mathscr{P}(\mathrm{d}\omega) &= \lim_{m\to\infty} \mathscr{E}(f(X_m, Y_m)) = \lim_{m\to\infty} \iint_{\mathscr{R}^2} f_m(x, y) \mu^2(\mathrm{d}x, \mathrm{d}y) \\ &= \iint_{\mathscr{R}^2} \lim_{m\to\infty} f_m(x, y) \mu^2(\mathrm{d}x, \mathrm{d}y) = \iint_{\mathscr{R}^2} f(x, y) \mu^2(\mathrm{d}x, \mathrm{d}y).\end{aligned}$$

□

Exercise 3.2.10. Prove that if $0 \leq r < r'$ and $\mathscr{E}(|X|^{r'}) < \infty$, then $\mathscr{E}(|X|^r) < \infty$. Also that $\mathscr{E}(|X|^r) < \infty$ if and only if $\mathscr{E}(|X - a|^r) < \infty$ for every a.

Proof. Use Liapounov inequality, we have

$$\mathscr{E}(|X|^r) \leq \mathscr{E}(|X|^{r'})^{\frac{r}{r'}} < \infty.$$

For the second, we shall prove the "only if" part. The "if" part is trivial. We have

$$\begin{aligned}\mathscr{E}(|X-a|^r) &= \int_{\{|X|\leq|a|\}} |X-a|^r \mathrm{d}\mathscr{P} + \int_{\{|X|>|a|\}} |X-a|^r \mathrm{d}\mathscr{P} \\ &\leq |2a|^r + \int_{\{|X|>|a|\}} (2|X|)^r \mathrm{d}\mathscr{P} \\ &\leq 2^r \mathscr{E}(|X|^r) + 2^r \cdot |a|^r < \infty.\end{aligned}$$

□

Exercise 3.2.11. If $\mathscr{E}(X^2) = 1$ and $\mathscr{E}(|X|) \geq a > 0$, then $\mathscr{P}\{|X| \geq \lambda a\} \geq (1-\lambda)^2 a^2$ for $0 \leq \lambda \leq 1$.

Proof. Let $\Lambda = \{|X| \geq \lambda a\}$, then

$$\mathscr{E}(|X|) = \mathscr{P}(|X|1_\Lambda) + \mathscr{P}(|X|1_{\Lambda^c}) \geq a.$$

Use Cauchy-Schwarz inequality for the first part, we have

$$\mathscr{E}(|X|1_\Lambda) \leq \left[\mathscr{E}(X^2)\mathscr{E}(1_\Lambda^2)\right]^{\frac{1}{2}} = \sqrt{\mathscr{P}(\Lambda)}.$$

For the second part, $\mathscr{P}(|X|1_{\Lambda^c}) < \lambda a$. Hence $\sqrt{\mathscr{P}(\Lambda)} \geq (1-\lambda)a$, squaring on the both sides, we get

$$\mathscr{P}\{|X| \geq \lambda a\} \geq (1-\lambda)^2 a^2, \quad 0 \leq \lambda \leq 1.$$

□

Exercise 3.2.12. If $X \geq 0$ and $Y \geq 0, p \geq 0$, then $\mathscr{E}\{(X+Y)^p\} \leq 2^p\{\mathscr{E}(X^p) + \mathscr{E}(Y^p)\}$. If $p > 1$, the factor 2^p may be replaced by 2^{p-1}. If $0 \leq p \leq 1$, it may be replaced by 1.

Proof. We shall directly prove the stronger case. As $\varphi(x) = x^p$ is a convex function when $p > 1$, we have

$$\left(\frac{x+y}{2}\right)^p \leq \frac{x^p + y^p}{2}.$$

Substitute the number x and y by r.v.'s X and Y, then integrate on Ω, we have $\mathscr{E}\{(X+Y)^p\} \leq 2^{p-1}\{\mathscr{E}(X^p) + \mathscr{E}(Y^p)\}$. For the second, we shall prove $(x+y)^p \leq x^p + y^p$ for $0 \leq p \leq 1$. We can assume $y > 0$ as this is the trivial case, then let $t = x/y \geq 0$, the question is to prove $g(t) = (1+t)^p - t^p - 1 \leq 0$. As $g(0) = 0$ and

$$g'(t) = p(1+t)^{p-1} - pt^{p-1} = p\left[(t+1)^{-(1-p)} - t^{-(1-p)}\right] \leq 0,$$

we have $g(t) \leq 0, t \geq 0$. Hence the inequality hold. Then use the similar operation and we get the inequality concerning r.v.'s. 2^p is a looser bound as $\max\{1, 2^{p-1}\} \leq 2^p, p \geq 0$. □

Exercise 3.2.13. If $X_j \geq 0$, then

$$\mathscr{E}\left\{\left(\sum_{j=1}^n X_j\right)^p\right\} \leq \quad \text{or} \quad \geq \sum_{j=1}^n \mathscr{E}(X_j^p)$$

according as $p \leq 1$ or $p \geq 1$.

Proof. We only need to prove the inequality when $n = 2$, the case when $n > 2$ can be finished by deduction method. But when $n = 2$, we can use g in Exercise 3.2.12 and the proof is quite easy. (**Note: we shall restrict $p \geq 0$ to avoid the nonexistence of power.**) □

Exercise 3.2.14. If $p > 1$, we have

$$\left|\frac{1}{n}\sum_{j=1}^{n} X_j\right|^p \leq \frac{1}{n}\sum_{j=1}^{n} |X_j|^p$$

and so

$$\mathscr{E}\left\{\left|\frac{1}{n}\sum_{j=1}^{n} X_j\right|^p\right\} \leq \frac{1}{n}\sum_{j=1}^{n} \mathscr{E}(|X_j|^p);$$

we have also

$$\mathscr{E}\left\{\left|\frac{1}{n}\sum_{j=1}^{n} X_j\right|^p\right\} \leq \left\{\frac{1}{n}\sum_{j=1}^{n} \mathscr{E}(|X_j|^p)^{1/p}\right\}^p.$$

Compare the inequalities.

Proof. The first inequality is obvious using Jensen's inequality on $f(x) = x^p (p > 1)$, and we get the following inequality about expectation. The last should use Minkowski's inequality:

$$\mathscr{E}\left\{\left|\frac{1}{n}\sum_{j=1}^{n} X_j\right|^p\right\}^{\frac{1}{p}} \leq \sum_{j=1}^{n} \mathscr{E}\left\{\left|\frac{X_j}{n}\right|^p\right\}^{\frac{1}{p}} = \frac{1}{n}\sum_{j=1}^{n} \mathscr{E}\left\{|X_j|^p\right\}^{\frac{1}{p}}.$$

Then raise both sides to the power of p and we get the inequality. □

Exercise 3.2.15. If $p > 0, \mathscr{E}(|X|^p) < \infty$, then $x^p \mathscr{P}\{|X| > x\} = o(1)$ as $x \to \infty$. Conversely, if $x^p \mathscr{P}\{|X| > x\} = o(1)$, then $\mathscr{E}(|X|^{p-\varepsilon}) < \infty$ for $0 < \varepsilon < p$.

Proof. As $\mathscr{E}(|X|^p) < \infty$, we have

$$\int_{\{|X|>x\}} |X|^p \mathrm{d}\mathscr{P} \geq \int_{\{|X|>x\}} x^p \mathrm{d}\mathscr{P} = x^p \mathscr{P}\{|X| > x\} \to 0. \ \ (x \to \infty)$$

Hence $x^p \mathscr{P}\{|X| > x\} = o(1)$. Conversely, we shall use Exercise 3.2.5, and the question transfer to prove

$$\sum_{n=1}^{\infty} n^{p-\varepsilon-1} \mathscr{P}(|X| \geq n) < \infty.$$

As $x^p \mathscr{P}\{|X| > x\} = o(1), \exists C > 0 \mathscr{P}\{|X| > x\} < C/x^p$. Let $x = n$ and we get

$$\sum_{n=1}^{\infty} n^{p-\varepsilon-1} \mathscr{P}(|X| \geq n) < C \cdot \sum_{n=1}^{\infty} \frac{1}{n^{1+\varepsilon}} < \infty.$$

Note that $>$ can be substituted with $\geq$, which does not affect the final result, as $\mathscr{P}(|X| \geq n) \leq \mathscr{P}(|X| > n - 1) < C/(n-1)^p < C'/n^p$. □

Exercise 3.2.16. For any d.f. and any $a \geq 0$, we have

$$\int_{-\infty}^{\infty} [F(x + a) - F(x)]\mathrm{d}x = a.$$

Proof. We can write $F(x+a) - F(x)$ to

$$\mu((x, x+a]) = \int_{\mathscr{R}^1} 1_{(x,x+a]}(y) \mathrm{d}\mathscr{P},$$

then we use Fubini's theorem, we have

$$\begin{aligned}\int_{-\infty}^{\infty} [F(x+a) - F(x)]\mathrm{d}x &= \int_{-\infty}^{\infty} \int_{\mathscr{R}^1} 1_{(x,x+a]}(y) \mathrm{d}\mathscr{P}\mathrm{d}x \\ &= \int_{\mathscr{R}^1} \int_{-\infty}^{\infty} 1_{(x,x+a]}(y) \mathrm{d}x\mathrm{d}\mathscr{P} \\ &= \int_{\mathscr{R}^1} a\mathrm{d}\mathscr{P} = a.\end{aligned}$$

□

Exercise 3.2.17. If F is a d.f. such that $F(0-) = 0$, then

$$\int_0^{\infty} \{1 - F(x)\}\mathrm{d}x = \int_0^{\infty} x\mathrm{d}F(x) \leq +\infty.$$

Thus if X is a positive r.v., then we have

$$\mathscr{E}(X) = \int_0^{\infty} \mathscr{P}\{X > x\}\mathrm{d}x = \int_0^{\infty} \mathscr{P}\{X \geq x\}\mathrm{d}x.$$

Proof. Consider the r.v. X having d.f. F, then

$$1 - F(x) = \mathscr{P}(X > x) = \int_0^{\infty} 1_{\{y>x\}}\mathrm{d}F(y).$$

Then we use Fubini's theorem,

$$\begin{aligned}\int_0^{\infty} \{1 - F(x)\}\mathrm{d}x &= \int_0^{\infty} \int_0^{\infty} 1_{\{y>x\}}\mathrm{d}F(y)\mathrm{d}x \\ &= \int_0^{\infty} \int_0^{\infty} 1_{\{x<y\}}\mathrm{d}x\mathrm{d}F(y) \\ &= \int_0^{\infty} y\mathrm{d}F(y) < \infty \leq +\infty.\end{aligned}$$

According to the form we get, $\mathscr{E}(X) = \int_0^{\infty} \mathscr{P}\{X > x\}\mathrm{d}x$ by letting $f = 1$ in Theorem 3.2.2. Now consider the case $X(\omega) = x$, and

$$\int_0^{\infty} \mathscr{P}\{X = x\}\mathrm{d}x = 0,$$

as F has at most countable jump points, hence the integral is 0. So we can change $>$ to $\geq$. □

Exercise 3.2.18. Prove that $\int_{-\infty}^{\infty} |x|\mathrm{d}F(x) < \infty$ if and only if

$$\int_{-\infty}^{0} F(x)\mathrm{d}x < \infty \text{ and } \int_0^{\infty} \{1 - F(x)\}\mathrm{d}x < \infty.$$

Proof. The integral $\int_{-\infty}^{\infty}|x|\mathrm{d}F(x)$ can be rewritten as

$$\int_{-\infty}^{+\infty}|x|\mathrm{d}F(x)=\int_0^\infty x\mathrm{d}F(x)-\int_{-\infty}^0 x\mathrm{d}F(x).$$

The first term is exactly what is shown in Exercise 3.2.17. For the second,

$$\begin{aligned}\int_{-\infty}^0 F(x)\mathrm{d}x &= \int_{-\infty}^0\int_{-\infty}^0 1_{\{y\le x\}}\mathrm{d}F(y)\mathrm{d}x\\ &= \int_{-\infty}^0\int_{-\infty}^0 1_{\{x\ge y\}}\mathrm{d}x\mathrm{d}F(y)\\ &= \int_{-\infty}^0 (-y)\mathrm{d}F(y).\end{aligned}$$

The last $-y$ is because $y<0$ and the Lebesgue measure $-y>0$. Hence

$$\int_{-\infty}^{\infty}|x|\mathrm{d}F(x)=\int_{-\infty}^0 F(x)\mathrm{d}x+\int_0^\infty\{1-F(x)\}\mathrm{d}x.$$

Then we have the equivalence of those finite relations. □

Exercise 3.2.19. If $\{X_n\}$ is a sequence of identically distributed r.v.'s with finite mean, then

$$\lim_n\frac{1}{n}\mathscr{E}\left\{\max_{1\le j\le n}|X_j|\right\}=0.$$

[HINT: Use Exercise 17 to express the mean of the maximum.]

Proof. Let $M_n=\max_{1\le j\le n}|X_j|$, then M_n is a positive r.v. By Exercise 3.2.17, we have $\mathscr{E}(M_n)=\int_0^\infty\mathscr{P}(M_n>x)\mathrm{d}x$. Let $f_n(t)=\frac{1}{n}\mathscr{P}(M_n>t)$, then

$$f_n(t)=\frac{1}{n}\mathscr{P}\left(\bigcup_{j=1}^n\left\{|X_j|>t\right\}\right)\le\frac{1}{n}\cdot n\mathscr{P}\{|X_1|>t\}=\mathscr{P}\{|X_1|>t\}.$$

As $\int_0^\infty\mathscr{P}(|X_1|>t)\mathrm{d}t=\mathscr{E}(|X_1|)<\infty$, by dominated convergence theorem, we have

$$\lim_n\frac{1}{n}\mathscr{E}\{M_n\}=\int_0^\infty\lim_n f_n(t)\mathrm{d}t=0.$$ □

Exercise 3.2.20. For $r>1$, we have

$$\int_0^\infty\frac{1}{u^r}\mathscr{E}(X\wedge u^r)\mathrm{d}u=\frac{r}{r-1}\mathscr{E}(X^{1/r}).$$

[HINT: By Exercise 17,

$$\mathscr{E}(X\wedge u^r)=\int_0^{u^r}\mathscr{P}(X>x)\mathrm{d}x=\int_0^u\mathscr{P}(X^{1/r}>v)rv^{r-1}\mathrm{d}v,$$

substitute and invert the order of the repeated integrations.]

Proof. Note that $X \wedge u^r \geq 0$ as the domain of integral about u is $(0, \infty)$ (**Note: we must assume $X \geq 0$, otherwise the equality does not hold.**). Hence by Exercise 3.2.17, we have

$$\mathscr{E}(X \wedge u^r) = \int_0^\infty \mathscr{P}\{(X \wedge u^r) > x\}\mathrm{d}x = \int_0^{u^r} \mathscr{P}(X > x)\mathrm{d}x = \int_0^u \mathscr{P}(X^{1/r} > v)rv^{r-1}\mathrm{d}v.$$

Then

$$\begin{aligned}\int_0^\infty \frac{1}{u^r}\mathscr{E}(X \wedge u^r)\mathrm{d}u &= \int_0^\infty \int_0^u \frac{1}{u^r}\mathscr{P}(X^{1/r} > v)rv^{r-1}\mathrm{d}v\mathrm{d}u \\ &= \int_0^\infty \int_v^\infty \frac{1}{u^r}\mathscr{P}(X^{1/r} > v)rv^{r-1}\mathrm{d}u\mathrm{d}v \\ &= \frac{r}{r-1}\int_0^\infty \mathscr{P}(X^{1/r} > v)\mathrm{d}v = \frac{r}{r-1}\mathscr{E}(X^{1/r}).\end{aligned}$$ □

3.3 Independence

Exercise 3.3.1. Show that two r.v.'s on $(\Omega, \mathscr{F})$ may be independent according to one p.m. $\mathscr{P}$ but not according to another!

Proof. Let $\Omega = \{1, 2\}, \mathscr{F} = \{\varnothing, \{1\}, \{2\}, \Omega\}$, we define two p.m.'s μ and ν, where $\mu(\{1\}) = 1, \mu(\{2\}) = 0, \nu(\{1\}) = 0.5, \nu(\{2\}) = 0.5$, and two r.v. $X_1(\{1\}) = 1, X_1(\{2\}) = 2, X_1(\{1, 2\}) = 3$, also X_2. Then under μ, X_1 and X_2 are independent, but under ν they are not. □

Exercise 3.3.2. If X_1 and X_2 are independent r.v.'s each assuming the values $+1$ and -1 with probability $\frac{1}{2}$, then the three r.v.'s $\{X_1, X_2, X_1X_2\}$ are pairwise independent but not totally independent. Find an example of n r.v.'s such that every $n-1$ of then are independent but not all of them. Find an example where the events Λ_1 and Λ_2 are independent, Λ_1 and Λ_3 are independent, but Λ_1 and $\Lambda_2 \cup \Lambda_3$ are not independent.

Proof. We only need to prove X_1, X_1X_2 are independent. We have

$$\mathscr{P}(X_1 = 1, X_1X_2 = 1) = \mathscr{P}(X_1 = 1, X_2 = 1) = \frac{1}{4} = \mathscr{P}(X_1 = 1)\mathscr{P}(X_1X_2 = 1),$$

and other cases are similar. The last equation is because $\{X_1X_2 = 1\} = \{X_1 = 1, X_2 = 1\} \cup \{X_1 = -1, X_2 = -1\}$. But all of them are not independent, as

$$\mathscr{P}(X_1 = 1, X_2 = 1, X_1X_2 = -1) = 0 \neq \frac{1}{8} = \mathscr{P}(X_1 = 1)\mathscr{P}(X_2 = 1)\mathscr{P}(X_1X_2 = -1).$$

For n r.v.'s, we still use r.v.'s having the same properties, naming them $X_1, X_2, \cdots, X_{n-1}$, and let $X_n = X_1X_2\cdots X_{n-1}$. Then

$$\mathscr{P}(X_1 = 1, X_2 = 1, \cdots, X_{n-1} = 1, X_n = -1) = 0 < \prod_{i=1}^{n-1} \mathscr{P}(X_i = 1) \times \mathscr{P}(X_n = -1),$$

and any $n-1$ of them are independent as X_n is entirely determined by $X_1, X_2, \cdots, X_{n-1}$. And we have $\mathscr{P}(X_n = 1) = 1/2$, by symmetry.

For the third, choose $\Lambda_1 = \{1, 2\}, \Lambda_2 = \{1, 3\}, \Lambda_3 = \{1, 4\}$ and

$$\mathscr{P}(\Lambda_i) = 1/2, i = 1, 2, 3, \mathscr{P}(\{1\}) = 1/4,$$

then we can verify that

$$\begin{aligned}
\mathscr{P}(\Lambda_1)\mathscr{P}(\Lambda_2) &= \frac{1}{4} = \mathscr{P}(\Lambda_1\Lambda_2),\\
\mathscr{P}(\Lambda_1)\mathscr{P}(\Lambda_3) &= \frac{1}{4} = \mathscr{P}(\Lambda_1\Lambda_3),\\
\mathscr{P}(\Lambda_1)\mathscr{P}(\Lambda_2 \cup \Lambda_3) &= \frac{3}{8} \neq \mathscr{P}(\Lambda_1 \cap (\Lambda_2 \cup \Lambda_3)) = \frac{1}{4}.
\end{aligned}$$

□

Exercise 3.3.3. If the events $\{E_\alpha, \alpha \in A\}$ are independent, then so are the events $\{F_\alpha, \alpha \in A\}$, where each F_α may be E_α or E_α^c; also if $\{A_\beta, \beta \in B\}$, where B is an arbitrary index set, is a collection of disjoint countable subsets of A, then the events

$$\bigcup_{\alpha \in A_\beta} E_\alpha, \quad \beta \in B,$$

are independent.

Proof. First, we observe a simple equation:

$$\mathscr{P}(E_{\alpha_1} \cap E_{\alpha_2}^c) = \mathscr{P}(E_{\alpha_1}) - \mathscr{P}(E_{\alpha_1}E_{\alpha_2}) = \mathscr{P}(E_{\alpha_1}) - \mathscr{P}(E_{\alpha_1})\mathscr{P}(E_{\alpha_2}) = \mathscr{P}(E_{\alpha_1})\mathscr{P}(E_{\alpha_2}^c).$$

Then by some simple algebra, we know this equation holds for any number of events, no matter they have the form E or E^c, and it is just the collection of events $\{F_\alpha, \alpha \in A\}$. From this conclusion we know when replacing E_α with E_α^c for all E_α does not harm the assumption of independence. Then for disjoint countable subsets of A, it is clear that

$$\bigcap_{i=1}^{n} \left(\bigcup_{\alpha \in A_{\beta_i}} E_\alpha \right)^c = \bigcap_{\alpha \in \bigcup_{i=1}^{n} A_{\beta_i}} E_\alpha^c$$

as A_β are disjoint. And the independence should use Exercise 3.3.7. □

Exercise 3.3.4. Fields or B.F.'s $\mathscr{F}_\alpha (\subset \mathscr{F})$ of any family are said to be independent iff any collection of events, one from each $\mathscr{F}_\alpha$, forms a set of independent events. Let $\mathscr{F}_\alpha^0$ be a field generating $\mathscr{F}_\alpha$. Prove that if the fields $\mathscr{F}_\alpha^0$ are independent, then so are the B.F.'s $\mathscr{F}_\alpha$. Is the same conclusion true if the fields are replaced by arbitrary generating sets? Prove, however, that the conditions (1) and (2) are equivalent. [HINT: Use Theorem 2.1.2.]

Proof. Without loss of generality, we just consider the case when the index set is finite, call them $\mathscr{F}_1^0, \mathscr{F}_2^0, \cdots, \mathscr{F}_n^0$. Let

$$\mathscr{L} = \{E \in \mathscr{F}_1 : \mathscr{P}(E \cap E_2 \cap \cdots \cap E_n) = \mathscr{P}(E)\mathscr{P}(E_2)\cdots\mathscr{P}(E_n), E_j \in \mathscr{F}_j^0, j = 2, \cdots, n\},$$

where $\mathscr{F}_1$ is generated by $\mathscr{F}_1^0$ and $scrF_i^0 (i = 1, 2, \cdots, n)$ are independent. We can verify like Exercise 3.3.3 that $E^c \in \mathscr{L}$, and use the continuity of probability, we know $\bigcup_k E_k \in \mathscr{L}, E_k \in \mathscr{F}_1^0, E_k \subset (\supset) E_{k+1}$. Hence $\mathscr{L}$ is an M.C. Then by Theorem 2.1.2, we know $\mathscr{L} = \mathscr{F}_1$. This means $\mathscr{F}_1, \mathscr{F}_2^0, \cdots, \mathscr{F}_n^0$ are independent. Repeat this operation for $n-1$ times and we know $\mathscr{F}_1, \mathscr{F}_2, \cdots, \mathscr{F}_n$ are independent. For conditions (1) and (2), (1) $\Rightarrow$ (1) is trivial, and as $\{X^{-1}((-\infty, x]), x \in \mathscr{R}^1\}$ is a field generating $\{X^{-1}(B), B \in \mathscr{B}^1\}$, which is a B.F., use the conclusion we just proved, the conditions (1) and (2) are indeed equivalent. However, if the fields are replaced by arbitrary generating sets, the conclusion may be false. We still use the example in Exercise 3.3.2, and let $\mathscr{G}_1^0 = \{\{1,2\}\}, \mathscr{G}_2^0 = \{\{1,3\},\{1,4\}\}$, then it is easy to verify $\mathscr{G}_1^0$ and $\mathscr{G}_2^0$ are independent, but the B.F.'s generated by them are not independent, as

$$\mathscr{P}(\{3,4\} \cap \{2,3,4\}) = \frac{1}{2} \neq \mathscr{P}(\{3,4\})\mathscr{P}(\{2,3,4\}) = \frac{3}{8}. \qquad \square$$

Exercise 3.3.5. If $\{X_\alpha\}$ is a family of independent r.v.'s, then the B.F.'s generated by disjoint subfamilies are independent. [Theorem 3.3.2 is a corollary to this proposition.]

Proof. We shall consider the finite subfamilies because of the definition of independence. No matter each subfamily has countable or uncountable r.v.'s, we always have

$$\mathscr{P}\left(\bigcap_{j=1}^n \{\boldsymbol{X}_j \in B_j\}\right) = \prod_{j=1}^n \mathscr{P}\{\boldsymbol{X}_j \in B_j\}, \ \ \forall B_j \in \mathscr{B}_1,$$

where $\boldsymbol{X}_j$ is the vector containing all r.v.'s in the j th subfamily. We shall notice that $\mathscr{F}_j = \{\boldsymbol{X}_j^{-1}(B), B \in \mathscr{B}^1\}$, so we can arbitrarily choose event from each $\mathscr{F}_j$ and use the independence of $\boldsymbol{X}_j$'s. $\square$

Exercise 3.3.6. The r.v. X is independent of itself if and only if it is constant with probability one. Can X and $f(X)$ be independent where $f \in \mathscr{B}^1$?

Proof. Let $B_1 = B_2 = B \in \mathscr{B}^1$, then

$$\mathscr{P}(X \in B, X \in B) = \mathscr{P}(X \in B) = \mathscr{P}^2(X \in B), \ \ \forall B \in \mathscr{B}^1,$$

which implies $\mathscr{P}(X \in B) = 0$ or $\mathscr{P}(X \in B) = 1$. Now let $B = (-\infty, x], x \in \mathscr{R}^1$, then the d.f.

$$F(x) = \mathscr{P}(X \in B) = \mathscr{P}\{X \leq x\}$$

is increasing, and only takes values 0 or 1. Hence there exists point c such that $\mathscr{P}(X = c) = F(c) - F(c-) = 1$, which implies $X = c$ a.e. The converse is trivial. X and $f(X)$ can be independent when $X = c$ a.e., as X can be constant. $\square$

Exercise 3.3.7. If $\{E_j, 1 \leq j < \infty\}$ are independent events then

$$\mathscr{P}\left(\bigcap_{j=1}^\infty E_j\right) = \prod_{j=1}^\infty \mathscr{P}(E_j),$$

where the infinite product is defined to be the obvious limit; similarly

$$\mathscr{P}\left(\bigcup_{j=1}^\infty E_j\right) = 1 - \prod_{j=1}^\infty (1 - \mathscr{P}(E_j)).$$

Proof. We shall use the continuity of probability. For $j = 1, 2, \cdots, n$, we have

$$\mathscr{P}\left(\bigcap_{j=1}^n E_j\right) = \prod_{j=1}^n \mathscr{P}(E_j),$$

Let $F_n = \bigcap_{j=1}^n E_j$, then $F_n \supset F_{n+1}$. Let $F = \lim_{n\to\infty} F_n$, then $F_n \downarrow F$. Hence

$$\mathscr{P}\left(\bigcap_{j=1}^\infty E_j\right) = \mathscr{P}(F) = \mathscr{P}\left(\lim_{n\to\infty} F_n\right) = \lim_{n\to\infty} \mathscr{P}(F_n) = \prod_{j=1}^\infty \mathscr{P}(E_j).$$

Then with De Morgan's law, we get the second equation. $\square$

Exercise 3.3.8. Let $\{X_j, 1 \le j \le n\}$ be independent with d.f.'s $\{F_j, 1 \le j \le n\}$. Find the d.f. of $\max_j X_j$ and $\min_j X_j$.

Proof. We have

$$\mathscr{P}\left\{\max_j X_j \le x\right\} = \mathscr{P}\left(\bigcap_{j=1}^n \{X_j \le x\}\right) = \prod_{j=1}^n \mathscr{P}\{X_j \le x\} = \prod_{j=1}^n F_j(x),$$

$$\mathscr{P}\left\{\min_j X_j \le x\right\} = 1 - \mathscr{P}\left(\min_j X_j \le x\right) = 1 - \mathscr{P}\left(\bigcap_{j=1}^n \{X_j > x\}\right) = 1 - \prod_{j=1}^n [1 - F_j(x)]. \quad \square$$

Exercise 3.3.9. If X and Y are independent and $\mathscr{E}(X)$ exists, then for any Borel set B, we have

$$\int_{\{Y \in B\}} X \mathrm{d}\mathscr{P} = \mathscr{E}(X)\mathscr{P}(Y \in B).$$

Proof. We can write the integral to $\int_\Omega X \cdot 1_{\{Y\in B\}} \mathrm{d}\mathscr{P}$. As X and Y are independent, X and $1_{\{Y\in B\}}$ are independent by Theorem 3.3.1. Thus

$$\int_{\{Y \in B\}} X \mathrm{d}\mathscr{P} = \mathscr{E}(X \cdot 1_{\{Y\in B\}}) = \mathscr{E}(X)\mathscr{E}(1_{\{Y\in B\}}) = \mathscr{E}(X)\mathscr{P}(Y \in B). \quad \square$$

Exercise 3.3.10. If X and Y are independent and for some $p > 0$: $\mathscr{E}(|X + Y|^p) < \infty$, then $\mathscr{E}(|X|^p) < \infty$ and $\mathscr{E}(|Y|^p) < \infty$.

Proof. We shall consider the integral representation, which is

$$\mathscr{E}(|X + Y|^p) = \iint_{\mathscr{R}^2} |x + y|^p \mu^2(\mathrm{d}x, \mathrm{d}y) < \infty.$$

As $f(x, y) = |x + y|^p$ is measurable and integrable with respect to μ^2, and X and Y are independent,then by Fubini's theorem, the integral

$$\int_{\Omega_1} f(x, y) \mathrm{d}F_1(x) = \mathscr{E}(|X + y|^p) < \infty$$

as $f(\cdot, y)$ is integrable with respect to μ_1 a.e., where $F_1(x)$ is the d.f. of X and μ_1 is the p.m. of X. Hence we proved that $\forall y \in \mathscr{R}^1, \mathscr{E}(|X + y|^p) < \infty$. Then by Exercise 3.2.10, we know $\mathscr{E}(|X|^p) < \infty$. Similarly we get $\mathscr{E}(|Y|^p) < \infty$. $\square$

Exercise 3.3.11. If X and Y are independent, $\mathscr{E}(|X|^p) < \infty$ for some $p \ge 1$, and $\mathscr{E}(Y) = 0$, then $\mathscr{E}(|X + Y|^p) \ge \mathscr{E}(|X|^p)$. [This is a case of Theorem 9.3.2; but try a direct proof!]

Proof. When $p \ge 1$, $\varphi(x) = |x|^p$ is convex. Then by Fubini's theorem and Jensen's inequality, along with the independence of X and Y, we have

$$\begin{aligned}
\mathscr{E}(|X + Y|^p) &= \iint_{\mathscr{R}^2} |x + y|^p \mu^2(\mathrm{d}x, \mathrm{d}y) = \iint_{\mathscr{R}^2} |x + y|^p \mathrm{d}F_1(x)\mathrm{d}F_2(y) \\
&= \int_{\mathscr{R}^1} \left(\int_{\mathscr{R}^1} |x + y|^p \mathrm{d}F_2(y)\right) \mathrm{d}F_1(x) \\
&\ge \int_{\mathscr{R}^1} \left|\int_{\mathscr{R}^1} (x + y) \mathrm{d}F_2(y)\right|^p \mathrm{d}F_1(x) \\
&= \int_{\mathscr{R}^1} |x|^p \mathrm{d}F_1(x) = \mathscr{E}(|X|^p).
\end{aligned} \quad \square$$

Exercise 3.3.12. The r.v.'s $\{\varepsilon_j\}$ in Example 4 are related to the "Rademacher functions":

$$r_j(x) = \operatorname{sgn}(\sin 2^j \pi x).$$

What is the precise relation?

Proof. When expanding x in binary form, we have

$$r_j(x) = \operatorname{sgn}\left[\sin\left\{2^j\pi \cdot \left(\sum_{n=1}^{\infty} \frac{\varepsilon_n}{2^n}\right)\right\}\right] = \operatorname{sgn}\left[\sin\left\{\pi\varepsilon_j + \pi \sum_{n=j+1}^{\infty} \frac{\varepsilon_n}{2^n}\right\}\right].$$

It is clear that the rest part (summation from $n = j+1$) is less than $1/2$, by discussing two cases we get

$$r_j(x) = \begin{cases} 1, & \varepsilon_j = 0, \\ -1, & \varepsilon_j = 1. \end{cases} \quad j \geq 1.$$

Hence $r_j(x) = 1 - 2\varepsilon_j(x)$, note that we ignore the case in which $\exists k, \varepsilon_n = 0, \forall n \geq k$, and the point like this forms a countable set, and we do not care this condition. □

Exercise 3.3.13. Generalize Example 4 by considering any s-ary expansion where $s \geq 2$ is an integer:

$$x = \sum_{n=1}^{\infty} \frac{\varepsilon_n}{s^n}, \quad \text{where } \varepsilon_n = 0, 1, \cdots, s-1.$$

Proof. Now the ε_j is an r.v. taking values $0, 1, \cdots, s-1$. And we allow only the expansion with infinitely many digits "$s-1$". The set

$$\{x : \varepsilon_j(x) = c_j, 1 \leq j \leq n\} = \bigcup_{j=1}^{n} \{x : \varepsilon_j(x) = c_j\}$$

is the set of numbers x whose first n digits are the given c_j's, with the digits from the $(n+1)$st on completely arbitrary. And this set is just an interval of length $1/s^n$, hence of probability $1/s^n$. On the other hand for each j, the set $x : \varepsilon_j(x) = c_j$ has probability $1/s$. We have therefore

$$\mathscr{P}\{\varepsilon_j = c_j, 1 \leq j \leq n\} = \frac{1}{s^n} = \prod_{j=1}^{n} \left(\frac{1}{s}\right) = \prod_{j=1}^{n} \mathscr{P}\{\varepsilon_j = c_j\}.$$

This being true for every choice of the c_j's, the r.v.'s $\{\varepsilon_j, j \geq 1\}$ are independent. Let $\{f_j, j \geq 1\}$ be arbitrary functions with domain $\{0, 1, \cdots, s-1\}$, then $\{f_j(\varepsilon_j), j \geq 1\}$ are also independent r.v.'s. □

Exercise 3.3.14. Modify Example 4 so that to each x in [0,1] there corresponds a sequence of independent and identically distributed r.v.'s $\{\varepsilon_n, n \geq 1\}$, each taking the values 1 and 0 with probabilities p and $1-p$, respectively, where $0 < p < 1$. Such a sequence will be referred to as a "coin-tossing game (with probability p for heads)"; when $p = \frac{1}{2}$ it is said to be "fair".

Proof. Now we shall use a partition of $(0, 1]$ with interval lengths p and $1-p$. If $x \in (0, 1-p]$, then let $\varepsilon_1 = 0$, otherwise $\varepsilon_1 = 1$. Then do the same partition for each subinterval. For example, for right side, if $x \in (1-p, 1-p+p\cdot(1-p)]$, then $\varepsilon_2 = 0$, otherwise $\varepsilon_2 = 1$. And apart from the end of each interval (a countable set), any $x \in (0,1]$ can be uniquely determined by $\{\varepsilon_n\}$. Let the interval determined by first n ε_j as $I(\varepsilon_1, \varepsilon_2, \cdots, \varepsilon_n)$, and we know the length of it is

$$\mu(I(\varepsilon_1, \varepsilon_2, \cdots, \varepsilon_n)) = \prod_{j=1}^{n} p^{\varepsilon_j}(1-p)^{1-\varepsilon_j}.$$

And it is easy to verify ε_j is an r.v. taking values 0 and 1 with probabilities $1-p$ and p, and they are independent, which can be verified using the interval length deduced above. □

Exercise 3.3.15. Modify Example 4 further so as to allow ε_n to take the values 1 and 0 with probabilities p_n and $1-p_n$, where $0 < p_n < 1$, but p_n depends on n.

Proof. In this case we just need to modify the partition at each step in Exercise 3.3.14, and now the interval length is

$$\mu(I(\varepsilon_1, \varepsilon_2, \cdots, \varepsilon_n)) = \prod_{j=1}^{n} p_n^{\varepsilon_j}(1-p_n)^{1-\varepsilon_j},$$

and the rest of proof is quite similar. □

Exercise 3.3.16. Generalize (14) to

$$\begin{aligned}\mathscr{P}(C) &= \int_{C(1,\cdots,k)} \mathscr{P}((C|\omega_1, \cdots, \omega_k))\mathscr{P}(\mathrm{d}\omega_1)\cdots\mathscr{P}(\mathrm{d}\omega_k) \\ &= \int_{\Omega} \mathscr{P}((C|\omega_1, \cdots, \omega_k))\mathscr{P}(\mathrm{d}\omega),\end{aligned}$$

where $C(1, \cdots, k)$ is the set of $\omega_1, \cdots, \omega_k$ which appears as the first k coordinates of at least one point in C and in the second equation $\omega_1, \cdots, \omega_k$ are regarded as functions of ω.

Proof. We only need to show for the two-dimensional case and use deduction for the first equation. As $(C|\omega_1, \omega_2) = ((C|\omega_1)|\omega_2)$, we have

$$\mathscr{P}((C|\omega_1)) = \int_{\Omega_2} \mathscr{P}((C|\omega_1, \omega_2))\mathscr{P}_2(\mathrm{d}\omega_2).$$

The reason we can restrict $\Omega_1 \times \Omega_2$ to $C(1,2)$ is that $(C|\omega_1, \omega_2) \neq \varnothing$ if and only if $(C|\omega_1, \omega_2) \in C(1,2)$. The second equation is because $\omega_j(\omega)$ takes the jth coordinate out of ω and $(C|\omega_1, \cdots, \omega_k) \neq \varnothing$ if and only if the first k coordinates of ω are in $(C|\omega_1, \cdots, \omega_k)$. □

Exercise 3.3.17. For arbitrary events $\{E_j, 1 \le j \le n\}$, we have

$$\mathscr{P}\left(\bigcup_{j=1}^{n} E_j\right) \ge \sum_{j=1}^{n} \mathscr{P}(E_j) - \sum_{1\le j<k\le n} \mathscr{P}(E_j E_k).$$

If $\forall n : \{E_j^{(n)}, 1 \le j \le n\}$ are independent events, and

$$\mathscr{P}\left(\bigcup_{j=1}^{n} E_j^{(n)}\right) \to 0 \text{ as } n \to \infty,$$

then

$$\mathscr{P}\left(\bigcup_{j=1}^{n} E_j^{(n)}\right) \sim \sum_{j=1}^{n} \mathscr{P}(E_j^{(n)}).$$

Proof. We shall use the indicators. We have

$$1_{\bigcup_{j=1}^n E_j} = 1 - \prod_{j=1}^{n}(1 - 1_{E_j}),$$

then by expanding the product, we get

$$1_{\bigcup_{j=1}^n E_j} = \sum_{j=1}^{n} 1_{E_j} - \sum_{1\le j<k\le n} 1_{E_j}1_{E_k} + \cdots + (-1)^{n-1}\prod_{j=1}^{n} 1_{E_j}.$$

and we should focus on the relation between

$$1_{\bigcup_{j=1}^n} \text{ and } \sum_{j=1}^{n} 1_{E_j} - \sum_{1\le j<k\le n} 1_{E_jE_k}.$$

When $\omega \notin \bigcup_{j=1}^n E_j$, both sides are 0. When ω is in k events, then the right side equals to $k - \binom{k}{2} = (3k - k^2)/2$. Let

$$f(k) = k^2 - 3k + 2, \quad k = 1, 2, \cdots, n,$$

then it is easy to observe $f(k) \ge 0$ for each k in the domain. Hence the relation is determined. Take expectation on both sides and we get the desired inequality. This is called *Bonferroni's inequality*. When $\{E_j^(n), 1 \le j \le n\}$ are independent events, it is obvious that

$$\mathscr{P}\left(\bigcup_{j=1}^{n} E_j^{(n)}\right) = 1 - \prod_{j=1}^{n}(1 - \mathscr{P}(E_j^{(n)})).$$

Use the inequality $1 - x \le \mathrm{e}^{-x}$, we get

$$1 - \mathscr{P}\left(\bigcup_{j=1}^{n} E_j^{(n)}\right) \le \exp\left[-\sum_{j=1}^{n}\mathscr{P}(E_j^{(n)})\right] \le 1,$$

and we know $\sum_{j=1}^n \mathscr{P}(E_j^{(n)}) \to 0$ from this inequality. now use the Bonferroni's inequality, we get

$$\mathscr{P}\left(\bigcup_{j=1}^{n} E_j^{(n)}\right) \ge \sum_{j=1}^{n}\mathscr{P}(E_j^{(n)}) - \sum_{1\le j<k\le n}\mathscr{P}(E_j^{(n)})\mathscr{P}(E_k^{(n)}) \ge \sum_{j=1}^{n}\mathscr{P}(E_j^{(n)}) - \frac{1}{2}\left[\sum_{j=1}^{n}\mathscr{P}(E_j^{(n)})\right]^2.$$

And as

$$\mathscr{P}\left(\bigcup_{j=1}^{n} E_j^{(n)}\right) \le \sum_{j=1}^{n}\mathscr{P}(E_j^{(n)}),$$

we get

$$1 - \frac{1}{2}\left[\sum_{j=1}^{n}\mathscr{P}(E_j^{(n)})\right] \le \frac{\mathscr{P}\left(\bigcup_{j=1}^n E_j^{(n)}\right)}{\sum_{j=1}^n \mathscr{P}(E_j^{(n)})} \le 1, \quad n \ge 1.$$

Hence we get the desired asymptotic relation. □

Exercise 3.3.18. Prove that $\overline{\mathscr{B}} \times \overline{\mathscr{B}} \neq \overline{\mathscr{B} \times \mathscr{B}}$, where $\overline{\mathscr{B}}$ is the completion of $\mathscr{B}$ with respect to the Lebesgue measure; similarly $\overline{\mathscr{B} \times \mathscr{B}}$.

Proof. Let V denote a Vitali set in $[0,1]$ then $V \times \{0\}$ is in $\overline{\mathscr{B} \times \mathscr{B}}$, as it is a set of measure zero in the 2-dimensional plain. However, it is not in $\overline{\mathscr{B}} \times \overline{\mathscr{B}}$. □

Exercise 3.3.19. If $f \in \mathscr{F}_1 \times \mathscr{F}_2$ and

$$\int_{\Omega_1 \times \Omega_2} |f| \mathrm{d}(\overline{\mathscr{P}_1 \times \mathscr{P}_2}) < \infty,$$

then

$$\int_{\Omega_1} \left[\int_{\Omega_2} f \mathrm{d}\overline{\mathscr{P}_2} \right] \mathrm{d}\overline{\mathscr{P}_1} = \int_{\Omega_2} \left[\int_{\Omega_1} f \mathrm{d}\overline{\mathscr{P}_1} \right] \mathrm{d}\overline{\mathscr{P}_2}$$

Proof. We should notice f is measurable with respect to the original probability space, so

$$\int_{\Omega_1 \times \Omega_2} |f| \mathrm{d}(\overline{\mathscr{P}_1 \times \mathscr{P}_2}) = \int_{\Omega_1 \times \Omega_2} |f| \mathrm{d}(\mathscr{P}_1 \times \mathscr{P}_2) < \infty$$

according to the definition of completion. Then by Fubini's theorem, we know

$$\int_{\Omega_1 \times \Omega_2} f \mathrm{d}(\mathscr{P}_1 \times \mathscr{P}_2) = \int_{\Omega_1} \left[\int_{\Omega_2} f \mathrm{d}\mathscr{P}_2 \right] \mathrm{d}\mathscr{P}_1,$$

and $f(\omega_1, \cdot)$ is measurable with respect to $\mathscr{F}_2$ and we have

$$g(x) = \int_{\Omega_2} f(\omega_1, \omega_2) \mathrm{d}\mathscr{P}_2 = \int_{\Omega_2} f(\omega_1, \omega_2) \mathrm{d}\overline{\mathscr{P}_2}, \ \forall \omega_1 \in \mathscr{F}_1.$$

Then as $g(\omega_1)$ is $\mathscr{F}_1$-measurable, we have

$$\int_{\Omega_1} g \mathrm{d}\mathscr{P}_1 = \int_{\Omega_1} g \mathrm{d}\overline{\mathscr{P}_1}.$$

The other side of the desired equation can be proved similarly. □

Exercise 3.3.20. A typical application of Fubini's theorem is as follows. If f is a Lebesgue measurable function of (x, y) such that $f(x, y) = 0$ for each $x \in \mathscr{R}^1$ and $y \notin \boldsymbol{N}_x$, where $m(\boldsymbol{N}_x) = 0$ for each x, then we have also $f(x, y) = 0$ for each $y \notin \boldsymbol{N}$ and $x \in \boldsymbol{N}'_y$, where $m(\boldsymbol{N}) = 0$ and $m(\boldsymbol{N}'_y) = 0$ for each $y \notin \boldsymbol{N}$.

Proof. From the first condition we have

$$\int_{\mathscr{R}^1} |f(x, y)| \mathrm{d}y = 0, \ \ \forall x \in \mathscr{R}^1,$$

hence we can use Fubini's theorem, and

$$\int_{\mathscr{R}^2} |f(x, y)| \mathrm{d}x \mathrm{d}y = \int_{\mathscr{R}^1} \left[\int_{\mathscr{R}^1} |f(x, y)| \mathrm{d}y \right] \mathrm{d}x = 0.$$

This shows f is integrable with respect to the 2-dimensional Lebesgue measure. Then we can use Fubini's theorem again to change the order of repeated integrals, we have

$$\int_{\mathscr{R}^1} g(y) \mathrm{d}y = \int_{\mathscr{R}^1} \left[\int_{\mathscr{R}^1} |f(x, y)| \mathrm{d}x \right] \mathrm{d}y = \int_{\mathscr{R}^1} \left[\int_{\mathscr{R}^1} |f(x, y)| \mathrm{d}y \right] \mathrm{d}x = 0.$$

This implies $g(y) = 0$ a.e. So $\exists \boldsymbol{N} \subset \mathscr{R}^1, m(\boldsymbol{N}) = 0$ such that $g(y) = 0$. As g is also a form of integral, for each $y \notin \boldsymbol{N}$, $\exists \boldsymbol{N}_x \subset \mathscr{R}^1, m(\boldsymbol{N}_x) = 0$ such that $f(x, y) = 0$. □

4 Convergence concepts

4.1 Various modes of convergence

Exercise 4.1.1. $X_n \to +\infty$ a.e. if and only if $\forall M > 0 : \mathscr{P}\{X_n < M \text{ i.o.}\} = 0$.

Proof. Let $\Omega_0 = \{\omega : \lim_{n\to\infty} X_n = +\infty\}\backslash\mathbf{N}$ and we have $\mathscr{P}(\Omega_0) = 1$ for $X_n \to \infty$ a.e. Define

$$A_m(M) = \bigcup_{n=m}^{\infty}\{X_n < M\},$$

then $A_m(M)$ is decreasing with m. For each $\omega_0 \in \Omega_0, \forall M > 0, \exists m(\omega_0, M)$ such that

$$n \geq m(\omega_0, M) \Rightarrow |X_n(\omega_0)| > M.$$

This means $\omega_0 \notin A_m(M)$, then $\omega_0 \notin \bigcap_{m=1}^{\infty} A_m(M)$. Thus we have

$$\bigcap_{m=1}^{\infty} A_m(M) \subset \Omega_0^c \Rightarrow \lim_{m\to\infty} \mathscr{P}(A_m(M)) = \mathscr{P}\{X_n < M \text{ i.o.}\} = 0.$$

Conversely, let $A(M) = \bigcap_{m=1}^{\infty} A_m(M)$, then the set

$$A = \bigcup_{n=1}^{\infty} A(n)$$

represents the event $X_n \nrightarrow +\infty$, because for these points, $\exists n$ such that $X_n(\omega_0) < n$ i.o. And A still has probability zero by monotone convergence property. This implies $X_n \to \infty$ a.e. $\square$

Exercise 4.1.2. If $0 \leq X_n, X_n \leq X \in L^1$ and $X_n \to X$ in pr., then $X_n \to X$ in L^1.

Proof. From conditions given we know $-X \leq X_n - X \leq 0$, hence $X_n - X$ is dominated by X. As $X_n \to X$ in pr. is equivalent to $X_n - X \to 0$ in pr., we know $X_n \to X$ in L^1. $\square$

Exercise 4.1.3. If $X_n \to X, Y_n \to Y$ both in pr., then $X_n \pm Y_n \to X \pm Y, X_nY_n \to XY$, all in pr.

Proof. We shall notice the inequality

$$|X_n + Y_n - (X + Y)| \leq |X_n - X| + |Y_n - Y|,$$

and from this we know

$$\left\{|X_n - X| \leq \frac{\varepsilon}{2}\right\} \cap \left\{|Y_n - Y| \leq \frac{\varepsilon}{2}\right\} \subset \{|X_n + Y_n - (X + Y)| \leq \varepsilon\}.$$

Hence

$$\lim_{n\to\infty} \mathscr{P}\{|X_n + Y_n - (X + Y)| > \varepsilon\} \leq \lim_{n\to\infty} \left[\mathscr{P}\left\{|X_n - X| > \frac{\varepsilon}{2}\right\} + \mathscr{P}\left\{|Y_n - Y| > \frac{\varepsilon}{2}\right\}\right] = 0.$$

So $X_n + Y_n \to X + Y$ in pr. The case "-" is analogus. Now we consider the product case. As

$$|X_nY_n - XY| \leq |X_n - X||Y_n| + |Y_n - Y||X|,$$

We shall notice the definition of convergence is based on "finite", for this reason, we can assume $|X| \leq M, |Y_n| \leq M$ to avoid complications. Now we have

$$\left\{|X_n - X| \leq \frac{\varepsilon}{2M}, |Y_n - Y| \leq \frac{\varepsilon}{2M}\right\} \subset \left\{|X_n - X||Y_n| \leq \frac{\varepsilon}{2}, |Y_n - Y||X| \leq \frac{\varepsilon}{2}\right\},$$

and the rest of the proof is similar to the sum case. $\square$

Exercise 4.1.4. Let f be a bounded uniformly continuous function on $\mathscr{R}^1$. Then $X_n \to 0$ in pr. implies $\mathscr{E}\{f(X_n)\} \to f(0)$. [Example:

$$f(x) = \frac{|X|}{1+|X|}$$

as in Theorem 4.1.5.]

Proof. The result we want to prove is equivalent to $\mathscr{E}(|f(X_n) - f(0)|) = 0$. Now assume $|f(x)| \leq M$,

$$\begin{aligned}\mathscr{E}(|f(X_n) - f(0)|) &= \int_{\{|X_n|>\delta\}} |f(X_n) - f(0)|\mathrm{d}\mathscr{P} + \int_{\{|X_n|\leq\delta\}} |f(X_n) - f(0)|\mathrm{d}\mathscr{P} \\ &\leq 2M\mathscr{P}\{|X_n| > \delta\} + \int_{\{|X_n|\leq\delta\}} |f(X_n) - f(0)|\mathrm{d}\mathscr{P}.\end{aligned}$$

As f is uniformly continuous, $\forall \varepsilon > 0, \exists\delta(\varepsilon) > 0, |X_n| \leq \delta \Rightarrow |f(X_n) - f(0)| < \varepsilon$. Hence we first choose an ε, then we have $\int_{\{|X_n|\leq\delta(\varepsilon)\}} |f(X_n) - f(0)| \leq \varepsilon$. Now let $n \to \infty$, we have $\mathscr{E}(|f(X_n) - f(0)|) < \varepsilon$. Then we get the result. □

Exercise 4.1.5. Convergence in L^p implies that in L^r for $r < p$.

Proof. Use Liapounov's inequality,

$$\lim_{n\to\infty} \mathscr{E}(|X_n - X|^r) \leq \lim_{n\to\infty} \mathscr{E}(|X_n - X|^p)^{r/p} = 0.$$

$|X|$ and $|X_n|$ are in L^r, which is also guaranteed by Liapounov's inequality, so $X_n \to X$ in L^r. □

Exercise 4.1.6. If $X_n \to X, Y_n \to Y$, both in L^p, then $X_n \pm Y_n \to X \pm Y$ in L^p. If $X_n \to X$ in L^p and $Y_n \to Y$ in L^q, where $p > 1$ and $1/p + 1/q = 1$, then $X_nY_n \to XY$ in L^1.

Proof. The first is trivial by Minkowski's inequality. For the second, first X_nY_n and XY are in L^1 by Hölder's inequality. Then

$$\begin{aligned}\mathscr{E}(|X_nY_n - XY|) &\leq \mathscr{E}(|X_n||Y_n - Y|) + \mathscr{E}(|Y||X_n - X|) \\ &\leq \mathscr{E}(|X_n|^p)^{1/p}\mathscr{E}(|Y_n - Y|^q)^{1/q} + \mathscr{E}(|Y|^q)^{1/q}\mathscr{E}(|X_n - X|^p)^{1/p} \to 0.\end{aligned}$$

Hence $X_nY_n \to XY$ in L^1. □

Exercise 4.1.7. If $X_n \to X$ in pr. and $X_n \to Y$ in pr., then $X = Y$ a.e.

Proof. We have the relation

$$E_m = \left\{|X - Y| > \frac{1}{m}\right\} \subset \left\{|X_n - X| > \frac{1}{2m}\right\} \cup \left\{|X_n - Y| > \frac{1}{2m}\right\}, \quad \forall m, n \geq 1.$$

So

$$\mathscr{P}(E_m) \leq \lim_{n\to\infty} \mathscr{P}\{|X_n - X| > \frac{1}{2m}\} + \mathscr{P}\{|X_n - Y| > \frac{1}{2m}\} = 0.$$

Then by monotone convergence property we have

$$\mathscr{P}\{|X - Y| \neq 0\} = \mathscr{P}\left(\bigcup_{m=1}^{\infty} E_m\right) = \lim_{m\to\infty} \mathscr{P}(E_m) = 0,$$

which is to say $X = Y$ a.e. □

Exercise 4.1.8. If $X_n \to X$ a.e. and μ_n and μ are p.m.'s of X_n and X, it does not follow that $\mu_n(P) \to \mu(P)$ even for all intervals P.

Proof. Let $X_n = \frac{1}{n}$, then $X_n \to 0$ everywhere. But for interval $P = (-\infty, 0]$,

$$\mu_n((-\infty, 0]) = 0, \mu((-\infty, 0]) = 1.$$

This can be explained by $0 \notin \left\{\frac{1}{n}, n = 1, 2, \cdots\right\}$. □

Exercise 4.1.9. Give an example in which $\mathscr{E}(X_n) \to 0$ but there does not exist any subsequence $\{n_k\} \to \infty$ such that $X_{n_k} \to 0$ in pr.

Proof. We use the coin tossing model, suppose X_n take the value 1 and -1 with probability $1/2$, then $\mathscr{E}(X_n) = 0$. However, $\mathscr{P}\{|X_n| > 0.5\} = 1 \neq 0, \forall n \geq 1$, so there does not exist any subsequence $\{n_k\} \to \infty$ such that $X_{n_k} \to 0$ in pr. □

Exercise 4.1.10. Let f be a continuous function on $\mathscr{R}^1$. If $X_n \to X$ in pr., then $f(X_n) \to f(X)$ in pr. The result is false if f is merely Borel measurable. [HINT: Truncate f at $\pm A$ for large A.]

Proof. We should truncate f at $\pm A$. As f is continuous, f is uniformly continuous in $[-A, A]$. In this interval, $\forall \varepsilon > 0, \exists \delta(\varepsilon) > 0, |X_n - X| \leq \delta \Rightarrow |f(X_n) - f(X)| \leq \varepsilon$. Hence we have the relation

$$\{|X| > A\} \cup \{|X_n - X| > \delta(\varepsilon)\} \supset \{|f(X_n) - f(X)| > \varepsilon\}.$$

Hence

$$\mathscr{P}\{|f(X_n) - f(X)| > \varepsilon\} \leq \mathscr{P}\{|X_n - X| > \delta(\varepsilon)\} + \mathscr{P}\{|X| > A\}, \forall A > 0.$$

Let $A \to \infty, n \to \infty$, then we have $\mathscr{P}\{|f(X_n) - f(X)| > \varepsilon\} = 0$, so $f(X_n) \to f(X)$ in pr. Let $f = 1_{\{0\}}$ is a Borel measurable function and $X_n = 1/n, X = 0$, then $X_n \to X$ in pr. However, $f(X_n) = 0, \forall n \geq 1$ and $f(X) = 1$, so $f(X_n) \nrightarrow f(X)$ in pr. □

Exercise 4.1.11. The extended-valued r.v. X is said to be *bounded in pr.* iff for each $\varepsilon > 0$, there exists a finite $M(\varepsilon)$ such that $\mathscr{P}\{|X| \leq M(\varepsilon)\} \geq 1 - \varepsilon$. Prove that X is bounded in pr. if and only if it is finite a.e.

Proof. If X is bounded in pr., let $\Omega_0 = \{\omega : \exists M, |X(\omega)| \leq M\}$, which consists of the finite point of X. for $\varepsilon_n = \frac{1}{n}$,

$$\exists M\left(\frac{1}{n}\right), \mathscr{P}\left\{A\left(\frac{1}{n}\right)\right\} = \mathscr{P}\left\{|X| \leq M\left(\frac{1}{n}\right)\right\} \geq 1 - \frac{1}{n}.$$

Hence $\forall \omega_0 \in A\left(\frac{1}{n}\right), x \in \Omega_0$, and we can deduce that

$$\bigcup_{n=1}^{\infty} A\left(\frac{1}{n}\right) \subset \Omega_0 \Rightarrow 1 \geq \mathscr{P}(\Omega_0) \geq \mathscr{P}\left(\bigcup_{n=1}^{\infty} A\left(\frac{1}{n}\right)\right) = \lim_{n\to\infty} \mathscr{P}\left(A\left(\frac{1}{n}\right)\right) = 1$$

by monotone convergence property. So X is finite a.e. Conversely, consider $E_n = \{|X| \leq n\}$, then $\bigcup_{n=1}^{\infty} E_n = \Omega_0$. As is proved, $\mathscr{P}(\Omega_0) = 1$, then use monotone convergence property, $\forall \varepsilon > 0, \exists N = M(\varepsilon) > 0$,

$$\mathscr{P}\left(\bigcup_{k=1}^{N} E_k\right) = \mathscr{P}\{|X| \leq M(\varepsilon)\} > 1 - \varepsilon.$$ □

Exercise 4.1.12. The sequence of extended-valued r.v. $\{X_n\}$ is said to be bounded in pr. iff $\sup_n |X_n|$ is bounded in pr.; $\{X_n\}$ is said to *diverge to* $+\infty$ *in pr.* iff for each $M > 0$ and $\varepsilon > 0$ there exists a finite $n_0(M, \varepsilon)$ such that if $n > n_0$, then $\mathscr{P}\{|X_n| > M\} > 1 - \varepsilon$. Prove that if $\{X_n\}$ diverges to $+\infty$ in pr. and $\{Y_n\}$ is bounded in pr., then $\{X_n + Y_n\}$ diverges to $+\infty$ in pr.

Proof. We shall notice the relation

$$\{|X_n| \le M + K\} \supset \{|X_n + Y_n| \le M, |Y_n| \le K\} \text{ for } M, K > 0.$$

Then $\forall \varepsilon > 0, \exists K(\varepsilon)$ such that $\mathscr{P}\{|Y_n| > K\} < \varepsilon/2$. And we can choose $M(\varepsilon) > K, \exists n_0(M, \varepsilon)$ such that $\mathscr{P}\{|X_n| > M + K\} > 1 - \varepsilon/2, \forall n > n_0$. Then we have

$$\mathscr{P}\{|X_n + Y_n| \le M\} \le \mathscr{P}\{|X_n| \le M + K\} + \mathscr{P}\{|Y_n| > K\} < \varepsilon.$$

This step uses the relation $A \cap C \subset B \Leftrightarrow A \subset B \cup C^c$. So $X_n + Y_n$ diverges to $+\infty$ in pr. $\square$

Exercise 4.1.13. If $\sup_n X_n = +\infty$ a.e., there need exist no subsequence $\{X_{n_k}\}$ that diverges to $+\infty$ in pr.

Proof. Replace the φ_{kj} by $k\varphi_{kj}$ in Example 1, and we get a sequence of X_n with $\sup_n X_n = +\infty$ a.e. and $X_n \to 0$ in pr. $\square$

Exercise 4.1.14. It is possible that for each ω, $\overline{\lim}_n X_n(\omega) = +\infty$, but there does not exist a subsequence $\{n_k\}$ and a set of Δ of positive probability such that $\lim_k X_{n_k}(\omega) = +\infty$ on Δ. [HINT: On $(\mathscr{U}, \mathscr{B})$ define $X_n(\omega)$ according to the nth digits of ω.]

Proof. On $(\mathscr{U}, \mathscr{B})$, let

$$\omega = 0.d_1 d_2 d_3 \cdots = \sum_{n=1}^{\infty} \frac{d_n(\omega)}{10^n}$$

is the decimal expansion of ω, define

$$X_n(\omega) = \begin{cases} n, & \exists m, \overline{d_1 d_2 \cdots d_m} = n, \\ 0, & \text{otherwise}, \end{cases}$$

The form of expansion cannot be all 0 after some digit. Then for each $\omega \in (0, 1]$, we can always find sequence $\{X_{\overline{d_1 d_2 \cdots d_m}}\} \to +\infty$, and others contain 0, so $\overline{\lim}_n X_n(\omega) = +\infty$. But for each $n = \overline{d_1 d_2 \cdots d_m}$, the ω making $X_n \ne 0$ has a measure of 10^{-m}. when $m \to \infty$, the measure is 0, so on no set with positive probability $\lim_k X_{n_k}(\omega) = +\infty$. $\square$

Exercise 4.1.15. Instead of the ρ in Theorem 4.1.5 one may define other metrics as follows. Let $\rho_1(X, Y)$ be the infimum of all $\varepsilon > 0$ such that

$$\mathscr{P}(|X - Y| > \varepsilon) \le \varepsilon.$$

Let $\rho_2(X, Y)$ be the infimum of $\mathscr{P}\{|X - Y| > \varepsilon\} + \varepsilon$ over all $\varepsilon > 0$. Prove that these are metrics and that convergence in pr. is equivalent to convergence according to either metric.

Proof. First we prove ρ_1, ρ_2 are metrics. The nonnegativity is obvious. And $\rho_1 = 0$ implies $X = Y$ a.e., so does ρ_2, if we identify two r.v.'s that are equal a.e., then the property for metric do hold. Assume $\rho_1(X, Z) = \varepsilon_1, \rho_1(Y, Z) = \varepsilon_2$. First, as

$$\mathscr{P}(|X - Y| > \varepsilon + \delta) \le \mathscr{P}(|X - Y| > \varepsilon) \le \varepsilon + \delta,$$

so $\forall\delta > 0, \varepsilon + \delta \in \{\varepsilon : \mathscr{P}(|X - Y| > \varepsilon) \le \varepsilon\}$. Now $\forall\delta > 0$,

$$\mathscr{P}(|X - Y| \ge \varepsilon_1 + \varepsilon_2 + 2\delta) \le \mathscr{P}(|X - Z| \ge \varepsilon_1 + \delta) + \mathscr{P}(|Y - Z| \ge \varepsilon_2 + \delta) \le \varepsilon_1 + \varepsilon_2 + 2\delta,$$

Now let $\delta \to 0^+$, and use the property of infimum, we have

$$\rho_1(X, Y) \le \rho_1(X, Z) + \rho_1(Y, Z),$$

as $\varepsilon_1 + \varepsilon_2$ is in $\{\varepsilon : \mathscr{P}(|X - Y| > \varepsilon) \le \varepsilon\}$. For ρ_2, let $f_{XY}(\varepsilon) = \mathscr{P}(|X - Y| > \varepsilon) + \varepsilon$, then

$$f_{XY}(\varepsilon_1 + \varepsilon_2) \le f_{XZ}(\varepsilon_1) + f_{YZ}(\varepsilon_2)$$

similar to the proof of ρ_1. As $\rho(X, Y) = \inf_\varepsilon f_{XY}(\varepsilon)$, we have $\rho(X, Y) \le f_{XZ}(\varepsilon_1) + f_{YZ}(\varepsilon_2)$. Then take infimum on the right side we get the final inequality.

For ρ_1, if $X_n \to X$ in pr., then $\forall\delta > 0, \lim_{n\to\infty} \mathscr{P}\{|X_n - X| > \delta\} = 0$. Hence $\exists N > 0, \mathscr{P}\{|X_n - X| > \delta\} \le \delta, \forall n > N$, so $\rho_1(X_n, X) \le \delta(n > N(\delta)) \Rightarrow \rho_1(X_n, X) \to 0$. Conversely, $\forall\delta > 0, \exists N > 0, \rho_1(X_n, X) < \delta(n > N)$. Hence $\exists\varepsilon(N) > 0, \mathscr{P}(|X_n - X| > \delta) \le \mathscr{P}(|X_n - X| > \varepsilon) \le \varepsilon < \delta$. Let $n \to \infty, \delta \to 0^+$, we get $X_n \to X$ in pr. Actually $\forall\varepsilon > 0$, we can always find $\delta < \varepsilon$ when $\delta \to 0$, so it is reasonable to let $\delta \to 0^+$ in the last step.

For ρ_2, if $X_n \to X$ in pr., $\forall\varepsilon, \rho_2(X_n, X) \le \mathscr{P}\{|X_n - X| > \varepsilon\} + \varepsilon$. Let $n \to \infty$, we get $\limsup_n \rho_2(X_n, X) \le \varepsilon$, then let $\varepsilon \to 0$, we get $\lim_{n\to\infty} \rho_2(X_n, X) = 0$. Conversely, $\exists\varepsilon_n$ such that $\mathscr{P}(|X_n - X| > \varepsilon_n) + \varepsilon \to 0$. Due to the nonnegativity, we have $\mathscr{P}(|X_n - X| > \varepsilon_n) \to 0, \varepsilon_n \to 0$. $\forall\delta > 0$, choose N such that $\varepsilon_n < \delta(n > N)$, then

$$\mathscr{P}\{|X_n - X| > \delta\} \le \mathscr{P}\{|X_n - X| > \varepsilon_n\} \to 0. \ \ (n \to \infty)$$

This tells $X_n \to X$ in pr. □

Exercise 4.1.16. Convergence in pr. for arbitrary r.v's may be reduced to that of bounded r.v.'s by the transformation

$$X' = \arctan X.$$

In a space of uniformly bounded r.v.'s, convergence in pr. is equivalent to that in the metric $\rho_0(X, Y) = \mathscr{E}|X - Y|$; this reduces to the definition given in Exercise 8 of Sec. 3.2 when X and Y are indicators.

Proof. As $\arctan x$ is a continuous function on $\mathscr{R}^1$, and $\tan x$ is continuous on $(-\pi/2, \pi/2)$, according to Exercise 4.1.10, we know $X_n \to X$ in pr. $\Leftrightarrow X_n' \to X'$ in pr. In the space where r.v.'s are uniformly bounded, say $|X| \le M$, if $X_n \to X$ in pr., then $\mathscr{P}\{|X_n - X| \le 2M\} = 1$. Hence

$$\mathscr{E}(|X_n - X|) = \int_{\{2M \ge |X_n - X| > \varepsilon\}} |X_n - X| \mathrm{d}\mathscr{P} + \int_{\{|X_n - X| \le \varepsilon\}} |X_n - X| \mathrm{d}\mathscr{P} \le 2M\mathscr{P}\{|X_n - X| > \varepsilon\} + \varepsilon.$$

Let $n \to \infty, \varepsilon \downarrow 0$, we get $\lim_{n\to\infty} \rho_0(X_n, X) = 0$. Conversely, by Markov's inequality, we have

$$\mathscr{P}\{|X_n - X| > \varepsilon\} \le \frac{\rho(X_n - X)}{\varepsilon},$$

let $n \to \infty$ and this finishes the proof. In fact, this is the corollary of Theorem 4.1.4. When $X = 1_{\Lambda_1}, Y = 1_{\Lambda_2}$, then $\rho_0(X, Y) = \mathscr{P}(\Lambda_1 \triangle \Lambda_2)$. □

Exercise 4.1.17. Unlike convergence in pr., convergence a.e. is not expressible by means of metric. [HINT: Metric convergence has the property that if $\rho(x_n, x) \not\to 0$, then there exists $\varepsilon > 0$ and $\{n_k\}$ such that $\rho(x_{n_k}, x) \ge \varepsilon$ for every k.]

Proof. Suppose convergence a.e. can be expressed by means of metric, consider the case $X_n \nrightarrow X$ a.e. but $X_n \to X$ in pr. There exists a subsequence X_{n_k} and $\varepsilon > 0$ such that $\rho(X_{n_k}, X) \geq \varepsilon$. As $X_n \to X$ in pr., we also have $X_{n_k} \to X$ in pr. And $\exists \{X_{n_{k_j}}\} \subset \{X_{n_k}\}$ such that $\rho(X_{n_{k_j}}, X) \to 0$, which is a contradiction. So convergence a.e. is not expressible by means of metric. □

Exercise 4.1.18. If $X_n \downarrow X$ a.s., each X_n is integrable and $\inf_n \mathscr{E}(X_n) > -\infty$, then $X_n \to X$ in L^1.

Proof. As $X_n \geq X$ a.s., We shall prove $\lim_{n\to\infty} \mathscr{E}(X_n - X) = 0$. Consider $Y_n = X_1 - X_n$, then $Y_n \geq 0, Y_n \uparrow X_1 - X$ a.s., hence

$$\lim_{n\to\infty} \mathscr{E}(Y_n) = \mathscr{E}(X_1) - \lim_{n\to\infty} \mathscr{E}(X_n) = \mathscr{E}(X_1) - \mathscr{E}(X).$$

As $\inf_n \mathscr{E}(X_n) > -\infty$, we know $X \in L^1$ and $\lim_{n\to\infty} \mathscr{E}(X_n) = \mathscr{E}(X)$, i.e. $X_n \to X$ in L^1. □

Exercise 4.1.19. Let $f_n(x) = 1 + \cos 2\pi nx$, $f(x) = 1$ in $[0, 1]$. Then for each $g \in L^1[0, 1]$ we have

$$\int_0^1 f_n g \mathrm{d}x \to \int_0^1 f g \mathrm{d}x,$$

but f_n does not converge to f in measure. [HINT: This is just the Riemann-Lebesgue lemma in Fourier series, but we have made $f_n \geq 0$ to stress a point.]

Proof. The convergence in the form of integral is just Riemann-Lebesgue lemma, consider the set

$$E_n = \left\{x \in [0,1] : |\cos 2\pi nx| \geq \frac{1}{2}\right\} = \left[0, \frac{1}{6n}\right] \cup \bigcup_{k=1}^{2n-1} \left[\frac{1}{3n} + \frac{k-1}{2n}, \frac{2}{3n} + \frac{k-1}{2n}\right] \cup \left[1 - \frac{1}{6n}, 1\right],$$

Then $m(E_n) = 2n \cdot \frac{1}{3n} = \frac{2}{3}, \forall n \geq 1$, so $f_n \nrightarrow f$ in measure. This exercise is quite like weak convergence does not imply convergence in pr. □

Exercise 4.1.20. Let $\{X_n\}$ be a sequence of independent r.v.'s with zero mean and unit variance. Prove that for any bounded r.v. Y we have $\lim_{n\to\infty} \mathscr{E}(X_n Y) = 0$. [HINT: Consider $\mathscr{E}\{[Y - \sum_{k=1}^n \mathscr{E}(X_k Y) X_k]^2\}$ to get Bessel's inequality $\mathscr{E}(Y^2) \geq \sum_{k=1}^n \mathscr{E}(X_k Y)^2$. The stated result extends to case where the X_n's are assumed only to be uniformly integrable (see Sec. 4.5) but now we must approximate Y by a function of $(X_1, \cdots, X_m)$, cf. Theorem 8.1.1.]

Proof. We have the inequality

$$\begin{aligned}
\mathscr{E}\{[Y - \sum_{k=1}^n \mathscr{E}(X_k Y) X_k]^2\} &= \mathscr{E}(Y^2) - 2\sum_{k=1}^n [\mathscr{E}(X_k Y)]^2 + \mathscr{E}\left(\sum_{k=1}^n \mathscr{E}(X_k Y) X_k\right)^2 \\
&= \mathscr{E}(Y^2) - 2\sum_{k=1}^n [\mathscr{E}(X_k Y)]^2 + \sum_{k=1}^n [\mathscr{E}(X_k Y)]^2 \mathscr{E}(X_k^2) \\
&= \mathscr{E}(Y^2) - \sum_{k=1}^n [\mathscr{E}(X_k Y)]^2 \geq 0.
\end{aligned}$$

The second "=" uses the independence of X_k, with $\mathscr{E}(X_k) = 0, k = 1, 2, \cdots, n$. As Y is bounded, we have $\mathscr{E}(Y^2) < \infty$, which leads to $\lim_{n\to\infty} \mathscr{E}(X_k Y) \to 0$. □

4.2 Almost sure convergence; Borel-Cantelli lemma

Exercise 4.2.1. Prove that

$$\mathscr{P}(\limsup_n E_n) \geq \overline{\lim_n} \mathscr{P}(E_n),$$
$$\mathscr{P}(\liminf_n E_n) \leq \underline{\lim_n} \mathscr{P}(E_n).$$

Proof. For the first, as $\mathscr{P}(\limsup_n E_n) = \lim_{m\to\infty} \mathscr{P}\left(\bigcup_{n=m}^{\infty} E_n\right)$, as $E_k \subset \bigcup_{n=m}^{\infty} E_n, \forall k \geq m$, we have

$$\mathscr{P}\left(\bigcup_{n=m}^{\infty} E_n\right) \geq \sup_{n\geq m} \mathscr{P}(E_n),$$

then put limit on both sides we get the first equation. The second is quite similar. □

Exercise 4.2.2. Let $\{B_n\}$ be a countable collection of Borel sets in $\mathscr{U}$. If there exists a $\delta > 0$ such that $m(B_n) \geq \delta$ for every n, then there is at least one point in $\mathscr{U}$ that belongs to infinitely many B_n's.

Proof. As $\forall n, m(B_n) \geq \delta$, then $m\left(\bigcup_{m=n}^{\infty} B_n\right) \geq \delta, \forall m \geq 1$. Let $m \to \infty$, then

$$m(B_n \text{ i.o.}) \geq \delta,$$

so there is at least one point in $\mathscr{U}$ that belongs to infinitely many B_n's. □

Exercise 4.2.3. If $\{X_n\}$ converges to a finite limit a.e., then for any ε there exist $M(\varepsilon) < \infty$ such that $\mathscr{P}\{\sup |X_n| \leq M(\varepsilon)\} \geq 1 - \varepsilon$.

Proof. As $\{X_n\}$ converges converges a.e., we have

$$\mathscr{P}\{\sup |X_n| < \infty\} = \mathscr{P}\left(\bigcap_{n=1}^{\infty} \{|X_n| < \infty\}\right) = 1.$$

Take events $E_k = \{\sup |X_n| \leq k\}, k = 1, 2, \cdots$, then

$$\lim_{k\to\infty} \mathscr{P}(E_k) = \mathscr{P}\left(\bigcup_{k=1}^{\infty} E_k\right) = \mathscr{P}\{\sup |X_n| < \infty\}.$$

according to the continuity. $\forall \varepsilon > 0, \exists k(\varepsilon)$ such that $\mathscr{P}\{\sup |X_n| \leq k\} \geq 1-\varepsilon$, letting $M(\varepsilon) = k$ is sufficient. □

Exercise 4.2.4. For any sequence of r.v.'s $\{X_n\}$ there exists a sequence of constants $\{A_n\}$ such that $X_n/A_n \to 0$ a.e.

Proof. As $\mathscr{P}\{|X_n| < \infty\} = 1, \forall n \geq 1$, for any $\varepsilon > 0$, we can find M_n such that $\mathscr{P}\{|X_n| > M_n\} < 1/2^n$. Let $A_n = nM_n, \forall \varepsilon > 0, \exists N(\varepsilon)$ such that $1/N < \varepsilon$. Hence we have

$$\begin{aligned}
\sum_{n=1}^{\infty} \mathscr{P}\left\{\left|\frac{X_n}{A_n}\right| > \varepsilon\right\} &= \sum_{n=1}^{N} \mathscr{P}\left\{\left|\frac{X_n}{A_n}\right| > \varepsilon\right\} + \sum_{n=N+1}^{\infty} \mathscr{P}\left\{\left|\frac{X_n}{A_n}\right| > \varepsilon\right\} \\
&\leq N + \sum_{n=N+1}^{\infty} \mathscr{P}\left\{\left|\frac{X_n}{A_n}\right| > \frac{1}{n}\right\} \\
&< N + \sum_{n=N+1}^{\infty} \frac{1}{2^n} < \infty.
\end{aligned}$$

Hence by Theorem 4.2.1, we know $\mathscr{P}\{|X_n/A_n| > \varepsilon \text{ i.o.}\} = 0$, by Theorem 4.2.2, $X_n/A_n \to 0$ a.e. $\square$

Exercise 4.2.5. If $\{X_n\}$ converges on the set C, then for any $\varepsilon > 0$, there exists $C_0 \subset C$ with $\mathscr{P}(C\backslash C_0) < \varepsilon$ such that X_n converges uniformly in C_0. [This is Egorov's theorem. We may suppose $C = \Omega$ and the limit to be zero. Let $F_{mk} = \bigcap_{n=k}^{\infty}\{\omega : |X_n(\omega)| \le 1/m\}$; then $\forall m, \exists k(m)$ such that

$$\mathscr{P}(F_{m,k(m)}) > 1 - \varepsilon/2^m.$$

Take $C_0 = \bigcap_{m=1}^{\infty} F_{m,k(m)}$.]

Proof. Without loss of generality, we can assume $C = \Omega$ and $X_n \to 0$. Let

$$F_{mk} = \bigcap_{n=k}^{\infty}\left\{\omega : |X_n(\omega)| \le \frac{1}{m}\right\},$$

as $X_n \to 0, \forall\omega \in \Omega, \exists N$ such that $|X_n(\omega)| \le 1/m, \forall n > N$. Hence $\omega \in F_{mN} \Rightarrow \omega \in \bigcup_{k=1}^{\infty} F_{mk}$, which means

$$\mathscr{P}\left(\bigcup_{k=1}^{\infty} F_{mk}\right) = \lim_{k\to\infty}\mathscr{P}(F_{mk}) = 1.$$

Hence $\forall m, \exists k(m)$ such that

$$\mathscr{P}(F_{m,k(m)}) > 1 - \frac{\varepsilon}{2^m}.$$

Let $C_0 = \bigcap_{m=1}^{\infty} F_{m,k(m)}, \forall\varepsilon, \exists m$ such that $1/m < \varepsilon$. Then $\forall\omega \in C_0, |X_n(\omega)| \le 1/m < \varepsilon$, which means $X_n \rightrightarrows 0$ on C_0. Then

$$\mathscr{P}(C_0^c) = \mathscr{P}\left(\bigcup_{k=1}^{\infty} F_{m,k(m)}^c\right) \le \sum_{m=1}^{\infty}\mathscr{P}(F_{m,k(m)}^c) < \sum_{m=1}^{\infty}\frac{\varepsilon}{2^m} = \varepsilon,$$

which is a desired result. $\square$

Exercise 4.2.6. Cauchy convergence of $\{X_n\}$ in pr. (or in L^p) implies the existence of an X (finite a.e.), such that X_n converges to X in pr. (or in L^p). [HINT: Choose n_k so that

$$\sum_k \mathscr{P}\left\{|X_{n_{k+1}} - X_{n_k}| > \frac{1}{2^k}\right\} < \infty;$$

cf. Theorem 4.2.3.]

Proof. As $\{X_n\}$ is of Cauchy convergence in pr., $\forall k = 1, 2, \cdots, \exists N > 0$ such that

$$\mathscr{P}\left\{|X_m - X_n| > \frac{1}{2^k}\right\} < \frac{1}{2^k}, \quad \forall m, n > N.$$

For $k = 1$, we set $m = n_2, n = n_1$; For $k = 2$, we set $m = n_3 > n_2, n = n_2$, …By theorem 4.2.1, we know $\mathscr{P}\{|X_{n_{k+1}} - X_{n_k}| > 1/2^k \text{ i.o.}\} = 0$, which means $\forall\omega \in \Omega\backslash\mathbf{N}$, where $\mathscr{P}(\mathbf{N}) = 0$, $X_{n_k}(\omega)$ converges. Now we find a sequence $\{X_{n_k}\}$ converges a.e., say $X_{n_k} \to X, X$ finite a.e. As

$$\mathscr{P}\{|X_n - X| > \varepsilon\} \le \mathscr{P}\{|X_n - X_{n_k}| > \varepsilon/2\} + \mathscr{P}\{|X_{n_k} - X| > \varepsilon/2\},$$

let $k \to \infty$, then let $n \to \infty$, we know $\lim_{n\to\infty}\mathscr{P}\{|X_n - X| > \varepsilon\} = 0$, hence $X_n \to X$ in pr. The case in L^p is similar, the only difference is we shall find a subsequence holding the same property by Chebyshev's inequality. $\square$

Exercise 4.2.7. $\{X_n\}$ converges in pr. to X if and only if every subsequence $\{X_{n_k}\}$ contains a further subsequence that converges a.e. to X. [HINT: Use Theorem 4.1.5.]

Proof. the "only if" part is by Theorem 4.2.3. Now we prove the "if" part. Without generality we can assume $X = 0$. suppose $\mathscr{E}(|X_n|/(1+|X_n|)) \nrightarrow 0$, then $\exists \varepsilon_0, \mathscr{E}(|X_n|/(1+|X_n|)) \geq \varepsilon_0$. We know $\exists \{X_{n_k}\} \to 0$ a.e., so do in pr, hence $\mathscr{E}(|X_{n_k}|/(1+|X_{n_k}|)) \to 0$, which is a contradiction, so $X_n \to X$ in pr. □

Exercise 4.2.8. Let $\{X_n, n \geq 1\}$ be any sequence of functions on Ω to $\mathscr{R}^1$ and let C denote the set of ω for which the numerical sequence $\{X_n(\omega), n \geq 1\}$ converges. Show that

$$C = \bigcap_{m=1}^{\infty} \bigcup_{n=1}^{\infty} \bigcap_{n'=n+1}^{\infty} \left\{|X_n - X_{n'}| \leq \frac{1}{m}\right\} = \bigcap_{m=1}^{\infty} \bigcup_{n=1}^{\infty} \bigcap_{n'=n+1}^{\infty} \bigwedge(m, n, n')$$

where

$$\bigwedge(m, n, n') = \left\{\omega : \max_{n<j<k\leq n'} |X_j(\omega) - X_k(\omega)| \leq \frac{1}{m}\right\}.$$

Hence if the X_n's are r.v.'s, we have

$$\mathscr{P}(C) = \lim_{m\to\infty} \lim_{n\to\infty} \lim_{n'\to\infty} \mathscr{P}\left(\bigwedge(m, n, n')\right).$$

Proof. For ω that $X_n(\omega)$ converges, according to Cauchy criterion, $\forall m, \exists N \geq 1, |X_n(\omega) - X_{n'}(\omega)| \leq 1/m, \forall n, n' > N$. From this we know $\omega \in \bigcap_{n'=n+1}^{\infty} \{|X_n - X_{n'}| \leq 1/m\}, n > N$, further $\omega \in \bigcup_{n=1}^{\infty} \bigcap_{n'=n+1}^{\infty} \{|X_n - X_{n'}| \leq 1/m\}$. As m is arbitrary, we have

$$C = \bigcap_{m=1}^{\infty} \bigcup_{n=1}^{\infty} \bigcap_{n'=n+1}^{\infty} \left\{|X_n - X_{n'}| \leq \frac{1}{m}\right\}.$$

For the second equation, we shall notice again n, n' is arbitrarily chosen from N on, so the Cauchy criterion is equivalent to

$$\forall m, \exists N \geq 1, \forall n' > n > N, \max_{n<j<k\leq n'} |X_j(\omega) - X_k(\omega)| \leq \frac{1}{m},$$

and for this equivalence we get the second equation. The last equation is directly deduced from monotone converge property. □

Exercise 4.2.9. As in Exercise 8 show that

$$\{\omega : \lim_{n\to\infty} X_n(\omega) = 0\} = \bigcap_{m=1}^{\infty} \bigcup_{k=1}^{\infty} \bigcap_{n=k}^{\infty} \left\{|X_n| \leq \frac{1}{m}\right\}.$$

Proof. For ω such that $X_n(\omega) \to 0$, for any $m \geq 1, \exists k(m)$ such that $|X_n| \leq 1/m, \forall n > k$, hence $\omega \in \bigcap_{n=k(m)}^{\infty} \{|X_n| \leq 1/m\}$, hence in $\bigcup_{k=1}^{\infty} \bigcap_{n=k}^{\infty} \{|X_n| \leq 1/m\}$. Then by the arbitrarity of m and the equivalence when stating the facts above, we have

$$\{\omega : \lim_{n\to\infty} X_n(\omega) = 0\} = \bigcap_{m=1}^{\infty} \bigcup_{k=1}^{\infty} \bigcap_{n=k}^{\infty} \left\{|X_n| \leq \frac{1}{m}\right\}.$$

□

Exercise 4.2.10. Suppose that for $a < b$, we have

$$\mathscr{P}\{X_n < a \text{ i.o. and } X_n > b \text{ i.o.}\} = 0;$$

then $\lim_{n\to\infty} X_n$ exists a.e. but it may be infinite. [HINT: Consider all pairs of rational numbers (a, b) and take a union over them.]

Proof. Consider the set where $\lim_{n\to\infty} X_n$ do not exist, which is $N = \{\omega : \limsup_n X_n(\omega) > \liminf_n X_n(\omega)\}$. And we can put two rational numbers $a, b (a < b)$ in $(X_*(\omega), X^*(\omega))$, where

$$X^*(\omega) = \limsup_n X_n(\omega), X_*(\omega) = \liminf_n X_n(\omega).$$

From the definition of upper and lower limit, we know there are infinite n such that $X_n(\omega) < a$ and $X_n(\omega) > b$. Now consider all rational intervals, we have

$$N \subset \bigcup_{a,b\in\mathbb{Q},a<b} \{X_n < a \text{ i.o. and } X_n > b \text{ i.o.}\}.$$

Then by subadditivity we get $\mathscr{P}(N) = 0$. Hence $\lim_{n\to\infty} X_n$ exists a.e., and can be infinite as $X^*(\omega), X_*(\omega)$ can be infinite. □

Below X_n, Y_n are r.v.'s, E_n events.

Exercise 4.2.11. Give a trivial example of dependent $\{E_n\}$ satisfying the hypothesis but not the conclusion of (6); give a less trivial example satisfying the hypothesis but with $\mathscr{P}(\limsup_n E_n) = 0$. [HINT: Let E_n be the event that a real number in [0,1] has its n-ary expansion begin with 0.]

Proof. Let $E_n = C, \forall n \geq 1$, then E_n are dependent and $\sum_n \mathscr{P}(E_n) = \infty$. However, $\mathscr{P}(E_n \text{ i.o.}) = \mathscr{P}(C)$. As long as $\mathscr{P}(C) < 1$, the conclusion does not hold. Next, let E_n be the event that a real number in [0,1] has its n-ary expansion begin with 0, then $E_n \supset E_{n+1}$, thus not dependent. As it is defined on $(\mathscr{U}, \mathscr{B})$, we have

$$\sum_{n=1}^{\infty} \mathscr{P}(E_n) = \sum_{n=1}^{\infty} \frac{1}{n} = \infty.$$

However, $\mathscr{P}(\limsup_n E_n) = \mathscr{P}(\lim_n E_n) = 0$, as $E_n \to \{0\}$. □

Exercise 4.2.12. Prove that the probability of a convergence of a sequence of independent r.v's is equal to zero or one.

Proof. Let $E_n(x) = \{X_n > x\}$. We don't need to care about whether $\sum_n \mathscr{P}(E_n)$ converges or diverges, use the corollary of Theorem 4.2.5, we have

$$\mathscr{P}(\limsup_n E_n) = \mathscr{P}\{\limsup_n X_n > x\} = 0 \text{ or } 1.$$

Hence $\limsup_n X_n$ is a constant a.e., let it be x^*, similarly $\liminf_n X_n = x_*$ a.e. According to $x^* = x_*$ and $x^* \neq x_*$, we have $\mathscr{P}(\omega : \lim_n X_n(\omega) \text{ exists}) = 1$ or 0. □

Exercise 4.2.13. If $\{X_n\}$ is a sequence of independent and identically distributed r.v.'s not constant a.e., than $\mathscr{P}\{X_n \text{ converges}\} = 0$.

Proof. Note that $\limsup_n X_n \neq \liminf_n X_n$ as X_n's are not constant a.e. and X_n's are identically distributed, then by Exercise 4.2.12, we know $\mathscr{P}\{X_n \text{ converges}\} = 0$. □

Exercise 4.2.14. If $\{X_n\}$ is a sequence of independent r.v.'s with d.f.'s $\{F_n\}$, then $\mathscr{P}\{\lim_n X_n = 0\} = 1$ if and only if $\forall \varepsilon > 0 : \sum_n \{1 - F_n(\varepsilon) + F_n(-\varepsilon)\} < \infty$.

Proof. The summation is actually

$$\sum_n \mathscr{P}(X_n > \varepsilon \text{ or } X_n \le -\varepsilon) \le \sum_n \mathscr{P}\{|X_n| \ge \varepsilon\}.$$

We can ignore the case $|X_n| = \varepsilon$ because ε is arbitrary. If $\mathscr{P}\{\lim_n X_n = 0\} = 1$, we have $\mathscr{P}\{|X_n| > \varepsilon \text{ i.o.}\} = 0$. Assume the summation diverges, then $\mathscr{P}\{|X_n| > \varepsilon \text{ i.o.}\} = 1$, which is a contradiction. Conversely, if the summation is finite, then $\mathscr{P}\{X_n > \varepsilon \text{ or } X_n \le -\varepsilon \text{ i.o.}\} = 0$, which is to say $\mathscr{P}\{|X_n| \le \varepsilon\} = 1$ from an n on. As ε is arbitrary, this tells $\mathscr{P}\{\lim_n X_n = 0\} = 1$. $\square$

Exercise 4.2.15. If $\sum_n \mathscr{P}(|X_n| > n) < \infty$, then

$$\limsup_n \frac{|X_n|}{n} \le 1 \text{ a.e.}$$

Proof. By Borel-Cantelli lemma, we know $\mathscr{P}(|X_n| > n \text{ i.o.}) = 0$, hence for each $\omega \in \{|X_n| > n \text{ i.o.}\}^c$, $\exists N(\omega)$ such that

$$|X_n| \le n \Leftrightarrow \frac{|X_n|}{n} \le 1, \forall n > N(\omega).$$

Hence $\limsup_n |X_n|/n \le 1$ a.e. as the events in the complement set is of probability one. $\square$

Exercise 4.2.16. Strengthen Theorem 4.2.4 by proving that

$$\lim_{n\to\infty} \frac{J_n}{\mathscr{E}(J_n)} = 1 \text{ a.e.}$$

[HINT: Take a subsequence $\{k_n\}$ such that $\mathscr{E}(J_{n_k}) \sim k^2$; prove the result first for this subsequence by estimating $\mathscr{P}\{|J_k - \mathscr{E}(J_k)| > \delta\mathscr{E}(J_k)\}$; the general case follows because if $n_k \le n < n_{k+1}$,

$$J_{n_k}/\mathscr{E}(J_{n_{k+1}}) \le J_n/\mathscr{E}(J_n) \le J_{n_{k+1}}/\mathscr{E}(J_{n_k}).]$$

Proof. As $\mathscr{E}(J_n) \to \infty$, we can choose a subsequence from $\{J_n\}$, say $\{J_{n_k}\}$, such that $\mathscr{E}(J_{n_k}) \sim k^2$. Now for this subsequence, by Chebyshev's inequality, we have

$$\begin{aligned}\mathscr{P}\{|J_{n_k} - \mathscr{E}(J_{n_k})| > \delta\mathscr{E}(J_{n_k})\} &\le \frac{1}{\delta^2} \cdot \frac{\sigma^2(J_{n_k})}{\mathscr{E}(J_{n_k})^2} = \frac{1}{\delta^2} \cdot \frac{\sum_{i=1}^{n_k} p_i(1-p_i)}{\left(\sum_{i=1}^{n_k} p_i\right)^2} \\ &\le \frac{1}{\delta^2 \mathscr{E}(J_{n_k})} \sim \frac{1}{\delta^2 k^2}.\end{aligned}$$

Then by Borel-Cantelli lemma, as $\sum_k \frac{1}{k^2} < \infty$, we know $J_{n_k}/\mathscr{E}(J_{n_k}) \to 1$ a.e. Then, for n such that $n_k \le n < n_{k+1}, J_{n_k} \le J_n \le J_{n_{k+1}}$, we have

$$\frac{J_{n_k}}{\mathscr{E}(J_{n_{k+1}})} \le \frac{J_n}{\mathscr{E}(J_n)} \le \frac{J_{n_{k+1}}}{\mathscr{E}(J_{n_k})}.$$

For the left side,

$$\frac{J_{n_k}}{\mathscr{E}(J_{n_{k+1}})} = \frac{J_{n_k}}{\mathscr{E}(J_{n_k})} \cdot \frac{\mathscr{E}(J_{n_k})}{\mathscr{E}(J_{n_{k+1}})} \sim \frac{k^2}{(k+1)^2} \to 1 \ (k \to \infty),$$

so do right side. Let $k \to \infty$, then $n \to \infty$, hence we have $\lim_{n\to\infty} J_n/\mathscr{E}(J_n) = 1$ a.e. Hence $\sum_{n=1}^{\infty} I_n = +\infty$ a.e. $\square$

Exercise 4.2.17. If $\mathscr{E}(X_n) = 1$ and $\mathscr{E}(X_n^2)$ is bounded in n, then

$$\mathscr{P}\{\overline{\lim_{n\to\infty}} X_n \geq 1\} > 0.$$

[This is due to Kochen and Stone. Truncate X_n at A to Y_n with A so large that $\mathscr{E}(Y_n) > 1-\varepsilon$ for all n; then apply Exercise 6 of Sec. 3.2.]

Proof. Let $Y_n = X_n 1_{\{X_n \leq A\}}$, then

$$1 - \mathscr{E}(Y_n) = \mathscr{E}(X_n) - \mathscr{E}(Y_n) = \mathscr{E}(X_n 1_{\{X_n > A\}}) \leq \mathscr{E}(X_n^2)^{\frac{1}{2}} \sqrt{\mathscr{P}(X_n > A)}.$$

As $\mathscr{E}(X_n^2)$ is bounded and $\mathscr{P}(X_n = \infty) = 0$, $\forall \varepsilon > 0$, we can find $A(\varepsilon) < \infty$ such that $\mathscr{E}(Y_n) > 1 - \varepsilon$ for all n. Now use Exercise 3.2.6, we get

$$\mathscr{E}(\overline{\lim_n} X_n) \geq \mathscr{E}(\overline{\lim_n} Y_n) \geq \overline{\lim_n} \mathscr{E}(Y_n) > 1 - \varepsilon.$$

As ε is arbitrary, we have $\mathscr{E}(\overline{\lim}_n X_n) \geq 1$. Assume $\mathscr{P}\{\overline{\lim}_{n\to\infty} X_n \geq 1\} = 0$, then $\overline{\lim}_n X_n < 1$ a.e. then $\mathscr{E}(\limsup_n X_n) < 1$, which is a contradiction. Hence $\mathscr{P}\{\overline{\lim}_{n\to\infty} X_n \geq 1\} > 0$. □

Exercise 4.2.18. Let $\{E_n\}$ be events and I_n their indicators. Prove the inequality

$$\mathscr{P}\left(\bigcup_{k=1}^n E_k\right) \geq \left\{\mathscr{E}\left(\sum_{k=1}^n I_k\right)\right\}^2 \Bigg/ \mathscr{E}\left\{\left(\sum_{k=1}^n I_k\right)^2\right\}.$$

Deduce from this that if (i) $\sum_n \mathscr{P}(E_n) = \infty$ and (ii) there exists $c > 0$ such that we have

$$\forall m < n : \mathscr{P}(E_m E_n) \leq c\mathscr{P}(E_m)\mathscr{P}(E_{n-m});$$

then

$$\mathscr{P}\{\limsup_n E_n\} > 0.$$

Proof. Let $S_n = \sum_{k=1}^n I_k$, then $\{S_n > 0\} = \bigcup_{k=1}^n E_k, S_n = S_n \cdot 1_{\{S_n>0\}}$. Take expectation on both sides, use Cauchy-Schwarz inequality we get

$$\left\{\mathscr{E}\left(\sum_{k=1}^n I_k\right)\right\}^2 = \mathscr{E}(S_n \cdot 1_{\{S_n>0\}})^2 \leq \mathscr{E}(S_n^2)\mathscr{E}(1_{\{S_n>0\}}) = \mathscr{P}\left(\bigcup_{k=1}^n E_k\right)\mathscr{E}\left\{\left(\sum_{k=1}^n I_k\right)^2\right\}.$$

This is a particular case of **Paley-Zygmund inequality**. Then we turn to second equation. We do an estimation on $\mathscr{E}(S_n^2)$:

$$\begin{aligned}
\mathscr{E}(S_n^2) &= \mathscr{E}(S_n) + 2\sum_{1\leq j<k\leq n} \mathscr{E}(I_j I_k) \\
&\leq \mathscr{E}(S_n) + 2c \sum_{1\leq j<k\leq n} \mathscr{E}(I_j)\mathscr{E}(I_{k-j}) \\
&= \mathscr{E}(S_n) + 2c\sum_{k=1}^{n-1} \mathscr{E}(I_k)\mathscr{E}(S_{n-k}) \\
&\leq \mathscr{E}(S_n) + 2c[\mathscr{E}(S_n)]^2.
\end{aligned}$$

By condition (1), we have

$$\mathscr{P}\left(\bigcup_{k=1}^{\infty} E_k\right) \geq \frac{\mathscr{E}(S_n)^2}{\mathscr{E}(S_n) + 2c\mathscr{E}(S_n)^2} = \frac{1}{1/\mathscr{E}(S_n) + 2c} \to \frac{1}{2c}(n \to \infty).$$

Now we also have $\mathscr{P}\left(\bigcup_{k=n}^{\infty} E_k\right) \geq \frac{1}{2c}$, by changing the beginning of the summation. Hence

$$\mathscr{P}\{\limsup_n E_n\} = \lim_{k\to\infty} \mathscr{P}\left(\bigcup_{n=k}^{\infty} E_n\right) \geq \frac{1}{2c} > 0. \qquad \square$$

Exercise 4.2.19. If $\sum_n \mathscr{P}(E_n) = \infty$ and

$$\varliminf_n \left\{\sum_{j=1}^{n}\sum_{k=1}^{n} \mathscr{P}(E_j E_k)\right\} \Bigg/ \left\{\sum_{k=1}^{n} \mathscr{P}(E_k)\right\}^2 = 1,$$

then $\mathscr{P}\{\limsup_n E_n\} = 1$. [HINT: Use (11) above.]

Proof. Denote I_k the indicator of E_k, then immediately we get

$$\left\{\sum_{j=1}^{n}\sum_{k=1}^{n} \mathscr{P}(E_j E_k)\right\} \Bigg/ \left\{\sum_{k=1}^{n} \mathscr{P}(E_k)\right\}^2 = \frac{\mathscr{E}(S_n^2)}{\mathscr{E}(S_n)^2},$$

where S_n has the same meaning as what is stated in Exercise 4.2.18. Hence $\varlimsup_{n\to\infty} \mathscr{E}(S_n)^2/\mathscr{E}(S_n^2) = 1$. By the inequality in Exercise 4.2.18, we get

$$1 \geq \mathscr{P}\left(\bigcup_{k=1}^{\infty} E_k\right) \geq \varlimsup_{n\to\infty} \mathscr{E}(S_n)^2/\mathscr{E}(S_n^2) = 1.$$

Now we consider the set $\bigcup_{k=m+1}^{n} E_k (m+1 < n)$. For each m, We have

$$\begin{aligned}\frac{\mathscr{E}\{(S_n - S_m)^2\}}{\{\mathscr{E}(S_n - S_m)\}^2} &\geq \frac{\mathscr{E}(S_n^2) - \mathscr{E}(S_m^2) - 2\mathscr{E}(S_m(S_n - S_m))}{\mathscr{E}(S_n)^2} \\ &\geq \frac{\mathscr{E}(S_n^2) - \mathscr{E}(S_m^2) - 2m\mathscr{E}(S_n - S_m)}{\mathscr{E}(S_n)^2}.\end{aligned}$$

As $\mathscr{E}(S_n) \to \infty$, We have

$$1 \geq \varliminf_{n\to\infty} \frac{\mathscr{E}\{(S_n - S_m)^2\}}{\{\mathscr{E}(S_n - S_m)\}^2} \geq \varliminf_{n\to\infty} \left[\frac{\mathscr{E}(S_n^2)}{\mathscr{E}(S_n)^2} - \frac{\mathscr{E}(S_m^2) - 2\mathscr{E}(S_m(S_n - S_m))}{\mathscr{E}(S_n)^2}\right] = 1.$$

Now we have $\mathscr{P}\left(\bigcup_{k=m+1}^{\infty} E_k\right) = 1$ for each m. Hence

$$\mathscr{P}(\limsup_n E_n) = \lim_{k\to\infty} \mathscr{P}\left(\bigcup_{n=k}^{\infty} E_n\right) = 1. \qquad \square$$

Exercise 4.2.20. Let $\{E_n\}$ be arbitrary events satisfying

$$\text{(i) } \lim_n \mathscr{P}(E_n) = 0, \quad \text{(ii) } \sum_n \mathscr{P}(E_n E_{n+1}^c) < \infty;$$

then $\mathscr{P}\{\limsup_n E_n\} = 0$. [This is due to Barndorff-Nielson.]

Proof. We shall notice the relation

$$\bigcup_{k=M}^{N} E_k \subset E_M \cup \bigcup_{k=M}^{N-1} E_k E_{k+1}^c.$$

Then we have

$$\mathscr{P}\left(\bigcup_{k=M}^{N} E_k\right) \le \mathscr{P}(E_M) + \sum_{k=M}^{N} \mathscr{P}(E_k E_{k+1}^c), \quad \forall M, N.$$

Let $N \to \infty$, we have

$$\mathscr{P}\left(\bigcup_{k=n}^{\infty} E_k\right) \le \mathscr{P}(E_n) + \sum_{k=n}^{\infty} \mathscr{P}(E_k E_{k+1}^c) < \infty.$$

Now as condition (i) and (ii) holds, let $n \to \infty$, we get

$$0 \le \mathscr{P}\{\limsup_n E_n\} \le 0 + 0 = 0.$$ □

4.3 Vague convergence

Exercise 4.3.1. Perhaps the most logical approach to vague convergence is as follows. The sequence $\{\mu_n, n \ge 1\}$ of s.p.m's is said to converge vaguely iff there exists a dense subset D of $\mathscr{R}^1$ such that for every $a \in D, b \in D, a < b$, the sequence $\{\mu_n(a, b), n \ge 1\}$ converges. The definition given before implies this, of course, but prove the converse.

Proof. Let $\mu_n(a, b) \to \mu_0(a, b), \forall a, b \in D$, as

$$(a, b] = (a, c)\backslash(b, c), \quad c \in D,$$

of course $\mu_n(a, b] \to \mu_0(a, b]$. Now let $F_n(x) \to G(x), \forall x \in \mathscr{R}^1$, where $F_n(x)$ is the s.d.f. corresponding to μ_n. Now let $F(x) = \inf_{x<r\in D} G(r)$, then F is a s.d.f., with a unique s.p.m. μ correponding to it. Let C denote the continuous point of F, as D is dense in $\mathscr{R}^1$, $\exists a', a'', b', b'' \in D, a, b \in C, a' < a < a'' < b'' < b < b'$. We have

$$\mu_n(a'', b''] \le \mu_n(a, b] \le \mu_n(a', b'].$$

Let $n \to \infty$, we have $\mu_n(a'', b''] \to \mu_0(a'', b''] = G(b'') - G(a''), \mu_n(a', b'] \to \mu_0(a', b'] = G(b') - G(a')$. Let $a' \to a, a'' \to a, b' \to b, b'' \to b$, as $a, b \in C$, we have

$$\lim_{n\to\infty} \mu_n(a, b] = F(b) - F(a) = \mu(a, b] = \mu_0(a, b].$$

Hence $\mu_n \xrightarrow{v} \mu_0$ on dense set C. □

Exercise 4.3.2. Prove that if (1) is true, then there exists a dense set D', such that $\mu_n(I) \to \mu(I)$ where I may be any of the four intervals $(a, b), (a, b], [a, b), [a, b]$ with $a \in D', b \in D'$.

Proof. Let D' denote the continuous point of μ, then we have $\mu(a, b] = \mu(a, b) = \mu[a, b) = \mu[a, b]$. We use $\mu(a, b)$ as an example, others are similar. Let $a, b \in D'$, then

$$\overline{\lim_{n\to\infty}} \mu_n(a, b) \le \lim_{n\to\infty} \mu_n(a, b] = \mu(a, b),$$

and let $a < b' < b, b' \in D'$, then we have

$$\underline{\lim_{n\to\infty}} \mu_n(a, b) \ge \lim_{n\to\infty} \mu_n(a, b'] = \mu(a, b').$$

Let $b' \to b$, we get $\lim_{n\to\infty} \mu_n(a, b) = \mu(a, b)$. □

Exercise 4.3.3. Can a sequence of absolutely continuous p.m.'s converge vaguely to a discrete p.m.? Can a sequence of discrete p.m.'s converge vaguely to an absolutely continuous p.m.?

Proof. Consider the uniform distribution on $(-1/n, 1/n)$, which has a absolutely continuous p.m. Let $f_n(t) = 2n \cdot 1_{(-1/n,1/n)}$, we have

$$F_n(x) = \int_{-\infty}^{x} f_n(t)\mathrm{d}t = \begin{cases} 2n\left(x + \frac{1}{n}\right), & -\frac{1}{n} < x < \frac{1}{n}, \\ 0, & x \le -\frac{1}{n}, \\ 1, & x \ge \frac{1}{n}. \end{cases}.$$

Let $F(x) = \delta_0(x)$, i.e. $\mu(\{0\}) = 1$, for each continuity interval $(a, b]$, if $0 \notin (a, b]$, then $\exists N(a, b)$ such that

$$\frac{1}{N} < \min\{|a|, |b|\},$$

which means $\mu_n(a, b] \to 0 = \mu(a, b]$. If $0 \in (a, b]$, similarly we can prove $\mu_n \to \mu$.

Now let $F_n(x) = \frac{1}{n}\sum_{k=1}^{n} \delta_{k/n}(x)$, which is a step function and is discrete, denote the d.f. of μ_n, and $F(x) = x, x \in [0, 1]$, with p.m. μ correponding to it. Now any $(a, b], a, b \in \mathscr{R}^1$ is the continuity interval of μ. Indeed,

$$|F_n(x) - x| \le \frac{1}{n}, \ \ \forall x \in [0, 1],$$

and $F_n(x) = F(x)$ on $(-\infty, 0) \cup (1, +\infty)$, hence $F_n(x) \rightrightarrows F(x)$, of course we have $\mu_n \xrightarrow{v} \mu$. □

Exercise 4.3.4. If a sequence of p.m.'s converges vaguely to an atomless p.m., then the convergence is uniform for all intervals, finite or infinite. (This is due to Pólya.)

Proof. Let $\mu_n \xrightarrow{v} \mu$. First as $\mathscr{R}^1$ construct the continuous point of μ, we have $\mu_n(a, b] \to \mu(a, b]$ for every interval. Let F_n and F denote the d.f. of μ_n and μ, then we have $F_n(x) \to F(x), \forall x$. Now choose x_j such that $F(x_j) = j/k, j = 1, 2, \cdots, k-1$, and let $x_0 = -\infty, x_k = +\infty$, $\forall x, \exists j$ such that $x \in (x_{j-1}, x_j], j = 1, 2, \cdots, k$. Then use monotonicity of F_n and F,

$$\begin{aligned} -\frac{1}{k} + F_n(x_{j-1}) - F(x_{j-1}) = F_n(x_{j-1}) - F(x_j) \le F_n(x) - F(x) \\ \le F_n(x_j) - F(x_{j-1}) = F_n(x_j) - F(x_j) + \frac{1}{k}. \end{aligned}$$

This can be rewritten as

$$|F_n(x) - F(x)| \le \max\{|F_n(x_{j-1}) - F(x_{j-1})|, |F_n(x_j) - F(x_j)|\} + \frac{1}{n}.$$

As this inequality holds for any x, we have

$$\sup |F_n(x) - F(x)| \le \max_{1\le j\le n-1} |F_n(x_j) - F(x_j)| + \frac{1}{k}.$$

$\forall \varepsilon > 0$, for each $x_j, \exists N_j$ such that $|F_n(x_j) - F_(x_j)| < \varepsilon/2, n > N_j$, and N such that $1/k < \varepsilon/2, k > N$. Let $N_0 = \max\{N, N_1, \cdots, N_{n-1}\}$, we have $\sup |F_n(x) - F(x)| < \varepsilon$. Hence $F_n(x) \rightrightarrows F(x)$. Take interval $(a, b]$ as an example, we have

$$|\mu_n(a, b] - \mu(a, b]| \le |F_n(a) - F(a)| + |F_n(b) - F(b)| \le 2 \sup_{x\in\mathscr{R}^1} |F_n(x) - F(x)| \to 0.$$

Hence the uniformly convergence also applies for each intervals. □

Exercise 4.3.5. Let $\{f_n\}$ be a sequence of functions increasing in $\mathscr{R}^1$ and uniformly bounded there: $\sup_{n,x}|f_n(x)| \leq M < \infty$. Prove that there exists an increasing function f on $\mathscr{R}^1$ and a subsequence $\{n_k\}$ such that $f_{n_k}(x) \to f(x)$ for every x. (This is a form of Theorem 4.3.3 frequently given; the insistence on "every x" requires an additional argument.)

Proof. Use the similar method as Theorem 4.3.3, we get $g(x)$ converging at each rationals $r_1, r_2, \cdots$. We assert that $g(x)$ is increasing, to see this, when constructing g, we get a sequence $f_{kk}(x)$, and for each k, we have

$$f_{kk}(r_1) \leq f_{kk}(r_2), \ \ \forall r_1 < r_2 \text{ and } k \geq 1.$$

Let $k \to \infty$ and the monotonicity is proved. Then we define

$$f(x) = \inf_{x<r\in\mathbb{Q}} g(r),$$

then f is increasing and bounded, hence with countable discontinuous points. Let C denote its continuous point, let $x \in C$, and $f = g$ on C. Now for those discontinuous points, we can use the diagonal method again for $\{f_{kk}(x)\}$, then substitute the value of these points in C with new limit, and we get the final sequence $f_{n_k}(x) \to f(x)$ for every x. □

Exercise 4.3.6. Let $\{\mu_n\}$ be a sequence of finite measures on $\mathscr{B}^1$. It is said to converge vaguely to a measure μ iff (1) holds. The limit μ is not necessarily a finite measure. But if $\mu_n(\mathscr{R}^1)$ is bounded in n, then μ is finite.

Proof. Consider the Lebesgue measure on $\mathscr{R}^1$, denote it μ, and let μ_n represent the constraint of Lebesgue measure on $[-n, n]$, then for sufficiently large n, $(a, b] \subset [-n, n], a, b < \infty$, hence $\mu_n \xrightarrow{v} \mu$, and μ is an infinite measure. When $\mu_n(\mathscr{R}^1)$ is bounded in n, as

$$\mu(\mathscr{R}^1) = \lim_{k\to\infty} \mu(a_k, b_k] = \lim_{k\to\infty} \lim_{n\to\infty} \mu_n(a_k, b_k],$$

and $\sup_{a,b\in\mathscr{R}^1, n\geq 1} \mu_n(a, b] \leq M$, we have $\mu(a_k, b_k) \leq M, \forall k \geq 1$, and let $k \to \infty$ we get $\mu(\mathscr{R}^1) \leq M$. □

Exercise 4.3.7. If $\mathscr{P}_n$ is a sequence of p.m.'s on $(\Omega, \mathscr{F})$ such that $\mathscr{P}_n(E)$ converges for every $E \in \mathscr{F}$, then the limit is a p.m. $\mathscr{P}$. Furthermore, if f is bounded and $\mathscr{F}$−measurable, then

$$\int_\Omega f \mathrm{d}\mathscr{P}_n \to \int_\Omega f \mathrm{d}\mathscr{P}.$$

(The first assertion is the Vitali-Hahn-Saks Theorem and rather deep, but it can be proved by reducing it to a problem of summability; see A. Rényi, [24].)

Proof. As $\mathscr{P}_n(\Omega) = 1$, we have $\mathscr{P}(\Omega) = 1$, and $\mathscr{P}(E) \geq 0$ is obvious. Then $\mathscr{P}$ has finite additivity. Suppose $\mathscr{P}$ do not bear countable additivity, then continuity do not hold. Hence for a sequence of sets $E_n \downarrow \varnothing, \exists \delta > 0$ such that $\mathscr{P}(E_n) \geq 2\delta, \forall n \geq 1$. As $\mathscr{P}_n(E_k) \to \mathscr{P}(E_k) \geq 2\delta$, choose a beginning index k_0 and n_1 such that $\mathscr{P}_{n_1}(E_{k_0}) > \delta$. As $E_k \to \varnothing, \exists k_1$ such that $\mathscr{P}_{n_1}(E_{k_1}) < \delta/5$. Let $G_1 = E_{k_0} \backslash E_{k_1}$, we have $\mathscr{P}_{n_1}(G_1) > 4\delta/5$. Now assume the first $j-1$ G_j is determined, for j th, let

$$F_j = \bigcup_{i<j,\ i \text{ even}} G_j,$$

and we can choose n_j such that $|\mathscr{P}_{n_j}(F_{j-1}) - \mathscr{P}(F_{j-1})| < \delta/5$ and $\mathscr{P}_{n_j}(E_{k_{j-1}}) > \delta$, we can again choose k_j such that $\mathscr{P}_{n_j}(E_{k_j}) < \delta/5$. And we have $\mathscr{P}_{n_j}(G_j) = \mathscr{P}_{n_j}(E_{k_{j-1}} \backslash E_{k_j}) > 4\delta/5$. Let $G = \bigcup_{j=1}^{\infty} G_{2j} = \lim_{k\to\infty} F_k$, for even j, we have

$$\mathscr{P}_{n_j}(G) = \mathscr{P}_{n_j}(F_{j-1}) + \mathscr{P}_{n_j}(G_j) + \mathscr{P}_{n_j}(G \backslash F_{j+1}) \geq \mathscr{P}_{n_j}(F_{j-1}) + \frac{4\delta}{5} > \mathscr{P}(F_{j-1}) + \frac{3\delta}{5},$$

for odd j,

$$\mathscr{P}_{n_j}(G) = \mathscr{P}_{n_j}(F_{j-1}) + \mathscr{P}_{n_j}(G \backslash F_j) \leq \mathscr{P}_{n_j}(F_{j-1}) + \frac{\delta}{5} < \mathscr{P}(F_{j-1}) + \frac{2\delta}{5}.$$

This is a contradiction. Hence we have countable additivity for $\mathscr{P}$.

Now for the second assertion, when f is a simple function, the conclusion holds (easy to verify), then for arbitrary bounded f and $\varepsilon > 0$, there exists a simple function s such that $|s - f| < \varepsilon/3$. For this ε, we can also find N such that $\left|\int_\Omega s \mathrm{d}\mathscr{P}_n - \int_\Omega s \mathrm{d}\mathscr{P}\right| < \varepsilon/3$. Then

$$\begin{aligned} \left|\int_\Omega f \mathrm{d}\mathscr{P}_n - \int_\Omega f \mathrm{d}\mathscr{P}\right| &\leq \int_\Omega |f - s| \mathrm{d}\mathscr{P}_n + \int_\Omega |f - s| \mathrm{d}\mathscr{P} + \left|\int_\Omega s \mathrm{d}\mathscr{P}_n - \int_\Omega s \mathrm{d}\mathscr{P}\right| \\ &\leq \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon. \end{aligned}$$

Hence we have the assertion. □

Exercise 4.3.8. If μ_n and μ are p.m.'s and $\mu_n(E) \to \mu(E)$ for every open set E, then this is also true for every Borel set. [HINT: Use (7) of Sec. 2.2.]

Proof. Obviously this convergence applies for all closed sets. Now for each Borel set B, there exist a closed set C and an open set U, $C \subset B \subset U$, such that

$$\begin{gathered} \mu_n(C) \leq \mu_n(B) \leq \mu_n(U), \\ \mu(U) - \varepsilon \leq \mu(B) \leq \mu(C) + \varepsilon. \end{gathered}$$

Let $n \to \infty$ in the first term, and take difference on two inequalities, then we get

$$\overline{\lim_{n\to\infty}} |\mu_n(B) - \mu(B)| < \varepsilon.$$

Let $\varepsilon \downarrow 0$, we get $\mu_n(B) \to \mu(B)$. □

Exercise 4.3.9. Prove a convergence theorem in metric space that will include both Theorem 4.3.3 for p.m.'s and the analogue for real numbers given before the theorem. [HINT: Use Exercise 9 of Sec. 4.4.]

Proof. □

4.4 Continuation

Exercise 4.4.1. Let μ_n and μ be p.m.'s such that $\mu_n \xrightarrow{v} \mu$. Show that the conclusion in (2) need not hold if (a) f is bounded and Borel measurable and all μ_n and μ are absolutely continuous, or (b) f is continuous except at one point and every μ_n is absolutely continuous. (To find even sharper counterexamples would not be too easy, in view of Exercise 10 of Sec. 4.5.)

Proof. For (a), let μ be the uniform distribution on $(0,1]$, and $g_n = n1_{A_n}$, where

$$A_n = \bigcup_{k=0}^{n-1}\left[\frac{k}{n}, \frac{k}{n} + \frac{1}{n^2}\right] \quad (n \geq 2)$$

and g_n is the density of μ_n. Now take any continuous function, by mean theorem of integration, we have

$$\int_{\mathscr{R}^1} f(x)g_n(x)\mathrm{d}x = n\sum_{k=0}^{n-1}\int_{k/n}^{k/n+1/n^2} f(x)\mathrm{d}x = \frac{1}{n}\sum_{k=0}^{n-1} f(\xi_k) \to \int_0^1 f(x)\mathrm{d}x = \int_{\mathscr{R}^1} f\mathrm{d}\mu,$$

which implies $\mu_n \xrightarrow{v} \mu$. However, let $f = 1_E$, where

$$E = \bigcup_{n=2}^{\infty} A_{2^n},$$

then

$$\int_{\mathscr{R}^1} f\mathrm{d}\mu_{2^n} = \mu_{2^n}(E) \geq \mu_{2^n}(A_{2^n}) = 1, \quad \int_{\mathscr{R}^1} f\mathrm{d}\mu \leq \sum_{n=2}^{\infty}\frac{1}{2^n} = \frac{1}{2},$$

so (2) do not hold. For (b), let μ_n be the uniform distribution on $(0, 1/n]$, then $\mu_n \xrightarrow{v} \delta_0$. Let $f = 1_{(0,\infty)}$, then

$$\int_{\mathscr{R}^1} f\mathrm{d}\mu_n = 1, \quad \int_{\mathscr{R}^1} f\mathrm{d}\mu = 0. \qquad \square$$

Exercise 4.4.2. Let $\mu_n \xrightarrow{v} \mu$ when the μ_n's are s.p.m.'s. Then for each $f \in C$ and each finite continuity interval I we have $\int_I f\mathrm{d}\mu_n \to \int_I f\mathrm{d}\mu$.

Proof. Assume $I = [a,b]$, then f is continuous on I, , let g be linear on $[a-1, a), (b, b+1]$, and 0 elsewhere. Then $g \in C_K$. $\forall \varepsilon > 0$, for each continuity interval I, there exists an step function g_ε such that $\sup_{x\in\mathscr{R}^1}|g(x) - g_\varepsilon(x)| \leq \varepsilon$. Then

$$\begin{aligned}
\left|\int_I f\mathrm{d}\mu_n - \int_I f\mathrm{d}\mu\right| &= \left|\int_I g\mathrm{d}\mu_n - \int_I g\mathrm{d}\mu\right| \\
&\leq \int_I |g - g_\varepsilon|\mathrm{d}\mu_n + \left|\int_I g_\varepsilon\mathrm{d}\mu_n - \int_I g_\varepsilon\mathrm{d}\mu\right| + \int_I |g - g_\varepsilon|\mathrm{d}\mu \\
&\leq 2\varepsilon + \left|\int_I g_\varepsilon\mathrm{d}\mu_n - \int_I g_\varepsilon\mathrm{d}\mu\right|.
\end{aligned}$$

The last term tends to 0 when we choose the dense subset D such that $\mu_n(a,b] \to \mu(a,b], a, b \in D$ and let g_ε is D-valued. As ε is arbitrary, we have $\int_I f\mathrm{d}\mu_n \to \int_I f\mathrm{d}\mu$. $\square$

Exercise 4.4.3. Let μ_n and μ be as in Exercise 1. If the f_n's are bounded continuous functions converging uniformly to f, then $\int f_n\mathrm{d}\mu_n \to \int f\mathrm{d}\mu$.

Proof. First $f \in C_B$, which is a wide-known fact. As $\mu_n \xrightarrow{v} \mu$ and they are all p.m.'s and the uniformly convergence, $\forall\varepsilon > 0, \exists N$ such that $|f_n - f| \leq \varepsilon, n > N$, and we have

$$\begin{aligned}
\left|\int f_n\mathrm{d}\mu_n - \int f\mathrm{d}\mu\right| &\leq \int |f_n - f|\mathrm{d}\mu + \left|\int f\mathrm{d}\mu_n - \int f\mathrm{d}\mu\right| \\
&\leq \varepsilon + \left|\int f\mathrm{d}\mu_n - \int f\mathrm{d}\mu\right|. \ (n > N)
\end{aligned}$$

The second term converges to 0 according to Theorem 4.4.2, hence $\int f_n\mathrm{d}\mu_n \to \int f\mathrm{d}\mu$. $\square$

Exercise 4.4.4. Give an example to show that convergence in dist. does not imply that in pr. However, show that convergence to the unit mass δ_a does imply that in pr. to the constant a.

Proof. Let $X_n = 1 - X$, where X is a Bernoulli r.v. with $p = 1/2$. Obviously $X_n \to X$ in dist, but $|X_n - X| = |1 - 2X| \geq 1$, hence X_n does not converge to X in pr. Let $X_n \to a$ in dist, then $\forall \varepsilon > 0$,

$$\mathscr{P}\{|X_n - a| > \varepsilon\} = 1 - F_n(a + \varepsilon) + F_n(a - \varepsilon - 0) \to 1 - 1 + 0 = 0,$$

because of the convergence in dist. Hence $X_n \to a$ in pr. □

Exercise 4.4.5. A set $\{\mu_\alpha\}$ of p.m.'s is tight if and only if the correponding d.f.'s $\{F_\alpha\}$ converge uniformly in α as $x \to -\infty$ and $x \to +\infty$.

Proof. If $\{\mu_\alpha\}$ is tight, then for any ε_0, there exists a finite interval I such that $\int_\alpha \mu_\alpha(I) > 1 - \varepsilon$, Let $I = (a, b]$ and $M = \max\{|a|, |b|\}$, then $\forall \alpha$, we have

$$\begin{aligned} \mu_\alpha(x, +\infty) &= 1 - F_\alpha(x) \leq 1 - \mu_\alpha(I) < \varepsilon, \forall x > M, \\ \mu_\alpha(-\infty, x] &= |F_\alpha(x)| \leq 1 - \mu_\alpha(I) < \varepsilon, \forall x < -M. \end{aligned}$$

This tells $F_\alpha(x)$ is uniformly convergent in α as $x \to -\infty$ and $x \to +\infty$. Conversely, we can find $M_1 > 0$ and $M_2 > 0$ such that $\mu_\alpha(M_1, +\infty) < \varepsilon/2, \mu_\alpha(-\infty, -M_2] < \varepsilon/2$ for any $\varepsilon > 0$, let $I = (-M_2, M_1]$ and we get the conclusion. □

Exercise 4.4.6. Let the r.v.'s $\{X_\alpha\}$ have the p.m.'s $\{\mu_\alpha\}$. If for some real $r > 0$, $\mathscr{E}\{|X_\alpha|^r\}$ is bounded in α, then $\{\mu_\alpha\}$ is tight.

Proof. As $\mathscr{E}\{|X_\alpha|^r\}$ is bounded in α, we have $\mathscr{E}\{|X_\alpha|^r\} \leq C, C > 0$. Now use Markov inequality, we have

$$\mathscr{P}\{|X_\alpha| > M\} \leq \frac{\mathscr{E}\{|X_\alpha|^r\}}{M^r} \leq \frac{C}{M^r}.$$

For any $\varepsilon > 0$, we can choose $M(\varepsilon)$ such that $C/M^r < \varepsilon$. Let $I = [-M, M]$, then $\mu_\alpha(I) < 1 - \varepsilon$, hence $\{\mu_\alpha\}$ is tight. □

Exercise 4.4.7. Prove the Corollary to Theorem 4.4.4.

Proof. Let $f = 1_O$, we can easily verify that f is a lower semicontinuous function, and we can easily get the form of corollary. The closed set part is similar. Conversely, suppose $f(x)$ is a lower semicontinuous function, we can construct a series of simple function by letting $O_{k,i} = \left\{f(x) > \frac{i}{2^k}\right\}, i = 1, , \cdots, k2^k$, and

$$f_k(x) = \sum_{i=1}^{k2^k} \frac{1}{2^k} 1_{O_{k,i}}(x),$$

we shall notice $O_{k,i}$ are all open sets due to the semicontinuity of f. Obviously $f_k \leq f$ and $f_k \uparrow f$, then we have

$$\begin{aligned} \underline{\lim}_n \int f \mathrm{d}\mu_n &\geq \underline{\lim}_n \int f_k(x) \mathrm{d}\mu_n = \underline{\lim}_n \sum_{i=1}^{k2^k} \frac{1}{2^k} \mu_n(O_{k,i}) \\ &\geq \sum_{i=1}^{k2^k} \frac{1}{2^k} \mu(O_{k,i}) = \int f \mathrm{d}\mu. \end{aligned}$$

The proof for upper semicontinuous function is similar. □

Exercise 4.4.8. If the r.v.'s X and Y satisfy

$$\mathscr{P}\{|X-Y|\geq\varepsilon\}\leq\varepsilon$$

for some ε, their d.f.'s F and G satisfying the inequalities:

$$\forall x\in\mathscr{R}^1: F(x-\varepsilon)-\varepsilon\leq G(x)\leq F(x+\varepsilon)+\varepsilon.$$

Derive another proof of Theorem 4.4.5 from this.

Proof. $\forall\varepsilon>0$, notice the relation

$$\{Y\leq x\}=\{Y\leq x,|X-Y|<\varepsilon\}\cup\{Y\leq x,|X-Y|\geq\varepsilon\}\subset\{X\leq x+\varepsilon\}\cup\{|X-Y|\geq\varepsilon\},$$

then we get the right side. For the left side, the operation is similar. Now let $Y=X_n$, as $X_n\to X$ in pr, for any ε,

$$F(x-\varepsilon)-\varepsilon\leq F_n(x)\leq F(x+\varepsilon)+\varepsilon\ (n>N).$$

Hence for any continuous point of F, let $n\to\infty,\varepsilon\downarrow 0$, we get $F_n(x)\to F(x)$, hence $X_n\to X$ in dist. □

Exercise 4.4.9. The *Lévy distance* of two d.f.'s F and G is defined to be the infimum of all $\varepsilon>0$ satisfying the inequalities in (15). Prove that this is indeed a metric in the space of s.d.f.'s, and that F_n converges to F in this metric if and only if $F_n\xrightarrow{v}F$ and $\int_{-\infty}^{\infty}\mathrm{d}F_n\to\int_{-\infty}^{\infty}\mathrm{d}F$.

Proof. □

Exercise 4.4.10. Find two sequences of p.m.'s $\{\mu_n\}$ and $\{\nu_n\}$ such that

$$\forall f\in C_K:\int f\mathrm{d}\mu_n-\int f\mathrm{d}\nu_n\to 0;$$

but for *no* finite (a,b) is it true that

$$\mu_n(a,b)-\mu(a,b)\to 0.$$

[HINT: Let $\mu_n=\delta_{r_n},\nu_n=\delta_{s_n}$ and choose $\{r_n\},\{s_n\}$ suitably.]

Proof. □

Exercise 4.4.11. Let $\{\mu_n\}$ be a sequence of p.m.'s such that for each $f\in C_B$, the sequence $\int_{\mathscr{R}^1}f\mathrm{d}\mu_n$ converges; then $\mu_n\xrightarrow{v}\mu$, where μ is a p.m. [HINT: If the hypothesis is Strengthened to include every f in C, and convergence of real numbers is interpreted as usual as convergence to a finite limit, the result is easy by taking an f going to ∞. In general one may proceed by contradiction using an f that oscillates at infinity.]

Proof. □

Exercise 4.4.12. Let F_n and F be d.f.'s such that $F_n\xrightarrow{v}F$. Define $G_n(\theta)$ and $G(\theta)$ as in Exercise 4 of Sec. 3.1. Then $G_n(\theta)\to G(\theta)$ in a.e. [HINT: Do this first when F_n and F are continuous and strictly increasing. The general case is obtained by smoothing F_n and F by convoluting with a uniform distribution in $[-\delta,\delta]$ and letting $\delta\downarrow 0$; see the end of Sec. 6.1 below.]

Proof. □

4.5 Uniform integrability; convergence of moments